

**ISLAND URBAN CLIMATOLOGY: MEASURING AND MODELING NOCTURNAL
HEAT IN LEEWARD COASTAL O‘AHU, HAWAI‘I**

A THESIS SUBMITTED TO THE GRADUATE DIVISION OF THE UNIVERSITY OF
HAWAI‘I AT MĀNOA IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE
DEGREE OF

MASTER OF ARTS

IN

GEOGRAPHY

AUGUST 2027

By

Justin Kellum

Thesis Committee:

Qi Chen, Chairperson

Aurora Kagawa-Viviani

Yuqin Jiang

Keywords: urban heat island, canopy-layer air temperature, urban form, low-cost sensor
network, mixed-effects models, Hawai‘i

Acknowledgments

This work was possible only because the communities of Honolulu and ‘Ewa stewarded this monitoring project, hosting the stations and helping to deploy and recover them. I thank my committee, Dr. Qi Chen, Dr. Aurora Kagawa-Viviani and Dr. Yuqin Jiang, for their guidance and for the detailed critique that shaped this thesis.

Abstract

Cities are warmer at night than their surroundings, and some streets warmer than others. The built-environment properties that control that pattern have been studied mostly in large continental cities; whether the same relationships hold in a small, humid, trade-wind island city was unknown, and no fixed air-temperature network had been operated in Hawai‘i. This thesis measured canopy-layer air temperature with 100 low-cost logger stations in two districts of leeward O‘ahu from November 2024 to January 2025: the compact, partly high-rise Honolulu core and the low-rise suburban plain of ‘Ewa–Kapolei (74 recovered sites). On 11 calm, clear nights identified from ERA5, each site’s night-mean departure from its district median (ΔT) was related by linear mixed-effects models to eight descriptors of urban form and setting from 1m land cover, building footprints and the coastline: impervious, tree-canopy, building-footprint and water fractions, mean building height, sky view factor, canyon aspect ratio and distance to the coast.

Site means of ΔT spanned 3.4°C in Honolulu and 3.0°C in ‘Ewa, and in Honolulu the pattern was highly repeatable between nights. Descriptors within 100m of each sensor explained it best or nearly so. Pooled across districts, impervious fraction (+0.16°C per 10 percentage points) and mean building height (+0.06°C per meter) were the strongest predictors, and ΔT fell with distance from the coast (−0.18°C per km). The drivers differed between districts ($p < 0.001$): coastal distance (−0.53°C per km) and impervious fraction dominated in Honolulu, building height (about 0.13°C per meter) and impervious fraction in ‘Ewa, where the coast had no effect. Urban form explained 25% of the variance of ΔT pooled and 33–39% within districts;

a further 40% pooled was persistent site-level variation the descriptors did not capture. The relationships held under every alternative night-selection threshold and, although weaker, on all nights. Impervious cover and building mass drive nocturnal heat retention in a tropical maritime city as elsewhere, but the weight of building height, and the role of the coast, differ between neighborhood types 25km apart. A single model should not be transferred across the island, and heat adaptation in Hawai‘i should draw on ground-level air-temperature measurements.

Table of Contents

Acknowledgments.....	ii
Abstract.....	iii
List of Tables.....	viii
List of Figures.....	x
1 Introduction.....	1
1.1 Urbanization and the urban heat island.....	1
1.2 Urbanization and land-use transformation in Hawai‘i.....	2
1.3 The research gap in Hawai‘i.....	4
1.4 Why Hawai‘i is scientifically different.....	5
1.5 What is known about urban form and nighttime air temperature.....	6
1.6 Coastal and island cities: is the evidence transferable?.....	10
1.7 Study design.....	11
1.8 Purpose and research questions.....	12
2 Study area.....	14
2.1 Location and rationale.....	14
2.2 Climate.....	16
2.3 The ‘Ewa plain.....	16
2.4 The Honolulu core.....	17
3 Data and methods.....	19
3.1 Overview.....	19
3.2 Measuring canopy-layer air temperature.....	20
3.3 Data processing.....	24

3.4 Selection of calm and clear nights.....	27
3.5 The response variable.....	30
3.6 Descriptors of urban form and setting.....	32
3.7 Statistical analysis.....	37
4 Results.....	41
4.1 The analyzed record.....	41
4.2 Magnitude and persistence of the within-district pattern.....	41
4.3 Source-area scale.....	44
4.4 Collinearity.....	45
4.5 Single descriptors.....	47
4.6 The pooled model.....	50
4.7 District models and the transferability test.....	52
4.8 Robustness.....	56
4.9 Unexplained site effects.....	62
4.10 Absolute comparison of the two districts on shared nights.....	64
5 Discussion.....	66
5.1 Answers to the research questions.....	66
5.2 The drivers in the context of the literature.....	67
5.3 Transferability between districts.....	71
5.4 What the unexplained component may be.....	72
5.5 Methodological lessons.....	74
5.6 Limitations.....	75
5.7 Implications for heat adaptation on O‘ahu.....	76
5.8 Future work.....	77

6 Conclusions.....	79
Declaration of the use of artificial intelligence.....	81
Appendix A. Site table.....	82
Appendix B. Supplementary maps and diagrams.....	86
Appendix C. Data and code.....	90
Glossary.....	94
References.....	97

List of Tables

Table 1. Summary of the two deployments and of the analyzed record.....	24
Table 2. Nights on which at least 75% of the available night bins were calm and clear.....	26
Table 3. Definitions of the eight site descriptors.....	33
Table 4. Site predictors at the adopted 100m source-area radius, by study area.....	37
Table 5. Variance components of the intercept-only (null) mixed models.....	43
Table 6. Source-area scale selection.....	45
Table 7. Variance inflation factors (VIF) of the eight predictors, for each district and pooled....	46
Table 8. Single-predictor mixed models (random intercept per site).....	48
Table 9. Full pooled mixed model.....	51
Table 10. District full mixed models.....	54
Table 11. The five best subsets of at most three predictors in each study area.....	55
Table 12. Relative importance of each predictor.....	55
Table 13. Study-area $\times$ predictor interactions.....	55
Table 14. Maximum-likelihood AIC of the candidate model structures.....	56
Table 15. Sensitivity of the results to the night-selection thresholds.....	57
Table 16. Terrain check.....	59
Table 17. Leave-one-site-out cross-validation of site-mean ΔT	62
Table 18. Absolute night-mean air temperatures of the two networks on the two calm, clear nights they share.....	65
Table 19. All 74 sites: coordinates, analyzed nights, site-mean ΔT and random intercept.....	82
Table 20. All 74 sites: the eight descriptors at the 100m source-area radius.....	83
Table 21. Software used for the analysis.....	91
Table 22. Agreement between the R analysis and the independent Python replication.....	93

List of Figures

Figure 1. The study area.....	15
Figure 2. The workflow of the study.....	20
Figure 3. The low-cost temperature station.....	22
Figure 4. Deployment and recovery of stations by the author and community volunteers.....	24
Figure 5. Night selection.....	26
Figure 6. The empirical basis for the calm–clear criterion.....	29
Figure 7. Distributions of nighttime air-temperature readings on all nights and on the selected calm, clear nights.....	30
Figure 8. How the response variable is defined, on the shared night of 21 December 2024.....	31
Figure 9. The eight site descriptors at the 100m source-area radius across the Honolulu network	36
Figure 10. The eight site descriptors at the 100m source-area radius across the ‘Ewa network...	36
Figure 11. Site-mean nocturnal departure from the district median, ΔT , on calm, clear nights....	42
Figure 12. Night-to-night repeatability of the spatial pattern.....	43
Figure 13. Source-area scale.....	45
Figure 14. Pairwise Pearson correlations between the eight descriptors at the site level.....	46
Figure 15. Site-mean ΔT against each descriptor at 100m, by district.....	48
Figure 16. Standardized fixed effects of the full models.....	51
Figure 17. The two descriptors whose effects differ significantly between districts.....	54
Figure 18. Sensitivity to the night-selection thresholds.....	57
Figure 19. The terrain check.....	59
Figure 20. Leave-one-site-out cross-validation.....	62
Figure 21. The unexplained site effects.....	64

Figure 22. Absolute comparison of the two networks on the shared calm, clear nights of 1 and 21 December 2024.....	65
Figure B1. Land cover from the 2021 NOAA C-CAP 1m product.....	87
Figure B2. Terrain from the USGS 10m digital elevation model of O‘ahu.....	88
Figure B3. How the sky view factor is calculated, for site ID26 in Honolulu.....	88

1 Introduction

1.1 Urbanization and the urban heat island

More than half of the world's people (55% in 2018) now live in urban areas, a share the United Nations projects to reach 68% by 2050, with nearly 90% of the increase in Asia and Africa ([United Nations, Department of Economic and Social Affairs, Population Division, 2019](#)). The land that houses them is growing faster still: global urban land cover was forecast to expand by about 1.2 million km² between 2000 and 2030, nearly tripling in three decades, with nearly half of the increase in Asia and the fastest rates in Africa ([Seto et al., 2012](#)). Each hectare converted replaces soil and vegetation with pavement, roofs and walls, and that substitution has a predictable thermal consequence. Because built surfaces store more heat by day, evaporate less, and release stored heat more slowly by night than the land they replace, cities are warmer than their surroundings. This is the urban heat island (UHI), first documented in London two centuries ago ([Howard, 1833](#)) and given its physical explanation by Oke ([1982](#)). The effect is measured in two distinct layers ([Oke, 1976](#)): the urban canopy layer, the air between the ground and roof level in which people walk, sleep and work, and the boundary layer above it. This thesis concerns the canopy layer, measured as air temperature at roughly head height.

Only the parts of the mechanism that bear on the variables measured here need to be stated. The surface energy balance of any land parcel is $Q^* + Q_F = Q_H + Q_E + \Delta Q_s$ ([Oke, 2002](#); [Oke et al., 2017](#)). The left side is the energy supplied to the surface: Q^* is the net all-wave radiation (the incoming solar and long-wave radiation minus the radiation the surface reflects and emits), and Q_F is the anthropogenic heat released by vehicles, buildings and people. The

right side is where that energy goes: Q_H is the sensible heat flux that warms the air, Q_E is the latent heat flux used to evaporate water from soil and plants, and ΔQ_s is the heat stored in, or released from, the ground, roads and buildings. Impervious materials suppress the latent term and enlarge the storage term, so that heat absorbed by day is returned to the canopy-layer air after sunset. Building walls add storage capacity in proportion to their height and multiply the surfaces that exchange radiation with each other rather than with the sky; the narrower and deeper a street canyon, the smaller the fraction of the sky hemisphere seen from the street (the sky view factor) and the slower its radiative cooling at night (Oke, 1981; Oke et al., 1991). Vegetation acts in the opposite direction through transpiration and shade (Bowler et al., 2010). Because these processes act on stored heat and radiative loss, the canopy-layer UHI is small at midday and largest in the hours after sunset, and it is strongest on calm, clear nights when advection and cloud do not mask the differences between surfaces (Morris et al., 2001; Oke, 1982). Nights also matter for health: mortality rises when hot days are followed by nights that do not cool (Laaidi et al., 2012; Murage et al., 2017), and the frequency of dangerous heat is projected to rise in the humid tropics (Mora et al., 2017).

1.2 Urbanization and land-use transformation in Hawai‘i

Hawai‘i has seen the same conversion of vegetated and farmed land to built surfaces, compressed onto a small island. Before contact, the ‘Ewa plain and the Honolulu lowlands were managed as ahupua‘a, valley-to-sea land units in which irrigated taro terraces and coastal fishponds occupied the same lowlands that are now paved (Handy et al., 1972); statewide, urban development has removed about 13% of the land once under these traditional agroecosystems (Kurashima et al., 2019). The privatization of land after the Māhele of 1848 and the plantation

century that followed converted those lowlands first to sugar and then, after the Second World War, to housing and commerce. Honolulu County held 1,016,508 people at the 2020 census ([U.S. Census Bureau, 2021](#)), most of them on the southern coastal plain of O‘ahu, and the ‘Ewa plain, planned as the island’s “second city” on former sugar-cane land, is where much of the island’s recent suburban growth has been directed. The agricultural base continues to shrink: land in farms statewide fell from 1,135,352 to 1,053,302 acres between the 2017 and 2022 Censuses of Agriculture, a loss of 82,000 acres in five years ([U.S. Department of Agriculture, National Agricultural Statistics Service, 2024](#)).

The extent of the resulting surface transformation can be measured directly, and it is mapped in Appendix B (Figure B1). Computed for this thesis from the 1m NOAA C-CAP land-cover product for 2021 ([NOAA Office for Coastal Management, 2024](#)) within the GSHHG land polygon of the island, 197km² of O‘ahu’s 1,571km² of land is impervious surface (12.6%), against 731km² of tree canopy (46.5%), most of it on the two mountain ranges. Within the two study districts examined here the proportions are very different: 53% of the land in the Honolulu study domain is impervious and 20% is under tree canopy, whereas the ‘Ewa domain is 27% impervious and 10% tree-covered, with much of the remainder still open former cane land awaiting development. The island is also warming. Station records show a statewide temperature rise that has accelerated since the 1970s and is concentrated in nighttime minimum temperatures rather than daytime maxima ([Giambelluca et al., 2008](#); [McKenzie et al., 2019](#)). Urban expansion on the leeward coast and a warming background climate therefore act on the same nights.

1.3 The research gap in Hawai‘i

Against this background, little is known about the urban climate of Hawai‘i’s only large city. No peer-reviewed study of the canopy-layer heat island of Honolulu, or of intra-urban air-temperature variation anywhere in the state, could be found. The most substantial observational effort to date was a single community “heat watch” campaign on 31 August 2019, in which volunteers drove sensor-equipped vehicles through O‘ahu neighborhoods during one morning, afternoon and evening ([CAPA Strategies, 2019](#)). That campaign gave a useful snapshot, but by design it did not cover the nocturnal hours, distinguish persistent from transient patterns, or relate temperature to measured urban form. The planning tools now in use, the Kaulunani Tree Canopy Viewer and the heat components of the City and County of Honolulu’s *Climate Ready O‘ahu* adaptation strategy ([City and County of Honolulu, 2024](#)), draw instead on satellite land-surface temperature. Surface temperature and canopy-layer air temperature are physically different quantities. Across 342 cities the satellite surface heat island averaged 1.45°C against 0.26°C for the crowdsourced canopy-layer heat island, a sixfold difference, and the two were poorly correlated in time at night ([Venter et al., 2021](#)). Mechanistic analysis shows that the two heat islands respond to the same drivers with different weights and can diverge in diurnal behavior ([Zhang et al., 2022](#)); in humid subtropical Miami, daytime surface temperature is a limited proxy for air temperature, its relationship with air temperature weakening to insignificance in the wet summer months ([Muse et al., 2024](#)), and a dense sensor network there found daily maximum air temperatures averaging 3.3°C above the airport station ([Clement et al., 2023](#)). Whether the maps guiding heat-mitigation investment in Honolulu correspond to where the night air is actually warmest is therefore an open empirical question.

1.4 Why Hawai‘i is scientifically different

Results from elsewhere might fill that gap, but there are reasons to doubt that they transfer. The magnitude of the heat island depends on the background climate as well as on the city. Across thousands of cities the surface heat island scales with population and, non-linearly, with mean annual precipitation ([Manoli et al., 2019](#)); the daytime surface heat island is amplified in humid climates by the lower efficiency of convective heat loss from cities, whereas the nighttime surface heat island is governed by heat release and surface properties rather than by precipitation ([Zhao et al., 2014](#)). In canopy-layer air temperature, the nighttime cooling delivered by vegetation shrinks with humidity, from 5.6°C in Las Vegas to 1.4°C in Miami across eight U.S. cities ([Ibsen et al., 2021](#)). Reviews of (sub)tropical urban climate find heat-island intensities generally lower than in temperate cities of comparable population, with lower intensities in the wet season because water availability changes how energy is partitioned ([Roth, 2007](#)), and strong seasonal modulation, with dry-season maxima in some cities and wet-season maxima in others ([Lefevre et al., 2025](#)), and the global literature remains heavily weighted toward large continental cities in East Asia, Europe and North America ([Cheval et al., 2024](#); [Fernandes et al., 2023](#); [Stewart, 2011](#)).

O‘ahu’s climate differs from those settings in five ways that bear directly on nocturnal heat. First, it is a trade-wind climate: north-easterly trade winds blow on the majority of days ([Schroeder, 1993](#)), although the number of north-east trade-wind days at Honolulu fell from 291 per year in 1973 to 210 in 2009 ([Garza et al., 2012](#)), so calm nights are less common than in continental interiors but do occur and may be becoming more frequent. Second, the trade winds are capped by the trade-wind inversion, a layer beginning about 2km above sea level (2.1km on

average) in which air temperature increases with height instead of decreasing. The inversion is present about 82% of the time and acts as a lid on the air below it: it limits how high clouds and rising air can reach and keeps the moist marine air, and with it high humidity, in the lowest 2km of the atmosphere (Cao et al., 2007). Third, the ocean is close to the whole city, because the leeward coastal plain is only a few kilometers wide. The daytime sea breeze and nighttime land breeze interact with the trade-wind flow in ways that depend on the size and height of the mountain barrier (Leopold, 1949; Schroeder, 1993), so the coast ventilates the city at some hours and not at others. Fourth, humidity is high year-round: moist air increases the downward long-wave radiation and so damps the radiative cooling of all surfaces, vegetated and paved alike (Oke, 2002), and the nighttime cooling that vegetation provides shrinks as humidity rises (Ibsen et al., 2021). Fifth, topography is extreme for a city of this size: the Ko‘olau and Wai‘anae ranges rise to 900–1,200m within 5–15km of the shoreline (Appendix B, Figure B2), channel the trade-wind flow through gaps and around ridges, and create a leeward rain shadow that leaves the southern coast dry and sunny. The two study districts of this thesis, Honolulu and ‘Ewa, lie on the same lee coast, but with very different urban form (Figure 1). Together they allow a controlled comparison of how form interacts with a shared maritime setting.

1.5 What is known about urban form and nighttime air temperature

Many studies have related temperature differences within cities to measurable properties of the built environment. They are reviewed below property by property, whether a study measured temperature from satellites or with sensors on the ground. The eight predictors examined in this thesis correspond to six of the seven geometric and surface-cover properties that define Stewart and Oke’s Local Climate Zones (sky view factor, canyon aspect ratio, building surface fraction,

impervious and pervious surface fractions, and height of roughness elements; the seventh, terrain roughness class, is not used) ([Stewart & Oke, 2012](#)), with tree canopy and water separated out of the pervious fraction and distance to the coast added as a geographic setting variable.

Impervious surface. Of all single descriptors, the fraction of impervious cover has the most consistent positive relationship with urban temperature. In satellite surface temperature it is nearly linear in every season ([F. Yuan & Bauer, 2007](#)), holds across most biomes in the 38 largest U.S. cities, reversing only in arid ones ([Imhoff et al., 2010](#)), and holds across 682 cities ([Q. Yang et al., 2021](#)). In canopy-layer air temperature the relationship is weaker and is usually expressed as the contrast between impervious-dominated and vegetated sites: in Baltimore, Maryland, green spaces were about 0.5°C cooler than impervious-dominated areas and green space was a stronger predictor than elevation ([Scott, Zaitchik, et al., 2017](#)); in Portland, Oregon, canopy cover, roadway density and commercial land use separated warm from cool sites ([Hart & Sailor, 2009](#)); and in Denver, Colorado, impervious and building cover warmed the air by day while irrigated turf cooled it by night ([Ibsen et al., 2022](#)).

Vegetation. Tree canopy cools by shading and transpiration; a meta-analysis of park studies found parks about 1°C cooler than their surroundings by day, on average ([Bowler et al., 2010](#)), and European cities show consistent surface cooling by trees ([Schwaab et al., 2021](#)). The effect on air temperature is scale-dependent and non-linear: in Madison, Wisconsin, daytime cooling was greatest where canopy cover exceeded about 40% within 60–90m of the sensor, and the effects of canopy were limited at night ([Ziter et al., 2019](#)). Nighttime vegetation cooling is larger in dry than in humid climates ([Ibsen et al., 2021](#)), so a weak or absent canopy effect at night is the expectation for O‘ahu.

Building footprint and density. The fraction of ground covered by buildings and the density of development raise nighttime temperature through storage and reduced ventilation. In simulated cities the heat-island intensity rises with building density and with the mutual amplification of neighboring built-up areas (Li et al., 2020), and across real cities the surface heat island scales with city size and form (B. Zhou et al., 2017); in Dutch cities the heat island rose with population density and fell with the fraction of green vegetation (Steenekveld et al., 2011), and in Rotterdam the nocturnal heat island was linked to vegetation, built-up area and water (Heusinkveld et al., 2014); in simulations of subtropical Brisbane, Australia, densification and loss of vegetation both raised night temperatures, vegetation being the stronger control (Chapman et al., 2018).

Building height and three-dimensional form. Studies that add vertical information consistently find that it adds explanatory power beyond two-dimensional cover, but the sign of the height effect depends on the time of day and on the quantity measured. For daytime surface temperature, taller buildings often cool by shading (Huang & Wang, 2019; B. Yuan et al., 2024; Zhu et al., 2023), whereas for air temperature at night three-dimensional form added warmth in residential neighborhoods (Tian et al., 2019), and in Hong Kong's high-rise estates it was reduced sky view rather than building height that warmed the night air (Giridharan et al., 2007); studies of daytime surface temperature in Beijing find three-dimensional form to matter less than two-dimensional cover (Guo et al., 2022), with effects that vary by season (Hu et al., 2020). In residential neighborhoods, three-dimensional metrics improved the prediction of canopy-layer air temperature beyond two-dimensional cover (Tian et al., 2019).

Sky view factor and canyon geometry. Oke's scale-model and field results established that the maximum nocturnal heat island of a canyon falls as its sky view factor rises, or equivalently rises with its height-to-width ratio (Oke, 1981, 1988). Field tests in mid-latitude cities find a moderate to fairly strong nocturnal relationship between sky view factor and air temperature (about half of the areal-mean variation in Szeged, Hungary (Unger, 2004); a fairly strong relationship in Göteborg, Sweden (Svensson, 2004)), although a two-canyon study in Adelaide, Australia, attributed nocturnal differences mainly to thermal mass rather than radiative geometry (Erell & Williamson, 2007); in Hong Kong's high-rise residential estates sky view factor was among the significant predictors of the nocturnal heat island (Giridharan et al., 2005; Giridharan et al., 2007). The relationship is clearest on calm, clear nights and weakens when ventilation dominates.

Distance to the coast and the sea breeze. In coastal cities the distance from the shoreline is often a first-order control on nighttime temperature that is independent of the built form. Lisbon's nocturnal heat island, 2.5°C on average, was modeled with sky view factor, building height, built-up fraction, altitude and distance from the Tagus estuary as terms (Alcoforado & Andrade, 2006); in humid tropical Colombo, Sri Lanka, maximum temperatures fell with proximity to the sea and with canyon height-to-width ratio, and street corridors perpendicular to the shore were recommended to let the sea breeze penetrate (Emmanuel & Johansson, 2006); in Adelaide, Australia, the integrated cooling delivered by the sea breeze declined by 0.7–0.9°C·h per kilometer inland and penetrated tens of kilometers (Y. Zhou et al., 2021); and in Shenzhen, China, the land–sea breeze and Local Climate Zone together explained the daytime and nocturnal heat island (J. Yang et al., 2022). The sea-breeze circulation itself interacts with the heat island:

the warm city accelerates the sea-breeze front toward its center (Freitas et al., 2007), the heat-island circulation intensifies the front and delays its advance inland (Yoshikado, 1992), and on sea-breeze days the coastal parts of a city stay cooler than the inland parts while urban hot spots persist (Ramamurthy et al., 2025).

Scale. Every one of these relationships depends on the area over which the predictor is measured. The screen-level air temperature at a site responds to a source area of the order of hundreds of meters upwind (Oke, 2006), the Local Climate Zone scheme is defined at that local scale of 10^2 to 10^4 m (Stewart & Oke, 2012), and empirical studies find the strongest relationships at radii between roughly 60 and 200m (60–90m in Madison (Ziter et al., 2019), 180m upwind in San Juan (Murphy et al., 2011); the Denver network adopted 60m on the same grounds (Ibsen et al., 2022)). The radius is therefore a choice that has to be tested.

1.6 Coastal and island cities: is the evidence transferable?

Two lines of evidence bear on whether relationships established in continental cities transfer to a tropical island. The first is that heat islands persist in tropical coastal cities despite the ocean: Singapore’s canopy-layer heat island peaks three to four hours after sunset and reached about 7°C in the commercial core under ideal conditions (Chow & Roth, 2006); that of Lagos, Nigeria, exceeds 7°C in the dry season, when wet rural soils no longer suppress it (Ojeh et al., 2016); and San Juan, Puerto Rico, which like Honolulu is a trade-wind tropical island city, has a nighttime heat island of about 2°C that has been growing by about 0.06°C per year (Velazquez-Lozada et al., 2006), modeled urban warming of up to 2.5°C (Comarazamy et al., 2010), and a spatial pattern predicted best by vegetation within about 180m upwind of each sensor (Murphy et al., 2011). Where the sea breeze penetrates, the ocean and green space can rival the built form: in

Aracaju, Brazil, the park cool island (1.5–2°C) exceeded the heat island itself (0.5–1.3°C) and the beach station was the coolest in the afternoon (Anjos & Lopes, 2017). On Xiamen Island, China, building height was negatively related to surface temperature and the surrounding sea altered the effect of water bodies relative to continental cities (Zhu et al., 2023).

The second line of evidence is about measurement. The studies summarized above measured one of two different quantities: the temperature of surfaces, observed from satellites, or the temperature of the air, measured by sensors on the ground. Surface temperatures do not translate directly into the air temperature people experience at night (Venter et al., 2021; Zhang et al., 2022). Fixed networks of low-cost loggers have made canopy-layer studies feasible in many cities (Ibsen et al., 2022; Muller et al., 2013; Scott, Zaitchik, et al., 2017), but such studies remain rare in the tropics, and none has been reported for a Pacific island city.

This thesis addresses two gaps. The *local empirical gap* is that no fixed, continuous, canopy-layer air-temperature record exists for any Hawaiian city, so that neither the size of the nighttime heat island nor its relationship to urban form is known for the setting in which the state’s heat-adaptation decisions are being made. The *transferability gap* is broader. The relationships between urban form and nighttime air temperature have been quantified mostly in continental cities, and nobody has tested whether the same predictors carry the same weight in a humid, trade-wind, coastal island setting, or in two different neighborhood types on the same island.

1.7 Study design

To address both gaps, this thesis established the first fixed sensor network for canopy-layer air temperature in Hawai‘i and analyzed it in a way that makes the two districts directly comparable.

One hundred low-cost logger stations were deployed at randomized sites in two districts of leeward O‘ahu that share the same regional weather but differ sharply in form: the compact, partly high-rise urban core of Honolulu, from Kaka‘ako through Waikīkī to Mānoa, and the low-rise, master-planned suburban plain of ‘Ewa–Kapolei, 25km to the west. Loggers recorded air temperature every 30 minutes from November 2024 to January 2025. Following the recommendation that heat-island analyses control for weather ([Stewart, 2011](#)), calm, clear nights were selected with ERA5 reanalysis wind and cloud cover. For each sensor and night the mean departure of its air temperature from the median of its district network, ΔT , was related to eight descriptors of urban form and setting computed at three source-area radii from 1m land cover and building footprints, using linear mixed-effects models that treat the sensor site as the unit of replication. The two-district design is the core of the study. Because both networks experience the same weather on any given night, differences between them in the drivers of ΔT reflect urban form and local setting rather than day-to-day weather.

1.8 Purpose and research questions

The purpose of this thesis is to quantify the nocturnal canopy-layer heat island of leeward O‘ahu and to identify which properties of urban form and geographic setting explain its within-district spatial pattern, in order to provide an empirical basis for heat adaptation in Hawai‘i and a test of whether relationships from continental cities transfer to a tropical island. Four research questions are addressed:

1. **How large and how persistent is the within-district variation of nighttime air temperature on calm, clear nights in the Honolulu core and on the ‘Ewa plain?**
2. **Which two- and three-dimensional properties of urban form, and which properties of geographic setting, explain that variation, and at what source-area scale (50, 100 or 200m) are they most informative?**
3. **Do the drivers differ between the compact urban core and the low-rise suburban plain, and does a model fitted in one district transfer to the other?**

4. **How much of the persistent between-site variation remains unexplained by the form descriptors, and where is it located?**

On the basis of the literature reviewed above, the expectations were that impervious fraction and building height would warm and tree canopy would cool (with a weak canopy effect at night in this humid climate); that lower sky view factor and higher canyon aspect ratio would warm on calm nights; that distance to the coast would matter, with a sign that could not be predicted with confidence, because daytime sea-breeze cooling argues for warmer air inland whereas the thermal inertia of the winter ocean argues for warmer air at the shore; and that the relative weights of these drivers would differ between the two districts. On the nights studied here, the water at the Honolulu tide station stayed between 25.7 and 26.5°C, several degrees warmer than the 20–23°C night air over land ([NOAA Center for Operational Oceanographic Products and Services, n.d.](#)). The following chapters describe the study area (Chapter 2), the measurements, data processing and statistical methods (Chapter 3), the results (Chapter 4), and their interpretation, limitations and implications (Chapters 5 and 6).

2 Study area

2.1 Location and rationale

The study was conducted in two districts on the leeward (southern) coast of O‘ahu, Hawai‘i (21.3°N, 158°W): the urban core of Honolulu and the ‘Ewa plain around Kapolei, 25km to the west (Figure 1). The two were chosen as a paired comparison. They lie on the same leeward coast, under the same trade-wind regime and within the same county. They differ in their built environment. Honolulu is compact and vertically developed, with high-rise corridors, mid-rise redevelopment, dense low-rise residential blocks, university campuses and scattered parks within a few kilometers of one another. ‘Ewa–Kapolei is a master-planned suburban landscape of one- and two-story housing, wide arterial roads, big-box commercial centers and industrial estates laid out since the 1990s on former sugar-cane fields, with large tracts still unbuilt. The settings of the two districts are not identical: the ‘Ewa plain is wider and flatter, and the Honolulu district reaches the mouths of the Ko‘olau valleys, where rainfall rises steeply inland ([Giambelluca et al., 2013](#)). On any given night, however, both lie under the same synoptic weather, so differences between them in how nighttime air temperature relates to urban form reflect urban form and local setting rather than day-to-day weather.

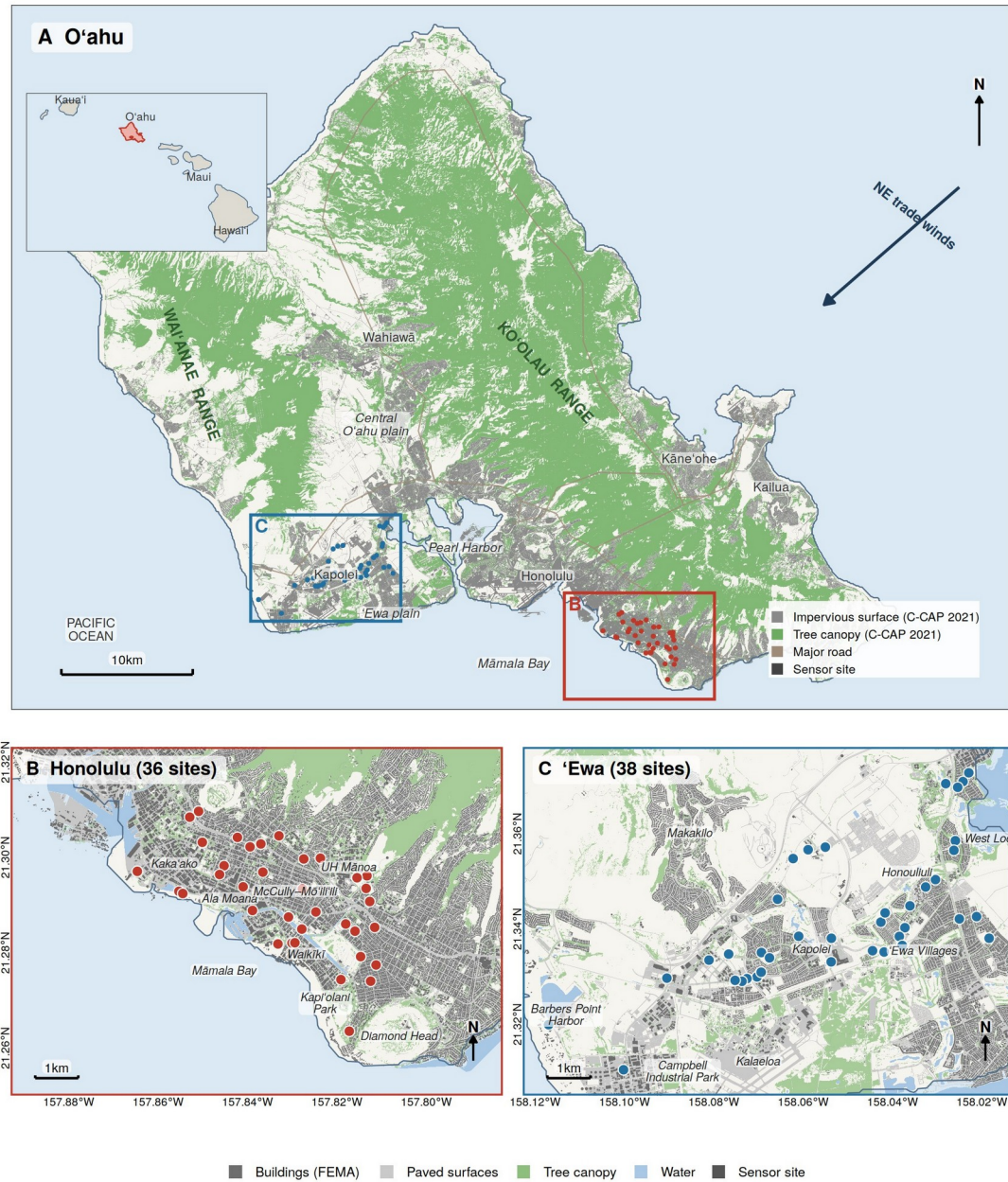


Figure 1. The study area. (A) O'ahu, with impervious surface and tree canopy from the 2021 NOAA C-CAP 1m land cover, the Ko'olau and Wai'anae ranges, the prevailing north-east trade-wind flow, and the two study domains. (B) The Honolulu core and (C) the 'Ewa plain at the same map scale, showing building footprints (FEMA), paved surfaces, tree canopy, water and the 74 recovered sensor sites. This locates the districts relative to the mountains, the coast and the trade-wind flow, and shows their contrasting urban form.

2.2 Climate

Both districts have a tropical, maritime, trade-wind climate. Mean annual rainfall on the leeward coastal plain is low, of the order of 500mm at the coast, rising steeply toward the Ko‘olau ridge, because the north-easterly trade winds deposit their moisture on the windward slopes and descend dry and warm onto the lee side ([Giambelluca et al., 2013](#); [Sanderson, 1993](#)). Trade winds prevail on most days but are interrupted by periods of light and variable winds (“Kona” conditions), most often in winter ([Schroeder, 1993](#); [Western Regional Climate Center, n.d.](#)); when the trades fail, a land–sea breeze circulation develops on the leeward coast, with an onshore sea breeze in the afternoon and a weak offshore land breeze at night ([Leopold, 1949](#)). The trade-wind inversion, the layer beginning about 2km above sea level in which air temperature increases with height (Section 1.4), is present about four days in five; it acts as a lid on the moist air below it and keeps the lowlands humid throughout the year ([Cao et al., 2007](#)). Winter (November–January), the period of this study, is the coolest and wettest part of the year on the leeward coast, with normal daily minimum temperatures at Honolulu Airport of 19–22°C and maxima of 27–29°C ([NOAA National Centers for Environmental Information, 2021](#)), and it is also the season in which the trade winds are least persistent: they blow 50–80% of the time from October to April, against 80–95% from May to September, and light, variable winds are most frequent in winter ([Western Regional Climate Center, n.d.](#)).

2.3 The ‘Ewa plain

The ‘Ewa district (Figure 1C) occupies a flat coastal plain of raised limestone and alluvium between the Wai‘anae Range to the north-west, Pearl Harbor’s West Loch to the east and the ocean to the south. The sensor network extends from the shoreline at ‘Ewa Beach and Kalaeloa

to the lower slopes below Makakilo; all sites are at low elevation. Development is low-rise almost without exception: single-family and townhouse subdivisions with tiled roofs and small lots, separated by arterial roads and drainage channels; a commercial core at Kapolei with large parking lots; the Campbell Industrial Park and Barbers Point Harbor at the south-western tip, with large industrial roofs, tank farms and a refinery; and the former naval air station at Kalaeloa, largely open runway and scrub. Within the 89km² study domain, 27% of the land is impervious and 10% is under tree canopy; the remaining pervious surface is mostly dry grass, bare soil and fallow cane land. Sensor sites span impervious fractions from 6 to 99% and mean building heights from 0 to 13m within 100m (Table 4), and lie between 0.2 and 4.8km from the shoreline.

2.4 The Honolulu core

The Honolulu district (Figure 1B) spans the urban core from Kaka‘ako and Ala Moana in the west, through the McCully–Mō‘ili‘ili residential blocks and the Waikīkī resort peninsula, to the University of Hawai‘i at Mānoa campus in the north-east and Kapi‘olani Park and Diamond Head in the south-east. It occupies the narrow coastal plain and the lower valley floors between Māmala Bay and the foothills of the Ko‘olau Range, so that sites range from the shoreline to about 2.7km inland, all at low elevation. The built environment is the most varied on the island: Waikīkī’s compact high-rise hotels and apartment towers, the mid-rise mixed-use redevelopment of Kaka‘ako and Ward, the dense two- and three-story walk-up blocks of McCully–Mō‘ili‘ili, the tree-lined institutional grounds of the university, and the roughly 80ha (200 acres) of turf and mature trees of Kapi‘olani Park ([City and County of Honolulu, Department of Parks and Recreation, n.d.](#)). Within the 52km² study domain, 53% of the land is impervious and 20% is under tree canopy. Sensor sites span impervious fractions from 1 to 98% and mean building

heights from 0 to 21m within 100m, and sky view factors from 0.70 to 1.0 apart from four sites whose recorded positions fall inside building footprints (Section 3.6; Table 4). The Ala Wai Canal, which separates Waikīkī from the residential blocks to its north, and the harbor at Kaka‘ako give some sites appreciable water fractions.

Compared with ‘Ewa, the Honolulu sites are on average more impervious, more built-up, taller, more enclosed and closer to the coast (Table 4); tree-canopy fractions are similar. The two districts therefore differ in vertical form and enclosure while having similar tree cover, which allows the effects of three-dimensional structure to be separated from those of surface cover.

3 Data and methods

3.1 Overview

The method has four stages (Figure 2): measuring canopy-layer air temperature with a fixed network of low-cost loggers in each district; processing the record into a per-site, per-night response variable on calm, clear nights; describing the urban form and setting of each site at three scales from public geospatial data; and relating the response to the descriptors with mixed-effects models that treat the site, not the reading, as the unit of replication. Each stage is described in turn, with the reasoning for every choice and a distinction between criteria adopted from the literature and criteria specific to this study. The entire chain, from the raw logger files to every table and figure in this thesis, is scripted and archived (Appendix C).

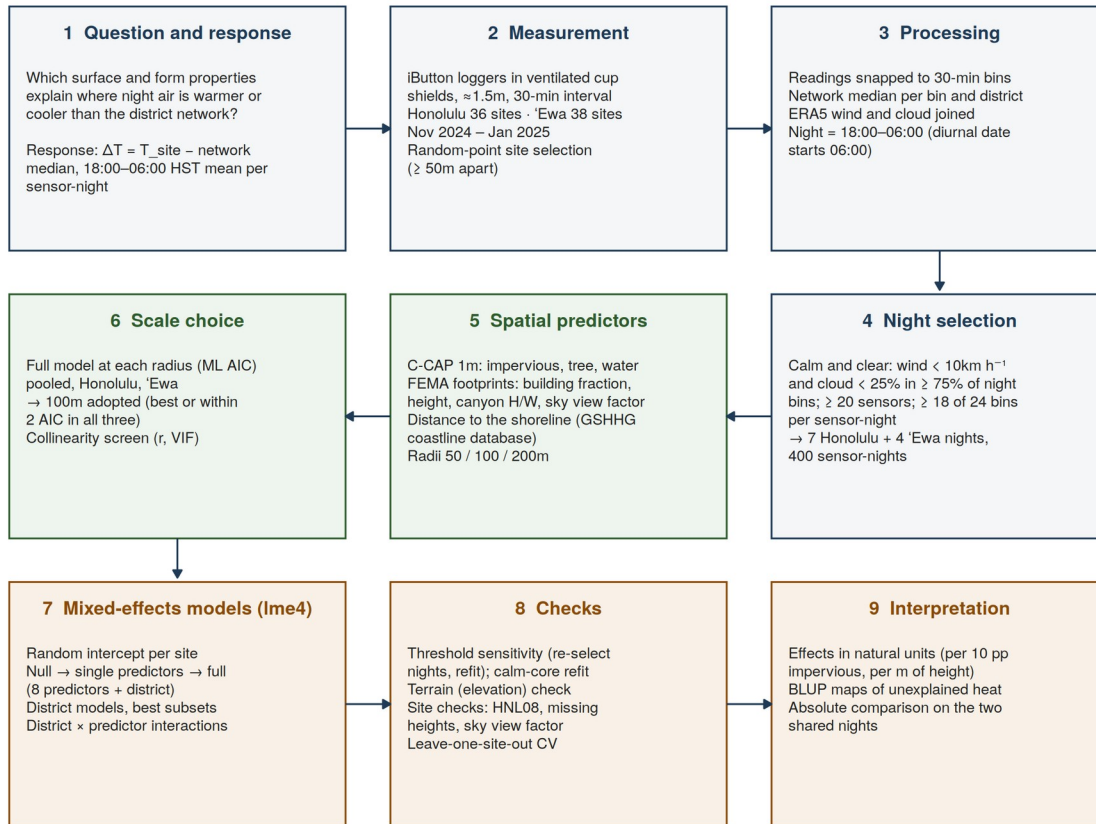


Figure 2. The workflow of the study, from the research question to the interpretation of the models. This gives the whole method in one view before its parts are described.

3.2 Measuring canopy-layer air temperature

Objective and variable. The objective was to measure, at many sites across each district and continuously through the night, the quantity to which the urban heat island literature and the physiological literature on nighttime heat refer: canopy-layer air temperature. The urban canopy layer is the layer of air between the ground and the mean roof height, in which the microclimates of individual streets, yards and buildings are formed (Oke, 1976; Oke et al., 2017); it is distinct from the boundary layer above roof level and from the surface (skin) temperature that satellites measure. The variable measured was therefore air temperature in shaded, ventilated conditions, and not surface temperature.

Why 1.5m. Standard meteorological practice places the air-temperature sensor between 1.25 and 2.0m above the ground ([World Meteorological Organization, 2024](#)); the World Meteorological Organization’s guidance for urban sites accepts that range for canopy-layer measurements, allows greater heights of 3–5m where the canopy is tall, and asks that the sensor be centered in a space of typical canyon geometry rather than placed against a wall ([Oke, 2006](#)). A height of about 1.5m is within that range and corresponds to the height at which people breathe and sense the air; comparable low-cost networks have mounted their sensors at about 2m ([Ibsen et al., 2022](#)). Stations were therefore mounted at approximately 1.5m.

Instrument. Each station used a Thermochron iButton temperature logger, model DS1921H (Analog Devices/Maxim), a 16mm stainless-steel can containing a thermometer, clock and memory. According to the manufacturer’s specifications, the DS1921H measures temperatures between +15 and +46°C at a resolution of 0.125°C with an accuracy of $\pm 1^\circ\text{C}$, and its memory can hold up to 2,048 readings, enough for 42.7 days at the 30-minute interval used here. These are properties of the instrument, not of the data collected. iButtons have been used widely for canopy-layer networks because they are inexpensive, robust and small ([Hubbart et al., 2005](#); [Ibsen et al., 2022](#); [Scott, Misiani, et al., 2017](#)); their principal weakness is radiative heating when exposed to sun or to warm surfaces, which can bias unshielded loggers by several degrees ([Terando et al., 2017](#)). The $\pm 1^\circ\text{C}$ accuracy specification means that individual loggers may carry constant offsets of the order of the between-site differences studied here. The statistical design (Section 3.7) absorbs such offsets into the site-level random effect, but they cannot be separated from true site effects without a side-by-side comparison of all loggers in one place before deployment (a co-location test), which was not performed (Section 5.6).

Shield. Each iButton was suspended inside an inverted white polystyrene cup that acted as a passive radiation shield (Figure 3). The logger hung from a paper-clip hook in a tea-bag sleeve so that it touched no surface, the cup’s open base allowed air to circulate around it, and the white outer surface reflected short-wave radiation. Inverted-cup and similar home-made shields have been evaluated against reference shields and perform adequately in shade and at night, with daytime errors that grow with solar loading and fall with wind ([Holden et al., 2013](#); [Tarara & Hoheisel, 2007](#)); and similar home-made shields were used for canopy-layer networks in Baltimore and in Nairobi ([Scott, Zaitchik, et al., 2017](#); [Scott, Misiani, et al., 2017](#)). Because the daytime error of this design is known to be large in direct sun, daytime readings were excluded from all analyses (Section 3.5).

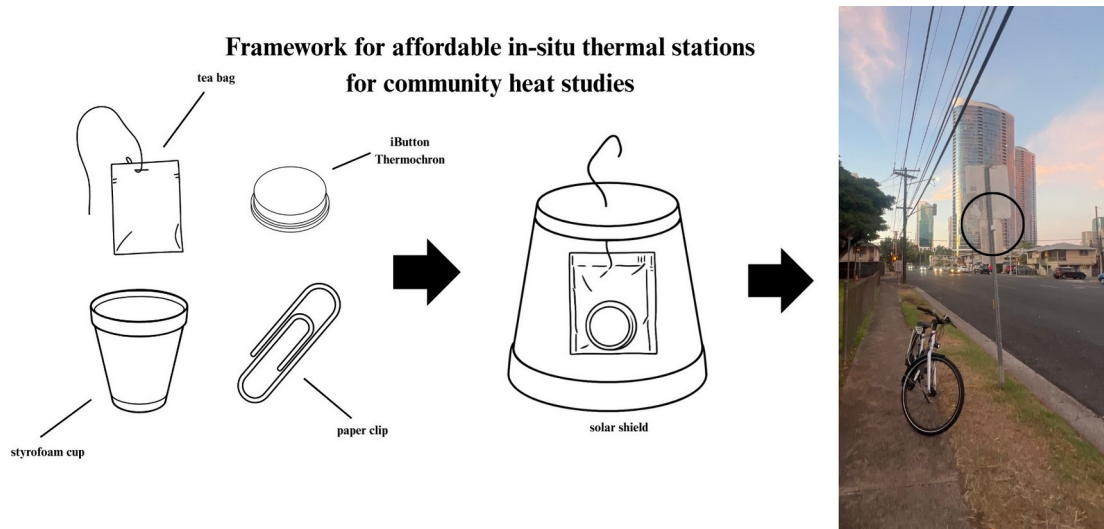


Figure 3. The low-cost temperature station: a Thermochron iButton DS1921H logger suspended in a tea-bag sleeve from a paper-clip hook inside an inverted white polystyrene cup that serves as a radiation shield, and a station mounted on a street-sign post in Honolulu at about 1.5m (circled). This documents the instrument, its shielding and its exposure.

Site selection and deployment. Fifty sites were generated at random within each district polygon, subject to a minimum separation of 50m, so that the network would sample the full

range of urban form rather than the places a person would choose, and would not cluster. Where a random point fell in an inaccessible location (private land, road median, water), the station was placed on the nearest publicly accessible fixed vertical structure (a utility pole, street-sign post or fence) that preserved the surroundings of the original point. Stations were attached with paper clips at approximately 1.5m, on the shaded side of the post where one existed, and away from walls, vents and vehicles. The ‘Ewa stations were deployed from 14 November 2024 and recovered on 27 December 2024; the Honolulu stations were deployed from 23 November 2024 and recovered in early January 2025, with a few loggers recovered as late as 20 January. Deployment took several days in each district, and readings made while loggers were being carried to their sites appear in the record as daytime spikes that reach 46°C, the upper limit of the loggers’ measurement range (Figure 5); no analyzed night falls in that period. Of the 100 stations, 74 were recovered with usable records, 36 in Honolulu and 38 in ‘Ewa. The remainder had been removed or lost, a loss rate (26 of 100) higher than the 12% (18 of 153) lost over one summer from Baltimore’s mostly tree-mounted network ([Scott, Zaitchik, et al., 2017](#)). At one site a passer-by mistook the station for a geocache and left a note inside the shield asking whether they had won a prize. Loggers recorded every 30 minutes, giving 73,728 readings in Honolulu and 77,824 in ‘Ewa (Table 1).



Figure 4. Deployment and recovery of stations by the author (left) and community volunteers. This documents how the network was installed on public infrastructure.

Table 1. Summary of the two deployments and of the analyzed record.

	Honolulu	‘Ewa
Loggers deployed / recovered	50 / 36	50 / 38
Record	23 Nov 2024 – 20 Jan 2025	14 Nov 2024 – 27 Dec 2024
Half-hourly readings	73,728	77,824
Calm, clear nights analyzed	7	4
Sensor-nights	248	152
Night-mean air temperature (°C)	22.2	21.3
Night-mean wind / cloud on analyzed nights	3.7km h ⁻¹ / 10%	3.3km h ⁻¹ / 13%

3.3 Data processing

Time binning. The loggers were started by hand one after another, so their clocks were offset from one another by up to two minutes: most sampled at :01 and :31 past the hour, some at :00/:30 and some at :02/:32. Every reading was assigned to the nearest 30-minute bin, so that all sensors of a district are compared at the same time step.

Reference and departure. For each district and each 30-minute bin, the network median air temperature was computed from all sensors reporting in that bin (a median rather than a mean so that a single faulty or sun-struck reading cannot shift the reference). At each 30-minute step, a sensor’s departure from its district is its temperature minus the district median:

$$\Delta T_i(t) = T_i(t) - \tilde{T}(t),$$

where $T_i(t)$ is the air temperature at sensor i in time step t and $\tilde{T}(t)$ is the median temperature of all sensors of the district in the same step.

Night window and diurnal date. Night was defined as 18:00–06:00 Hawai‘i Standard Time (24 bins), which brackets sunset (about 17:50) and sunrise (about 06:50–07:10) in November–January and thus covers the whole period of stored-heat release and radiative cooling. Each 24-hour cycle was labeled by the date on which it began at 06:00, so that a “night” is the evening of that date and the following early morning.

Weather data. Hourly 10m wind speed and total cloud cover from the ERA5 reanalysis ([Hersbach et al., 2020](#)), retrieved through the Open-Meteo archive interface ([Zippenfenig, 2023](#)) for the centroid of each district, were attached to every reading by nearest hour. ERA5 has a native resolution of about 31km and is distributed on a 0.25° grid, so the values describe regional conditions over the leeward coast rather than the wind in any street; that is the appropriate quantity for classifying nights as synoptically calm and clear, but it is not a measurement of local ventilation (Section 5.5).

Completeness rules. Two study-specific rules protect the reference. A night was analyzed only if at least 20 sensors of the district network reported, because the median of a handful of loggers is not a district reference; this rule excluded the night of 17 January 2025, when a single Honolulu logger was still recording. A sensor-night was retained only if at least 18 of its 24 night bins were present. Nights for which ERA5 coverage did not span all 24 bins (only the first night of each deployment) were not classified. Table 2 lists every night that met the weather criterion and the decision taken on it.

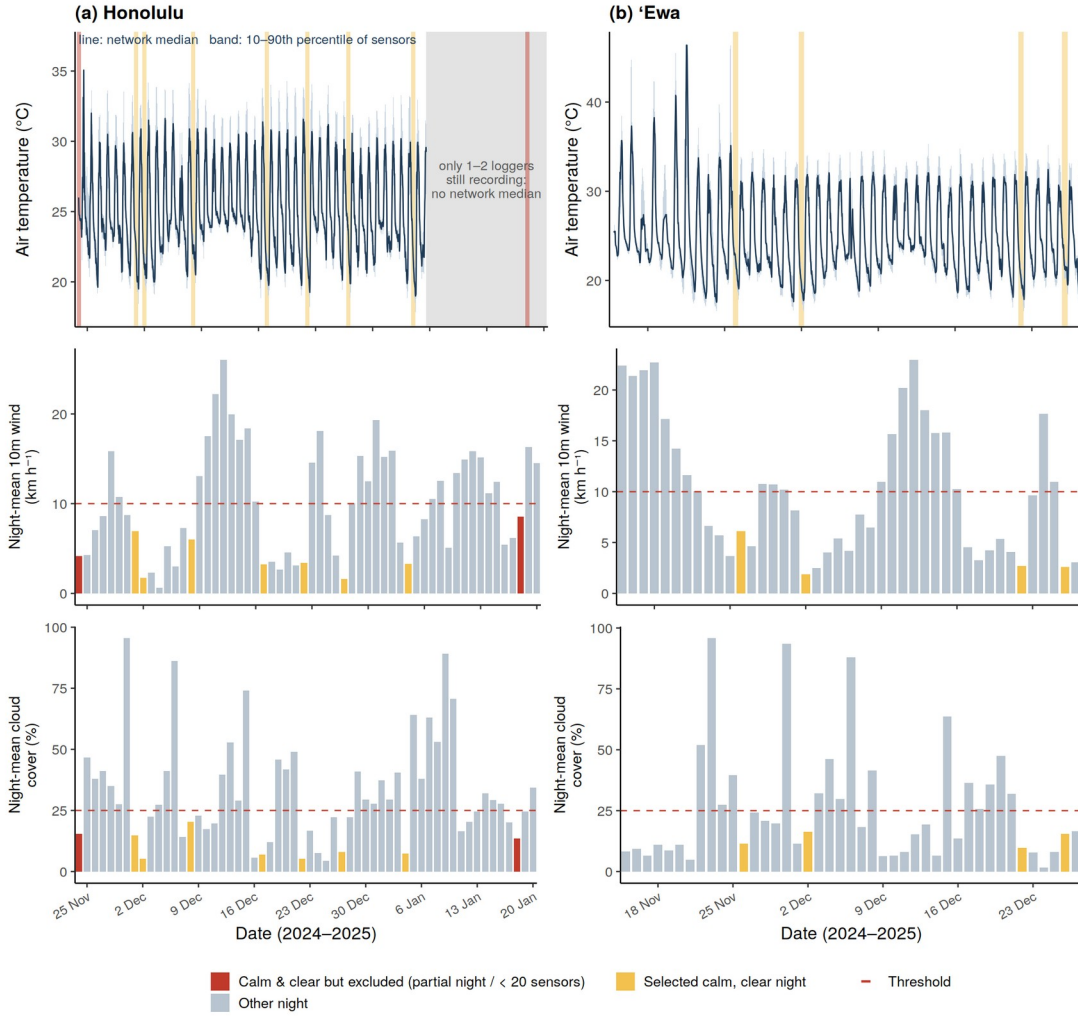


Figure 5. Night selection. Top: network median air temperature (line) and the 10th–90th percentile range of sensors (band) through each deployment, for (a) Honolulu and (b) ‘Ewa; selected calm, clear nights are shaded yellow and nights that met the weather criterion but were excluded by the completeness rules are shaded red. From 6 to 20 January only one or two Honolulu loggers were still recording, so no network median was computed (gray band). Middle and bottom: night-mean ERA5 10m wind speed and cloud cover for every night, with the thresholds of 10km h⁻¹ and 25% (dashed). The daytime spikes at the start of each record are readings made while loggers were being carried to their sites. This shows every night of the record and why each was retained or excluded.

Table 2. Nights on which at least 75% of the available night bins were calm and clear (ERA5 10m wind below 10km h⁻¹ and cloud cover below 25%), with the completeness decision (all 24 bins present; at least 20 loggers reporting). Diurnal dates start at 06:00 HST.

Study area	Night (18:00–06:00 HST)	Sensors reporting	Night-mean wind (km h ⁻¹)	Night-mean cloud (%)	Share of calm & clear bins	Decision
‘Ewa	25 Nov 2024	38	6.1	12	75%	analyzed
‘Ewa	1 Dec 2024	38	1.9	16	75%	analyzed
‘Ewa	21 Dec 2024	38	2.7	10	88%	analyzed

Study area	Night (18:00–06:00 HST)	Sensors reporting	Night-mean wind (km h ⁻¹)	Night-mean cloud (%)	Share of calm & clear bins	Decision
‘Ewa	25 Dec 2024	38	2.6	15	75%	analyzed
Honolulu	23 Nov 2024	34	4.1	16	88%	excluded: partial night (loggers still being deployed)
Honolulu	30 Nov 2024	35	6.9	15	75%	analyzed
Honolulu	1 Dec 2024	35	1.7	5	92%	analyzed
Honolulu	7 Dec 2024	35	6.0	20	79%	analyzed
Honolulu	16 Dec 2024	36	3.2	7	92%	analyzed
Honolulu	21 Dec 2024	36	3.4	5	100%	analyzed
Honolulu	26 Dec 2024	36	1.6	8	92%	analyzed
Honolulu	3 Jan 2025	36	3.3	7	96%	analyzed
Honolulu	17 Jan 2025	1	8.5	14	79%	excluded: one logger still recording

3.4 Selection of calm and clear nights

Why. The heat island arises from differences in surface energy balance between sites, and those differences show up in air temperature only under certain atmospheric conditions. Wind mixes air across the city and advects the warm air away; cloud reduces the long-wave radiative loss that drives nighttime cooling and thereby reduces the contrast between well- and poorly-ventilated surfaces. Both effects are well established: the maximum heat island falls with the inverse square root of wind speed (Oke, 1973); in Melbourne the nocturnal heat island was inversely proportional to about the fourth root of both wind speed and cloud amount (Morris et al., 2001); in Vancouver a weather factor built from wind and cloud explained much of the night-to-night variation in heat-island intensity (Runnalls & Oke, 2000), and in Seoul wind speed and cloudiness, together with humidity and the previous day’s value, were significant predictors of the daily maximum heat island (Kim & Baik, 2002). Accordingly, the methodological critique of the heat-island literature requires that weather be controlled, with measurements made in steady

conditions free of fronts, strong advection and precipitation (Stewart, 2011), and the Local Climate Zone framework recommends that the relationship between urban form and temperature be examined on calm, clear nights, when it is strongest (Stewart & Oke, 2012). Selecting such nights is therefore a criterion adopted from the literature.

Which thresholds, and why. The literature recommends the principle without fixing the numbers; definitions of “calm” and “clear” vary between studies and depend on the wind data available. The thresholds used here are study-specific choices, and were set as follows. Wind was required to be below 10 km h^{-1} (2.8 m s^{-1}) at 10m in the ERA5 grid cell. In the Beaufort scale this lies within the “light breeze” class ($1.6\text{--}3.3\text{ m s}^{-1}$); Oke’s classic St. Lawrence Lowland traverses were made under clear skies with regional winds averaging 2.3 m s^{-1} , within this class, while his compiled maximum heat islands refer to calm, clear conditions (Oke, 1973), and the Melbourne relationship implies that halving the wind speed from this level would raise the heat island by only about a fifth (a fourth-root dependence), so the threshold keeps most of the signal while excluding the trade-wind nights on which it is strongly mixed. Because ERA5 winds over a grid cell that includes ocean are systematically higher than winds within the urban canopy, the threshold is conservative with respect to the streets. Cloud cover was required to be below 25%, that is, no more than about a quarter of the sky covered. Weather observers report cloud cover in eighths of the sky, called oktas, and up to two oktas (25%) is reported as “few clouds”. Because a night can be interrupted briefly by a passing cloud band or gust without losing its character, both criteria were required to hold in at least 75% of the 24 night bins (at least 18 half-hours, or nine hours) rather than in every bin. These three numbers were fixed before the analysis; their consequences were then examined in two ways.

Empirical check. First, the effect of weather on the quantity of interest was examined in the record itself. Across all 42 complete nights of each deployment, the spread of ΔT across the network (its standard deviation among sites) falls steeply with ERA5 wind speed (Spearman $\rho = -0.61$ in both districts, $p < 0.001$; Figure 6), which is the expected signature of wind mixing away the between-site contrasts. Cloud cover showed no such monotonic relationship in this record ($\rho = -0.06$ in Honolulu and $+0.25$ in 'Ewa), which may reflect the difficulty of representing nocturnal cloud over a small island in a coarse reanalysis; the cloud criterion was retained because relaxing it to 50% reduced the explained variance of the models (Section 4.8). Second, the entire night selection was repeated for a grid of alternative thresholds (wind limits of 8 to 20 km h^{-1} , cloud limits of 15 to 50%, minimum shares of 60 to 75% of bins, and no weather filter at all), and the principal models were refitted on each alternative data set (Section 3.7; results in Section 4.8). The conclusions are reported with that sensitivity analysis alongside them.

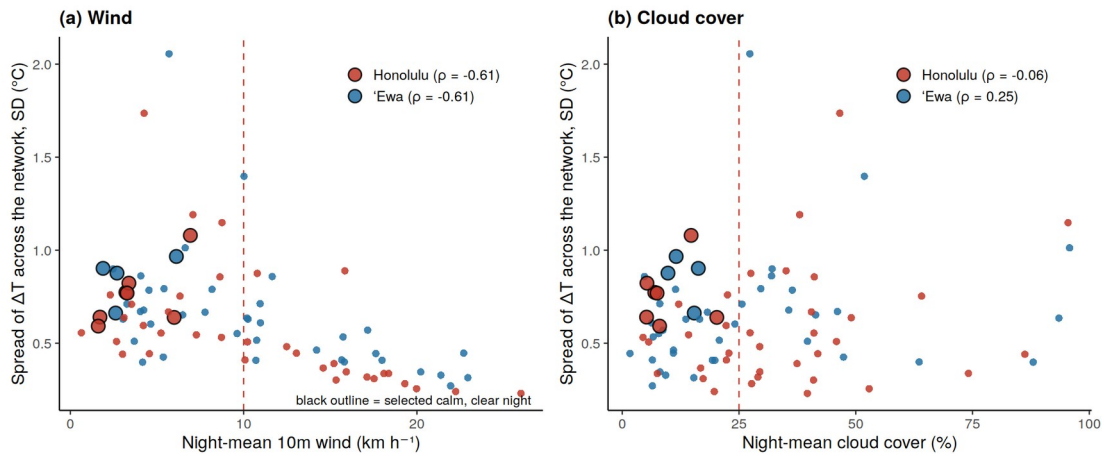


Figure 6. The empirical basis for the calm–clear criterion: the spread of ΔT across each network (standard deviation among sites) on every complete night, against (a) night-mean ERA5 wind speed and (b) cloud cover, with the adopted thresholds (dashed). Selected nights have black outlines. This shows that between-site contrasts collapse as wind increases in this record.

Result of the selection. Seven Honolulu nights and four ‘Ewa nights met all criteria (Table 2), two of which, 1 and 21 December 2024, were shared by both districts. The selected nights had night-mean ERA5 winds of 1.6–6.9km h⁻¹ and cloud cover of 5–20%. After the completeness rules, the analyzed data set comprises 400 sensor-nights: 248 in Honolulu (36 sites) and 152 in ‘Ewa (38 sites) (Table 1). The temperature distributions of the selected nights are compared with those of all nights in Figure 7.

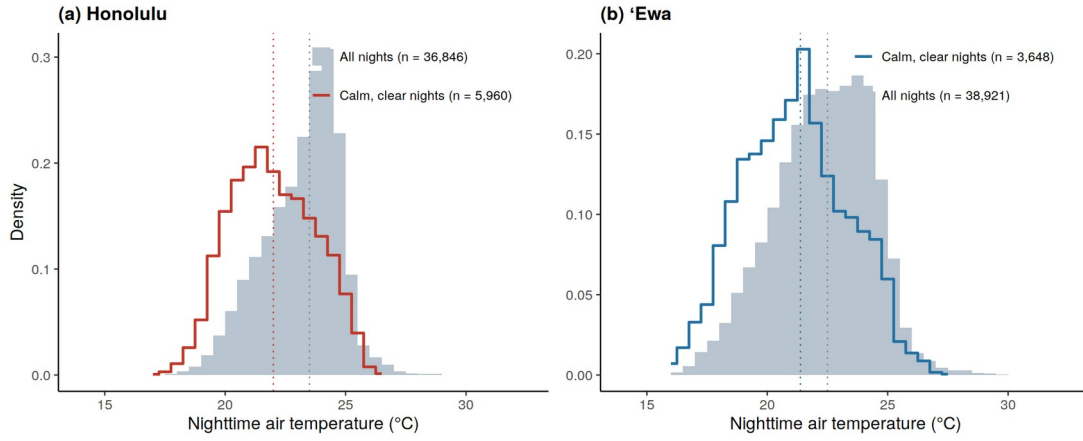


Figure 7. Distributions of nighttime (18:00–06:00 HST) air-temperature readings on all nights (filled) and on the selected calm, clear nights (outline) in (a) Honolulu and (b) ‘Ewa; dotted lines mark the medians. This shows what the night selection does to the sample.

3.5 The response variable

For each sensor and each selected night, the response variable is the mean departure from the network median over the 24 night bins:

$$\Delta T_{i,n} = \frac{1}{24} \sum_{t \in \text{night } n} [T_i(t) - \tilde{T}(t)],$$

in degrees Celsius. Figure 8 shows how it arises on one shared night: every sensor’s trace, the network median that serves as the reference, and the resulting departures. The departures are

large and erratic by day, when the cup shields respond to direct sun, and small and steady by night.

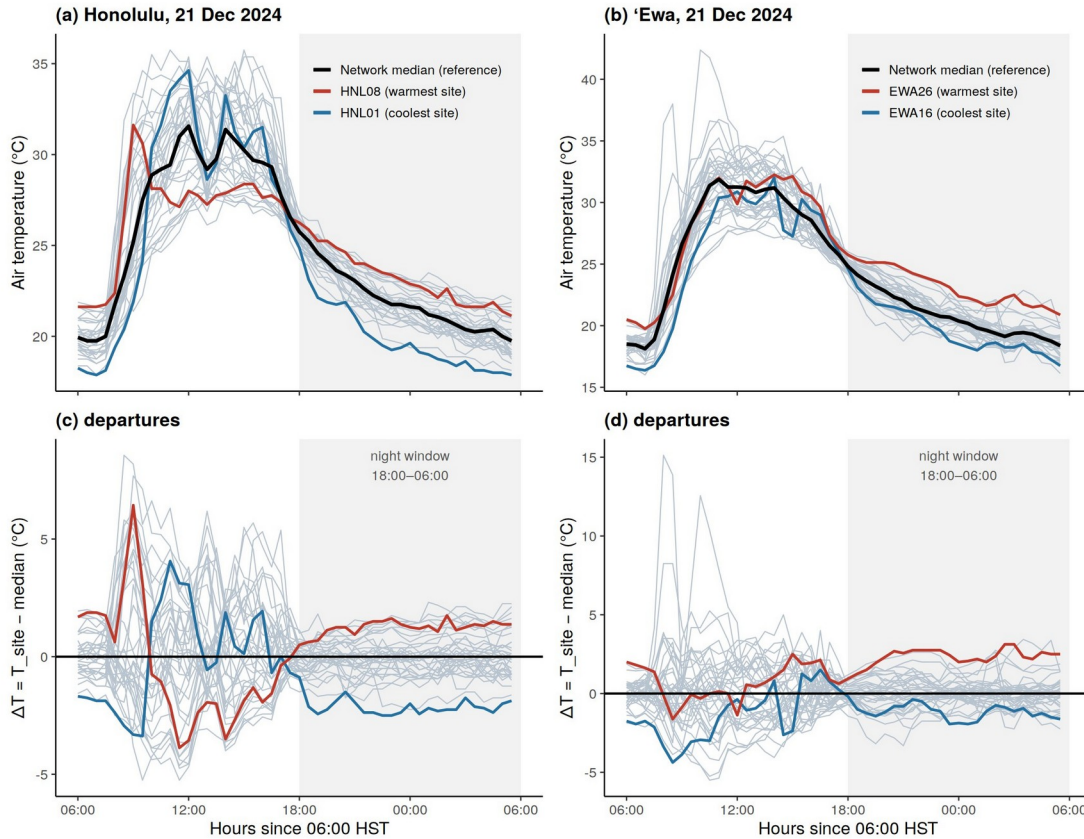


Figure 8. How the response variable is defined, on the shared night of 21 December 2024. (a, b) Every sensor’s air temperature through the 24-hour cycle in each district (gray), the network median used as the reference (black), and the warmest and coolest sites of the district (red, blue). (c, d) The corresponding departures $\Delta T = T - \text{median}$. The shaded band is the night window (18:00–06:00 HST). This makes the reference and the departure concrete and shows why only the night is analyzed: daytime departures are dominated by radiation errors of the shields.

An internal reference was used instead of a rural station. The classical heat island is the difference between an urban and a rural station, but the choice and description of the rural reference is a long-recognized source of inconsistency in the heat-island literature (Stewart, 2011; Stewart & Oke, 2012), and on leeward O’ahu there is no rural site that shares the coastal plain’s exposure without also being on a different part of the coast, at a different elevation or in a

different wind regime. The district median avoids that problem, but it has three properties that affect how the results should be read. First, it removes from each reading the regional temperature of that night, so that ΔT is a purely within-district, within-night quantity. If a whole district is warmer or cooler on one night than on another, ΔT does not change; only how each site compares with the rest of its district on that night enters the analysis. Second, because it is computed separately for each district, it also removes the mean difference between the districts; ΔT in Honolulu is a departure from the Honolulu median and ΔT in 'Ewa from the 'Ewa median, and the two are not on a common absolute scale. Any “district effect” in a pooled model must therefore be read as a difference in how the districts’ networks are centered, not as one district being warmer than the other. Third, the absolute comparison that the median removes can be recovered on the two nights the districts share, and it is reported separately (Section 4.10). ΔT is best described as the site’s nocturnal thermal departure from its district; positive values are sites that are warmer than the district as a whole and negative values sites that are cooler.

3.6 Descriptors of urban form and setting

Sources. Eight descriptors were computed for each site from three public data sets (Table 3): the NOAA Coastal Change Analysis Program (C-CAP) 2021 high-resolution land cover for Hawai‘i, a set of 1m rasters of impervious surface, tree canopy and water derived from aerial imagery ([NOAA Office for Coastal Management, 2024](#)); the FEMA USA Structures building-footprint inventory, which provides a polygon for each building and, where lidar was available, a height ([Federal Emergency Management Agency and Oak Ridge National Laboratory, 2025](#)) (the inventory’s source fields show that the O‘ahu heights are lidar measurements made in 2010–2011 and that the 23% of footprints added from satellite images between 2012 and 2016 have no

height; these were given 5m, and Section 4.8 shows that the results do not depend on that value; heights above 60m, 28 buildings on the island and none within 100m of a sensor, were capped at 60m; buildings completed after 2016 are not in the inventory); and, for the coastline, the full-resolution shoreline of the Global Self-consistent, Hierarchical, High-resolution Geography database (GSHHG), a widely used global coastline data set (Wessel & Smith, 1996). All calculations were made in the projected coordinate system of the C-CAP tiles (NAD83(PA11) / UTM zone 4N, meters).

Table 3. Definitions of the eight site descriptors. Area-based descriptors were computed within circular source areas of radius 50, 100 and 200m centered on the sensor; the two point descriptors do not depend on the radius.

Descriptor	Definition	Source	Type
Impervious surface fraction	Percentage of 1m pixels classified as impervious within the source area	C-CAP 2021	2-D cover
Tree canopy fraction	Percentage of pixels classified as tree canopy (shrub excluded)	C-CAP 2021	2-D cover
Water surface fraction	Percentage of pixels classified as open water	C-CAP 2021	2-D cover
Building footprint fraction	Percentage of the source area covered by building footprints (polygons clipped to the circle)	FEMA USA Structures	2-D form
Mean building height	Footprint-area-weighted mean height of the buildings intersecting the source area (m)	FEMA USA Structures	3-D form
Canyon aspect ratio (H/W)	Mean building height divided by an estimate of mean street width, $W = \sqrt{(A/n)} - \sqrt{(\bar{a})}$, where A is the source area, n the number of buildings and $\bar{a}$ their mean footprint; W is bounded below at 6m and H/W above at 5	FEMA USA Structures	3-D form
Sky view factor	Fraction of the sky hemisphere visible from the sensor at 1.5m above ground, from building obstructions within 200m in 36 directions: $SVF = (1/36) \sum \cos^2 \beta$, where β is the steepest angle up to a building top along each direction	FEMA USA Structures	3-D form, point
Distance to coast	Shortest distance from the sensor to the ocean shoreline (km)	GSHHG coastline	Setting, point

Scales. The area-based descriptors were computed at three radii, 50, 100 and 200m, chosen to bracket the range that the literature and the physics of the source area suggest: 50m describes the immediate surroundings of a street-level sensor, 200m approaches the lower limit of the Local Climate Zone scale, and 100m lies within the 60–180m range at which studies have found the strongest relationships with air temperature (60–90m in Madison (Ziter et al., 2019),

180m upwind in San Juan (Murphy et al., 2011), 60m in Denver (Ibsen et al., 2022)). The radius was then chosen by the models themselves (Section 3.7).

Sky view factor. Sky view factor was calculated directly from the building footprints and their heights. From a point 1.5m above each sensor, lines were traced outward for 200m in 36 directions, one every 10°. Along each line, the steepest angle from the sensor up to the top of a building (the elevation angle β) was found, and the sky view factor was taken as the average of $\cos^2 \beta$ over the 36 directions, because a horizontal surface bounded in a given direction by an obstruction of elevation angle β sees a fraction $\cos^2 \beta$ of the sky in that direction (Johnson & Watson, 1984); computing the sky view factor from the elevation angles of building tops follows the canyon studies of Oke (1981) and Unger (2004). A value of 1 means the sky is completely open; lower values mean that more of the sky is blocked by buildings. Figure B3 in Appendix B shows the calculation for an example site. Trees were not included as obstructions, so the descriptor measures enclosure by buildings alone; tree cover enters separately through the canopy fraction. Five recorded logger positions fall inside a building footprint, either because the post stood under a roof overhang or a covered structure or because the recorded position is off by up to about 13m. For these sites the calculation treats the building as an obstruction 1m away in every direction and returns a sky view factor below 0.1, whereas all other sites lie between 0.70 and 1.00. Section 4.8 shows that recomputing their sky view factor from a point just outside the footprint does not change the results.

Distance to coast was measured as the shortest straight-line distance from each sensor to the ocean shoreline in the GSHHG database. That shoreline includes the shores of Pearl Harbor, so for several ‘Ewa sites the nearest coast is West Loch rather than the open south shore. Smaller

water bodies inside the shoreline, such as the Ala Wai Canal, small boat basins and drainage channels, were not treated as coast.

Elevation is not one of the eight descriptors but was added as a check (Section 4.8), because in Honolulu the shore-to-valley gradient that distance to the coast describes also runs up-slope, and a coastal effect could in principle be an elevation effect. The bare-earth elevation of each logger position was read from the USGS 3D Elevation Program 1m lidar digital elevation model through the USGS Elevation Point Query Service ([U.S. Geological Survey, n.d.](#)). The Honolulu sites lie between 0.8 and 41m above sea level (median 4.8m) and the 'Ewa sites between 4 and 41m (median 17.5m); elevation is correlated with distance to the coast at $r = 0.61$ in both districts and only weakly with the built-form descriptors ($|r| \leq 0.39$).

Descriptive statistics of the eight descriptors by district are given in Table 4 and their spatial patterns in Figures 9 and 10.

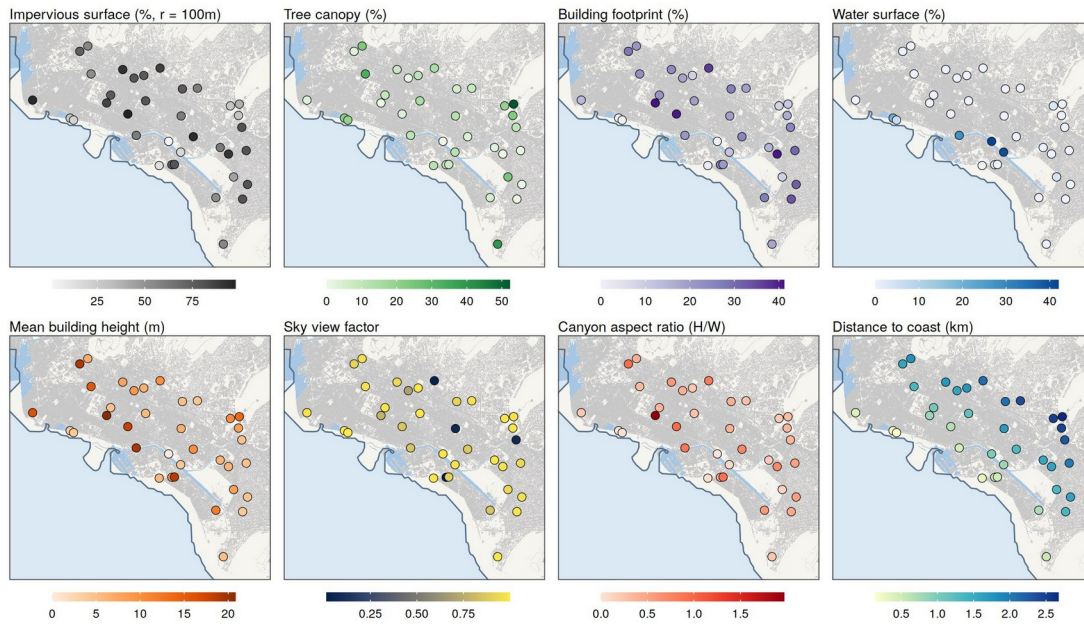


Figure 9. The eight site descriptors at the 100m source-area radius across the Honolulu network (sky view factor and distance to the coast are point properties). This shows the spatial pattern and range of each descriptor.

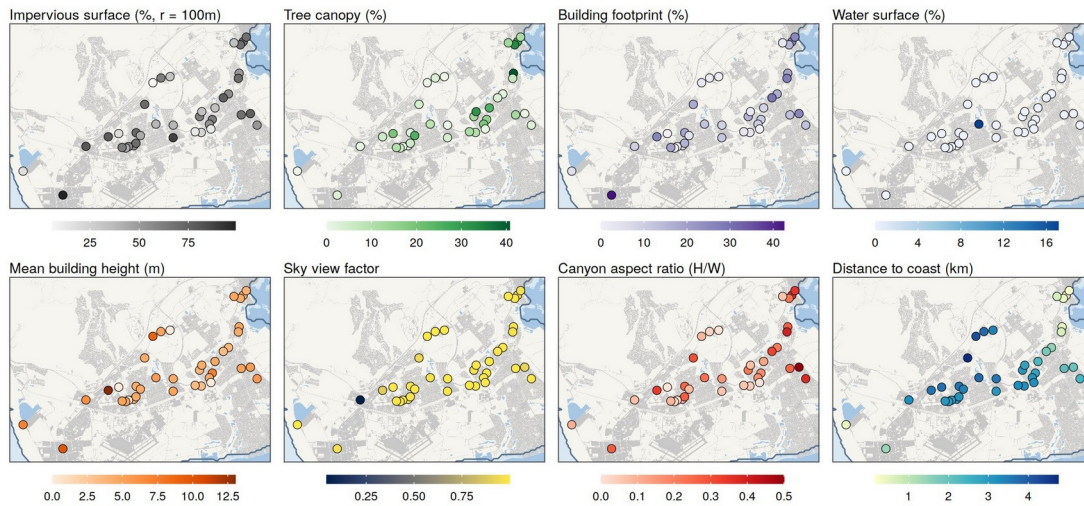


Figure 10. The eight site descriptors at the 100m source-area radius across the 'Ewa network. This shows the spatial pattern and range of each descriptor in 'Ewa, as Figure 9 does for Honolulu.

Table 4. Site predictors at the adopted 100m source-area radius, by study area (sky view factor and coastal distance are point properties of the site). The last column tests whether a descriptor differs between the two districts’ sites: it is the p-value of a Mann–Whitney (Wilcoxon rank-sum) test, and small values mean the districts differ.

Predictor	Honolulu mean $\pm$ SD	Honolulu range	‘Ewa mean $\pm$ SD	‘Ewa range	Honolulu vs ‘Ewa, p
Impervious surface fraction (%)	62.6 $\pm$ 26.1	1.3 – 97.7	53.0 $\pm$ 23.0	5.7 – 99.0	0.04
Tree canopy fraction (%)	11.9 $\pm$ 11.9	0.0 – 52.3	9.9 $\pm$ 10.8	0.0 – 40.8	0.40
Building footprint fraction (%)	20.4 $\pm$ 11.3	0.0 – 41.2	12.3 $\pm$ 10.4	0.0 – 42.6	0.003
Water surface fraction (%)	4.0 $\pm$ 10.8	0.0 – 42.0	0.5 $\pm$ 2.8	0.0 – 17.1	0.02
Mean building height (m)	8.9 $\pm$ 5.4	0.0 – 20.8	4.7 $\pm$ 2.5	0.0 – 13.0	< 0.001
Sky view factor (0–1)	0.84 $\pm$ 0.29	0.01 – 1.00	0.96 $\pm$ 0.15	0.04 – 1.00	< 0.001
Canyon aspect ratio (H/W)	0.45 $\pm$ 0.38	0.00 – 1.97	0.18 $\pm$ 0.13	0.00 – 0.50	< 0.001
Distance to coast (km)	1.33 $\pm$ 0.72	0.14 – 2.67	2.57 $\pm$ 1.22	0.16 – 4.76	< 0.001

3.7 Statistical analysis

Unit of analysis and model structure. The data have two levels: each of the 74 sites contributes several nights, giving 400 sensor-nights in all. Because urban form is a property of the site, treating every sensor-night as an independent observation would count each site’s form four to seven times and overstate how certain the form effects are, an error known as pseudoreplication (Hurlbert, 1984). All inference was therefore made with linear mixed-effects models that include a random intercept for each site (Bolker et al., 2009; Harrison et al., 2018; Zuur et al., 2009):

$$\Delta T_{i,n} = \beta_0 + \sum_k \beta_k x_{k,i} + u_i + \varepsilon_{i,n}, u_i \sim N(0, \sigma_u^2), \varepsilon_{i,n} \sim N(0, \sigma_\varepsilon^2).$$

The terms of the model are:

- $\Delta T_{i,n}$ is the response: the night-mean departure of site i from its district median on night n , in °C.
- β_0 is the intercept: the expected departure of a site whose descriptors all equal their average values.
- $x_{k,i}$ is the value of descriptor k (for example, impervious fraction) at site i , standardized to a mean of 0 and a standard deviation of 1, and β_k is its coefficient: the change in ΔT for a one-standard-deviation increase in that descriptor. These are the fixed effects, the part of each site’s departure that its urban form explains.
- u_i is the random intercept of site i : the part of its departure that repeats every night but is not explained by the descriptors, such as the exact placement of the station, surroundings that the descriptors do not capture, or a constant offset of the logger. The random intercepts are assumed to vary between sites around zero with variance σ_u^2 .
- $\varepsilon_{i,n}$ is the residual: the night-to-night scatter of a site around its own average, with variance σ_ε^2 .

Random slopes for the descriptors were not fitted because a descriptor that is constant within a site cannot have a slope that varies within that site; such a term is not separately identifiable from the random intercept. A random effect of night was not included because the reference removes the night mean of ΔT by construction.

Estimation and testing. Models were fitted with the `lmer` function of the `lme4` package (version 1.1-35) in R 4.3 (Bates et al., 2015; R Core Team, 2024). Following standard practice, fixed effects were tested with likelihood-ratio tests between nested models fitted by maximum likelihood (ML), and reported coefficients and variance components come from restricted maximum likelihood (REML) fits (Zuur et al., 2009). The p-value and confidence interval of an individual coefficient use the Satterthwaite approximation to the degrees of freedom as implemented in `lmerTest` (Kuznetsova et al., 2017), which recognizes that a descriptor that varies only between sites is effectively tested on 74 sites rather than 400 sensor-nights. Explained variance is reported as Nakagawa and Schielzeth's marginal R^2 (fixed effects only) and conditional R^2 (fixed and random effects together) (Nakagawa & Schielzeth, 2013), computed from the variance components and checked against the performance package (Lüdtke et al., 2021), and the intraclass correlation $ICC = \sigma^2_u / (\sigma^2_u + \sigma^2_\epsilon)$ gives the share of the variance of ΔT that is persistent between sites. Descriptors were standardized (mean 0, standard deviation 1, using the pooled 74-site values throughout so that coefficients are comparable between districts); coefficients are reported per standard deviation and converted to natural units (per 10 percentage points of cover, per meter of height, per kilometer of coastal distance, per 0.1 of sky view factor or aspect ratio). Model comparison among non-nested candidates used Akaike's information criterion computed from ML fits (Burnham & Anderson, 2002).

Sequence of models. The analysis proceeded in the order shown in Figure 2:

1. *Scale.* The full model (all eight descriptors, plus district in the pooled case) was fitted at each radius, pooled and separately for each district. The radius adopted was the one that was best, or within 2 AIC units of the best (Burnham & Anderson, 2002), in all three comparisons. As a cross-check, a forward-stepwise ordinary least squares regression on site means was also fitted at each radius.
2. *Collinearity.* Pairwise correlations and variance inflation factors (VIF) of the descriptors were examined at the adopted radius; VIF values were interpreted with the caution that they indicate imprecision, not bias (O'Brien, 2007).
3. *Null models.* Intercept-only models gave the variance partition (ICC) pooled and by district.
4. *Single-descriptor models,* pooled with district as a covariate and separately by district, gave each descriptor's effect when considered alone.
5. *Full models.* The pooled full model (district + eight descriptors) and the two district full models gave the conditional effects. Because the descriptors are collinear, the full models were complemented by an exhaustive comparison of all subsets of at most three descriptors within each district (93 candidate models), ranked by AIC; the Akaike weight of each model and the summed weight of every model containing a given descriptor were used as a measure of that descriptor's importance (Burnham & Anderson, 2002).
6. *Interactions.* Whether the effect of each descriptor differs between districts was tested by adding district $\times$ descriptor interactions to the pooled full model, one at a time and all together, with likelihood-ratio tests. This is the formal test of transferability between the two districts (Research Question 3).
7. *Checks.* The spatial pattern was compared between nights; the night-selection thresholds were varied as described above and the pooled models refitted; the district models were refitted on the calmest nights only (night-mean wind below 5 km h^{-1}); and the ability of the models to predict a site they had not seen was assessed by leave-one-site-out cross-validation, in which each site's mean ΔT was predicted from the fixed effects of a model fitted to the other sites and the root-mean-square error and correlation over all sites were computed.
8. *Unexplained site effects.* The random intercepts (best linear unbiased predictions, BLUPs) of the full pooled model were extracted with their conditional standard deviations and mapped (Research Question 4).
9. *Absolute comparison.* On the two nights shared by the districts, the absolute night-mean air temperatures of the two networks were compared directly.

Software and reproducibility. The whole analysis was written in R 4.3 (R Core Team, 2024).

The logger record was binned, referenced and screened with dplyr and lubridate (Grolemund & Wickham, 2011; Wickham et al., 2019); the descriptors were computed with sf for vector geometry (Pebesma, 2018) and terra for the 1m rasters (Hijmans, 2024); the mixed models were fitted with lme4 and lmerTest (Bates et al., 2015; Kuznetsova et al., 2017) with variance inflation factors from the car package (Companion to Applied Regression) (Fox & Weisberg, 2019); and the figures were drawn with ggplot2 (Wickham, 2016). Table 21 in Appendix C lists every package with its version, its role and the functions used. To guard against an error in any single implementation, the complete pipeline was written a second time, independently, in Python 3.11 (pandas, rasterio, Shapely, statsmodels (Seabold & Perktold, 2010)) and run on the same inputs; the two implementations agree on the sensor-night departures to machine precision, on every

coefficient and variance component to three decimals and on every AIC and R^2 to two (Appendix C). Every script, the processed data and the raw logger exports are archived with the thesis (Appendix C), and a single command regenerates every number, table and figure; an R Markdown walkthrough accompanies the archive and shows the code of each step with its output. The code, data and walkthrough are also available in the project's GitHub repository (<https://github.com/justinkellum-ship-it/JustinKellumThesis>).

4 Results

This chapter presents the results in the order of the analysis: the record and the checks on its structure (4.1), the size and persistence of the within-district pattern (4.2; Research Question 1), the scale and collinearity of the descriptors (4.3–4.4), the single-descriptor and pooled models (4.5–4.6; Research Question 2), the district models and the interaction tests (4.7; Research Question 3), the robustness checks (4.8), the unexplained site effects (4.9; Research Question 4) and the absolute comparison of the two districts (4.10).

4.1 The analyzed record

The analyzed data set comprises 400 sensor-nights on 11 calm, clear nights: 248 in Honolulu (36 sites, seven nights between 30 November 2024 and 3 January 2025) and 152 in ‘Ewa (38 sites, four nights between 25 November and 25 December 2024) (Tables 1 and 2). All 400 sensor-nights had the full 24 night bins. Night-mean air temperatures on the analyzed nights averaged 22.2°C in Honolulu and 21.3°C in ‘Ewa.

Because the reference removes the district’s median at every step, the mean of ΔT over a night’s sensors is close to zero on every night (largest absolute night mean 0.23°C, on the windiest Honolulu night). The night-to-night component of the variance is therefore residual scatter around each site’s persistent departure, not a shift common to all sites, and the random-intercept structure with a single residual term is appropriate.

4.2 Magnitude and persistence of the within-district pattern

The within-district contrasts are large relative to the district-wide differences between nights. Site-mean ΔT ranged from -1.87 to $+1.52^\circ\text{C}$ in Honolulu (a span of 3.4°C , standard deviation

among sites 0.73°C) and from -1.40 to $+1.61^{\circ}\text{C}$ in ‘Ewa (span 3.0°C , standard deviation 0.62°C) (Figure 11; Appendix Table 19). On individual nights the extremes were wider: -2.1 to $+3.9^{\circ}\text{C}$ in Honolulu and -2.0 to $+2.3^{\circ}\text{C}$ in ‘Ewa. The coolest Honolulu site (HNL01, -1.87°C) is a grassed, tree-bordered site on the University of Hawai‘i campus at the mouth of Mānoa Valley; the warmest (HNL08, $+1.52^{\circ}\text{C}$) and the high-rise Waikīkī and Ala Moana sites (HNL33, ID07; $+1.2$ to $+1.4^{\circ}\text{C}$) lie on the coastal plain. In ‘Ewa the extremes were an open, unbuilt site at the edge of Kalaeloa (EWA16, -1.40°C) and a fully paved site in the Campbell Industrial Park (EWA26, $+1.61^{\circ}\text{C}$), with the paved commercial and industrial sites of Kapolei warmest and the open and residential sites coolest.

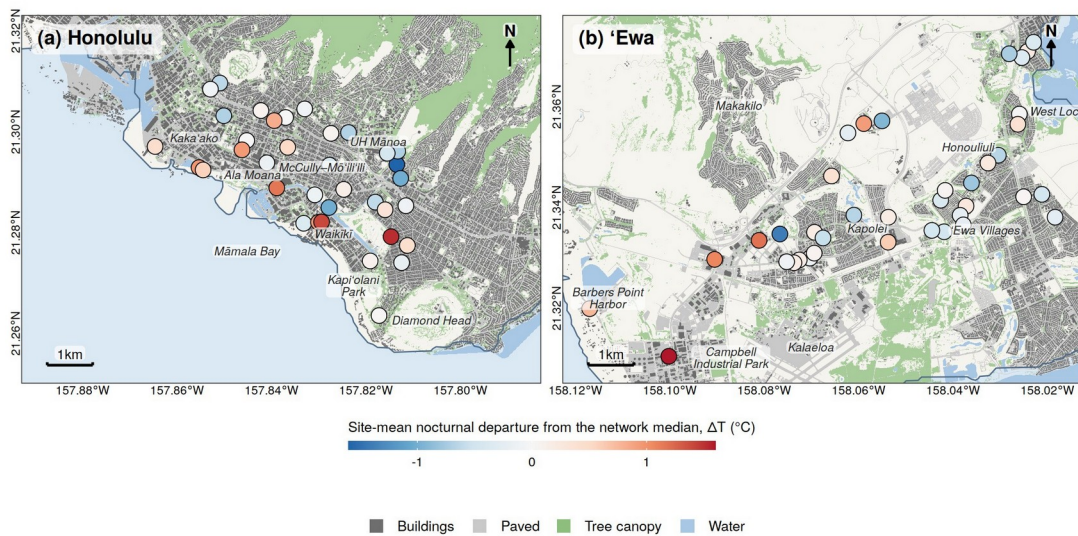


Figure 11. Site-mean nocturnal departure from the district median, ΔT , on calm, clear nights, in (a) Honolulu and (b) ‘Ewa, on the urban-form base map. This shows the observed pattern that the models explain (Research Question 1).

The null models quantify how persistent these contrasts are (Table 5). Pooled, 63% of the variance of ΔT is between sites (ICC = 0.63); the between-site standard deviation is 0.64°C and the night-to-night residual 0.49°C . The two districts differ markedly. In Honolulu the pattern is

highly persistent (ICC = 0.81; between-site SD 0.72°C, residual 0.35°C), whereas in ‘Ewa it is less so (ICC = 0.36; between-site SD 0.52°C, residual 0.68°C). The persistence can also be seen directly in the correlation of site ΔT between nights (Figure 12). In Honolulu the six other nights (night-mean winds 1.6–6.0km h⁻¹) reproduce each other’s pattern with correlations of 0.90–0.98; only the windiest night (30 November, 6.9km h⁻¹) departs, correlating at 0.58–0.67 with the others. In ‘Ewa the three nights of 1, 21 and 25 December are mutually consistent ($r = 0.80$ –0.97), but the first night, 25 November, the windiest of the four at 6.1km h⁻¹, shares none of their pattern ($r = -0.19$ to 0.00). The lower ICC in ‘Ewa therefore arises largely from one night on which the spatial pattern was different, not from noise spread evenly over all nights; this is examined further in Section 4.8.

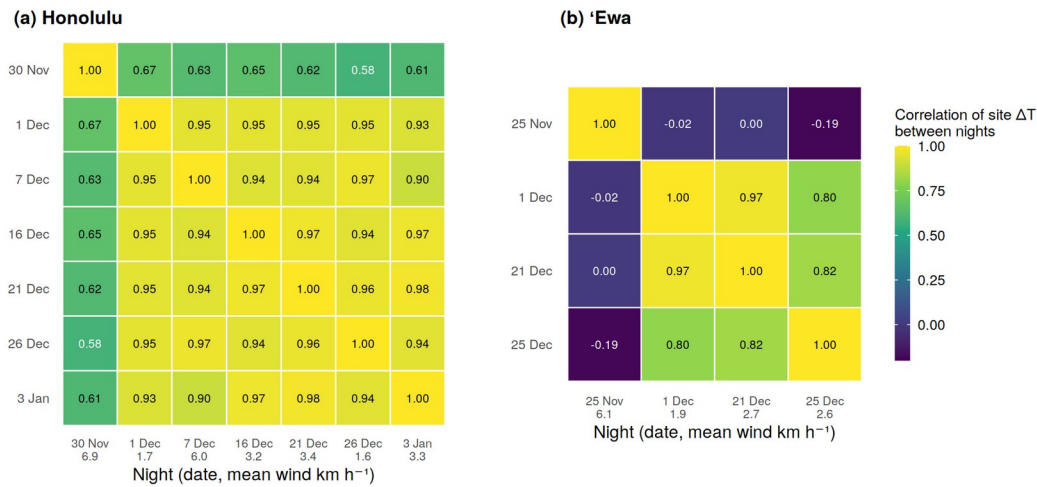


Figure 12. Night-to-night repeatability of the spatial pattern: correlation of site ΔT between each pair of analyzed nights in (a) Honolulu and (b) ‘Ewa, with each night’s mean ERA5 wind speed. This shows how persistent the pattern is and which nights depart from it.

Table 5. Variance components of the intercept-only (null) mixed models (REML) and the intraclass correlation coefficient $ICC = \sigma^2_u / (\sigma^2_u + \sigma^2_\epsilon)$.

Model	Between-site variance σ^2_u (°C ²)	Between-site SD (°C)	Residual variance σ^2_ϵ (°C ²)	Residual SD (°C)	ICC
Pooled (74 sites, 400 sensor-nights)	0.403	0.64	0.241	0.49	0.63

Model	Between-site variance σ^2_u (°C ²)	Between-site SD (°C)	Residual variance σ^2_ε (°C ²)	Residual SD (°C)	ICC
Honolulu (36 sites, 248)	0.512	0.72	0.120	0.35	0.81
‘Ewa (38 sites, 152)	0.267	0.52	0.465	0.68	0.36

4.3 Source-area scale

At the pooled level, the full model at 100m was within 1.7 AIC units of the model at 50m and had a marginal R^2 as high as any (0.27 against 0.27 and 0.21), while the model at 200m was 10 AIC units worse (Table 6; Figure 13). Within Honolulu the three radii were nearly equivalent (100m best, 50m within 0.4 and 200m within 3.3 AIC units); within ‘Ewa 100m was clearly best (50m worse by 12.3 and 200m by 5.1 AIC units). Under the pre-specified rule (the radius that is best or within 2 AIC units of the best in all three comparisons), 100m was adopted. The forward-stepwise regression on site means, fitted as a cross-check, gave the same answer for ‘Ewa (R^2 0.64 at 100m against 0.45 and 0.58) and a marginal preference for 50m in Honolulu (0.49 against 0.44 and 0.42), where the two radii are statistically indistinguishable in the mixed models. The single-descriptor results by radius (Figure 13b, c) show why: in ‘Ewa, building footprint gains explanatory power as the radius grows, building height peaks at 100m and impervious fraction loses a little, whereas in Honolulu impervious fraction is most informative at 50m and canyon aspect ratio at 200m. All subsequent results use the 100m descriptors; the point descriptors (sky view factor and distance to coast) are independent of the radius.

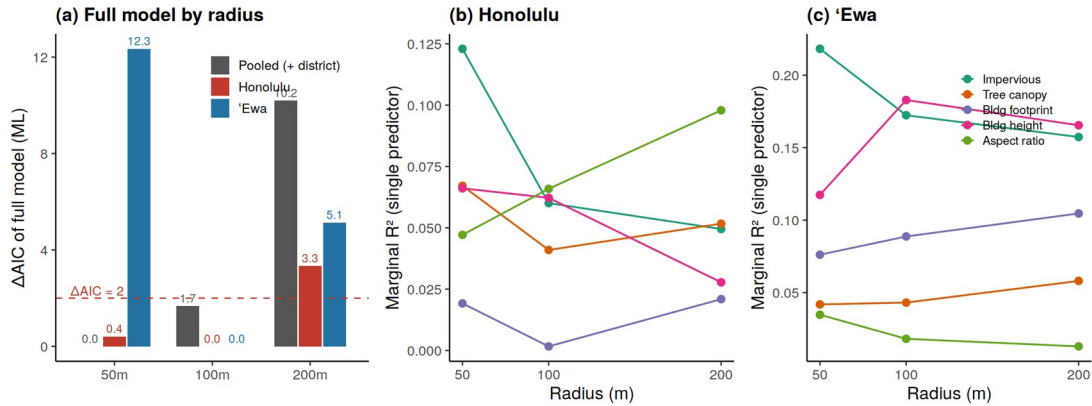


Figure 13. Source-area scale. (a) Δ AIC of the full mixed model at each radius, pooled and by district; the dashed line marks 2 AIC units. (b, c) Marginal R^2 of single-descriptor models against radius, by district. This justifies the 100m radius with the same models used in the analysis.

Table 6. Source-area scale selection. AIC and Δ AIC (difference from the best radius within each scope) of the full mixed model (all eight predictors, random intercept per site) fitted by maximum likelihood; R^2_m and R^2_c are its marginal and conditional R^2 . The last column gives the R^2 of a forward-stepwise ordinary least squares regression on site means, fitted as a cross-check.

Scope	Radius (m)	AIC (ML)	Δ AIC	R^2_m	R^2_c	Stepwise OLS R^2 (site means)
Pooled + study area	50	720.2	0.0	0.27	0.62	
	100	721.8	1.7	0.27	0.63	
	200	730.4	10.2	0.21	0.63	
Honolulu	50	294.8	0.4	0.43	0.80	0.49
	100	294.4	0.0	0.44	0.80	0.44
	200	297.7	3.3	0.41	0.80	0.42
'Ewa	50	352.0	12.3	0.28	0.36	0.45
	100	339.7	0.0	0.34	0.36	0.64
	200	344.8	5.1	0.32	0.36	0.58

4.4 Collinearity

The descriptors are correlated, as the physics of urban form implies (Table 7; Figure 14).

Impervious fraction and building footprint are strongly related in both districts ($r = 0.77$ and 0.75), and canyon aspect ratio, which is a function of building height and footprint, is correlated with footprint ($r = 0.77$ in Honolulu, 0.83 in 'Ewa) and, in Honolulu, with height ($r = 0.68$).

Variance inflation factors are highest for building footprint (5.1 and 5.8) and aspect ratio (5.4 and 3.4) and below 4 for the other six descriptors; none reaches the level at which estimation

becomes unstable, but the standard errors of footprint and aspect ratio in the full models are inflated by a factor of about two ($\sqrt{\text{VIF}}$ 1.8–2.4), and their conditional effects must be read with that in mind. Tree canopy, water, sky view factor and distance to the coast are nearly independent of the built-form cluster (VIF 1.1–1.9). In Honolulu, tree canopy is negatively related to impervious fraction ($r = -0.43$) and water fraction to impervious fraction ($r = -0.57$), reflecting the parks and the harbor; in ‘Ewa these correlations are absent.

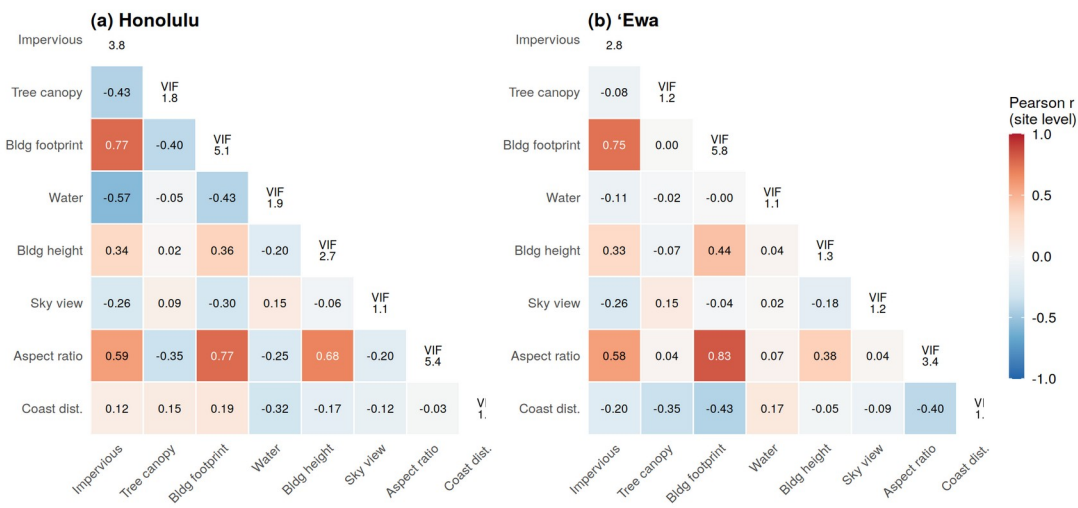


Figure 14. Pairwise Pearson correlations between the eight descriptors at the site level, with the variance inflation factor of each descriptor on the diagonal, in (a) Honolulu and (b) ‘Ewa. This shows the collinearity that limits the interpretation of the full models.

Table 7. Variance inflation factors (VIF) of the eight predictors, for each district and pooled, and the largest correlation of each predictor with any of the other seven in each district (absolute value of Pearson’s r), at the site level.

Predictor	VIF Honolulu	Largest correlation, Honolulu	VIF ‘Ewa	Largest correlation, ‘Ewa	VIF pooled
Impervious surface fraction (%)	3.8	0.77	2.8	0.75	3.0
Tree canopy fraction (%)	1.8	0.43	1.2	0.35	1.2
Building footprint fraction (%)	5.1	0.77	5.8	0.83	4.4
Water surface fraction (%)	1.9	0.57	1.1	0.17	1.3
Mean building height (m)	2.7	0.68	1.3	0.44	2.2
Sky view factor (0–1)	1.1	0.30	1.2	0.26	1.1
Canyon aspect ratio (H/W)	5.4	0.77	3.4	0.83	4.1

Predictor	VIF Honolulu	Largest correlation, Honolulu	VIF ‘Ewa	Largest correlation, ‘Ewa	VIF pooled
Distance to coast (km)	1.3	0.32	1.6	0.43	1.4

4.5 Single descriptors

Considered one at a time (Table 8; Figure 15), five descriptors are related to ΔT in the pooled, district-adjusted models: impervious fraction ($+0.11 \pm 0.03^\circ\text{C}$ per 10 percentage points; $p < 0.001$; $R^2_m = 0.11$), building height ($+0.06 \pm 0.02^\circ\text{C}$ per meter; $p = 0.002$; $R^2_m = 0.10$), tree canopy ($-0.15 \pm 0.07^\circ\text{C}$ per 10 percentage points; $p = 0.03$), canyon aspect ratio ($+0.06 \pm 0.03^\circ\text{C}$ per 0.1; $p = 0.04$) and distance to the coast ($-0.16 \pm 0.08^\circ\text{C}$ per kilometer; $p = 0.04$). Building footprint is marginal ($p = 0.07$) and water fraction and sky view factor are unrelated to ΔT ($p = 0.97$ and 0.34).

The districts, however, do not share this ranking. In Honolulu the single strongest descriptor is distance to the coast: sites are cooler by $0.57 \pm 0.14^\circ\text{C}$ for every kilometer inland ($p < 0.001$; $R^2_m = 0.26$), and no built-form descriptor is significant on its own (impervious fraction $p = 0.10$; height $p = 0.10$; aspect ratio $p = 0.09$; tree canopy $p = 0.18$). In ‘Ewa the strongest descriptors are building height ($+0.15 \pm 0.03^\circ\text{C}$ per meter; $p < 0.001$; $R^2_m = 0.18$) and impervious fraction ($+0.15 \pm 0.04^\circ\text{C}$ per 10 percentage points; $p < 0.001$; $R^2_m = 0.17$), followed by building footprint ($p = 0.008$) and sky view factor (warmer where the sky is more obstructed; $p = 0.03$), and distance to the coast has no effect at all ($p = 0.84$). The bivariate plots (Figure 15) show the same contrast: the ‘Ewa sites fall along a clear impervious and height gradient, whereas in Honolulu the strongest single gradient runs from the shore to the valley mouths.

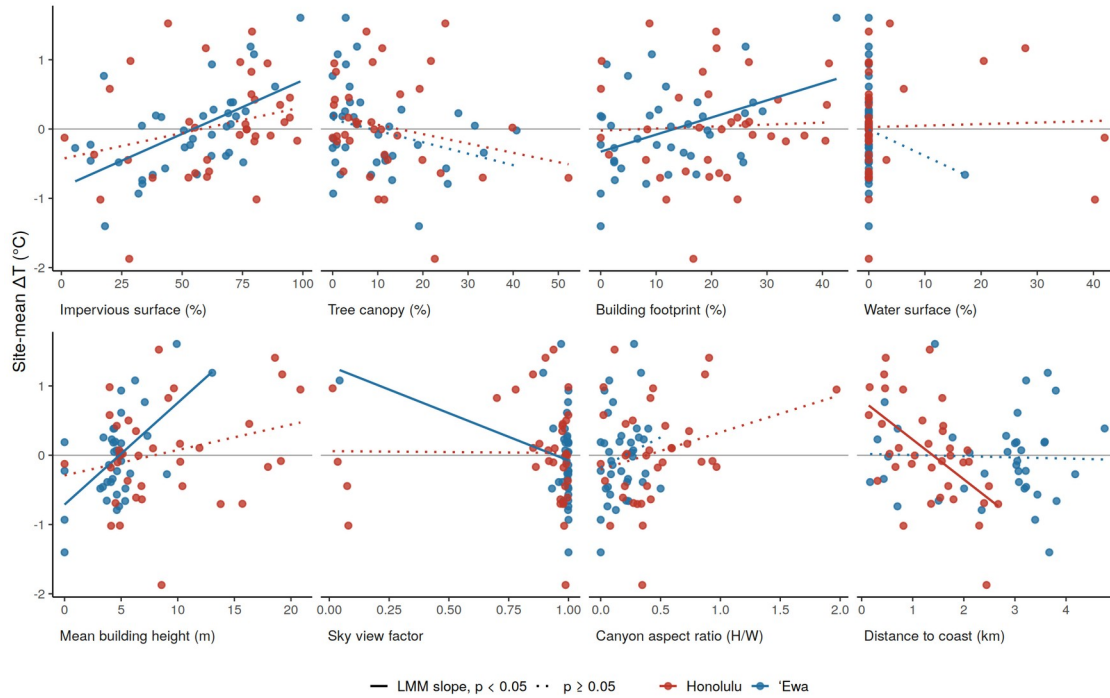


Figure 15. Site-mean ΔT against each descriptor at 100m, by district, with the slope of the single-descriptor mixed model (solid where $p < 0.05$, dotted otherwise). This shows the raw relationships that the models summarize.

Table 8. Single-predictor mixed models (random intercept per site). β is the change in ΔT per one standard deviation of the predictor ($\pm$ standard error, REML); p is from the likelihood-ratio test against the null model (ML). Bold: $p < 0.05$.

Model	Predictor	β per SD $\pm$ SE (°C)	Effect in natural units (°C)	p (LRT)	R^2_m
Pooled (study area as covariate)	Impervious surface fraction (%)	+0.27 $\pm$ 0.08	+0.109 $\pm$ 0.030 per +10 pp	< 0.001	0.11
	Tree canopy fraction (%)	-0.17 $\pm$ 0.08	-0.148 $\pm$ 0.068 per +10 pp	0.03	0.04
	Building footprint fraction (%)	+0.15 $\pm$ 0.08	+0.128 $\pm$ 0.073 per +10 pp	0.07	0.03
	Water surface fraction (%)	-0.00 $\pm$ 0.09	-0.004 $\pm$ 0.102 per +10 pp	0.97	0.00
	Mean building height (m)	+0.28 $\pm$ 0.09	+0.056 $\pm$ 0.018 per +1m	0.002	0.10
	Sky view factor (0–1)	-0.08 $\pm$ 0.09	+0.032 $\pm$ 0.034 per -0.1	0.34	0.01
	Canyon aspect ratio (H/W)	+0.19 $\pm$ 0.09	+0.057 $\pm$ 0.027 per +0.1	0.04	0.05
	Distance to coast (km)	-0.18 $\pm$ 0.09	-0.161 $\pm$ 0.078 per +1km	0.04	0.04
Honolulu	Impervious surface fraction (%)	+0.19 $\pm$ 0.12	+0.075 $\pm$ 0.046 per +10 pp	0.10	0.06
	Tree canopy fraction (%)	-0.15 $\pm$ 0.12	-0.135 $\pm$ 0.102 per +10 pp	0.18	0.04
	Building footprint fraction (%)	+0.03 $\pm$ 0.13	+0.029 $\pm$ 0.111 per +10 pp	0.79	0.00

Model	Predictor	β per SD $\pm$ SE (°C)	Effect in natural units (°C)	p (LRT)	R ² m
'Ewa	Water surface fraction (%)	+0.02 $\pm$ 0.10	+0.022 $\pm$ 0.116 per +10 pp	0.85	0.00
	Mean building height (m)	+0.18 $\pm$ 0.11	+0.037 $\pm$ 0.022 per +1m	0.10	0.06
	Sky view factor (0–1)	-0.01 $\pm$ 0.11	+0.003 $\pm$ 0.043 per -0.1	0.95	0.00
	Canyon aspect ratio (H/W)	+0.18 $\pm$ 0.11	+0.054 $\pm$ 0.032 per +0.1	0.09	0.06
	Distance to coast (km)	-0.64 $\pm$ 0.16	-0.574 $\pm$ 0.143 per +1km	< 0.001	0.26
	Impervious surface fraction (%)	+0.39 $\pm$ 0.09	+0.155 $\pm$ 0.037 per +10 pp	< 0.001	0.17
	Tree canopy fraction (%)	-0.19 $\pm$ 0.10	-0.165 $\pm$ 0.092 per +10 pp	0.07	0.04
	Building footprint fraction (%)	+0.28 $\pm$ 0.10	+0.247 $\pm$ 0.091 per +10 pp	0.008	0.09
	Water surface fraction (%)	-0.34 $\pm$ 0.32	-0.382 $\pm$ 0.365 per +10 pp	0.29	0.01
	Mean building height (m)	+0.72 $\pm$ 0.16	+0.147 $\pm$ 0.033 per +1m	< 0.001	0.18
	Sky view factor (0–1)	-0.35 $\pm$ 0.16	+0.136 $\pm$ 0.063 per -0.1	0.03	0.06
	Canyon aspect ratio (H/W)	+0.29 $\pm$ 0.25	+0.087 $\pm$ 0.076 per +0.1	0.25	0.02
	Distance to coast (km)	-0.02 $\pm$ 0.09	-0.017 $\pm$ 0.085 per +1km	0.84	0.00

4.6 The pooled model

With all eight descriptors and districts together (Table 9; Figure 16a), the descriptors explain 25% of the variance of ΔT (marginal $R^2 = 0.25$) and the descriptors plus the site intercepts explain 65% (conditional $R^2 = 0.65$). The eight descriptors together are highly significant given district ($\chi^2(8) = 34.9$, $p < 0.001$) and reduce the between-site variance from 0.408 to 0.275°C², i.e. by a third. Three effects are individually significant, with the expected signs: impervious fraction (+0.16 ± 0.05°C per 10 percentage points; $p < 0.001$), building height (+0.06 ± 0.02°C per meter; $p = 0.01$) and distance to the coast (−0.18 ± 0.07°C per kilometer, cooler inland; $p = 0.02$). Tree canopy is marginally cooling (−0.13 ± 0.07°C per 10 percentage points; $p = 0.06$). Building footprint takes a negative conditional coefficient ($p = 0.07$) that reverses its positive bivariate relationship; this follows from its collinearity with impervious fraction and aspect ratio and should not be read as a cooling effect. Water fraction, sky view factor and aspect ratio have no conditional effect.

The district term must be interpreted with care. Its coefficient (−0.34 ± 0.18°C for Honolulu relative to ‘Ewa; $p = 0.06$) does not mean that Honolulu is cooler. Because ΔT is centered on each district’s own median, a site of average form is by construction near zero in both districts; the Honolulu sites are, on average, more impervious, taller and closer to the coast than the ‘Ewa sites (Table 4), and those descriptors alone would predict a Honolulu site to be 0.40°C warmer than an ‘Ewa site. The district coefficient is the offset that cancels that difference (−0.34°C, so that the Honolulu mean returns to about zero). It reflects the way each network is centered, not a real difference in temperature, and it should not be read as evidence that one district is cooler than the other. The absolute comparison is made in Section 4.10. A

parsimonious pooled model with impervious fraction and height alone explains 15% of the variance ($p = 0.006$ and 0.03), significantly less than the full model ($\chi^2(6) = 17.3$, $p = 0.008$), the difference being carried mainly by the coastal-distance term.

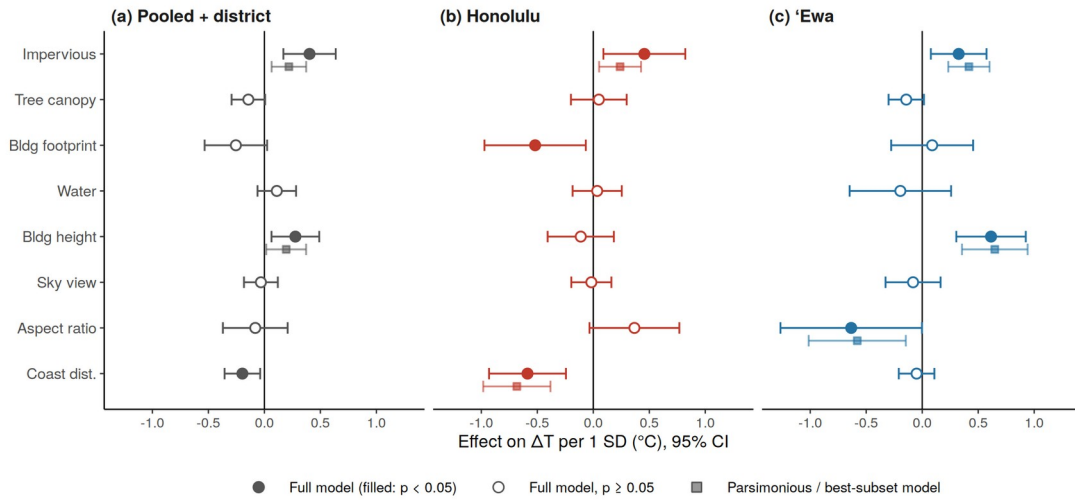


Figure 16. Standardized fixed effects (change in ΔT per one standard deviation of the descriptor, with 95% confidence intervals) of the full models, (a) pooled with district, (b) Honolulu and (c) 'Ewa; filled symbols are significant at $p < 0.05$, and the smaller squares show the parsimonious or best-subset model of each panel. This compares the conditional effects of all descriptors across the three models.

Table 9. Full pooled mixed model: $\Delta T \sim \text{study area} + \text{eight standardized predictors}$, random intercept per site (REML estimates; $n = 400$ sensor-nights, 74 sites; p -values from Satterthwaite t tests). Marginal $R^2 = 0.25$, conditional $R^2 = 0.65$; between-site variance 0.275°C^2 , residual variance 0.240°C^2 . Likelihood-ratio test of the eight predictors given study area: $\chi^2(8) = 34.9$, $p < 0.001$. Bold: $p < 0.05$.

Term	Estimate per SD ($^\circ\text{C}$)	SE	95% CI	p	Effect in natural units
Intercept ('Ewa, all predictors at their mean)	+0.231	0.120	[-0.01, +0.47]	0.06	
Study area: Honolulu (vs 'Ewa)	-0.338	0.177	[-0.69, +0.02]	0.06	
Impervious surface fraction (%)	+0.403	0.117	[+0.17, +0.64]	< 0.001	+0.161 $\pm$ 0.047 $^\circ\text{C}$ per +10 pp
Tree canopy fraction (%)	-0.143	0.075	[-0.29, +0.01]	0.06	-0.125 $\pm$ 0.066 $^\circ\text{C}$ per +10 pp
Building footprint fraction (%)	-0.254	0.140	[-0.53, +0.02]	0.07	-0.221 $\pm$ 0.121 $^\circ\text{C}$ per +10 pp
Water surface fraction (%)	+0.111	0.086	[-0.06, +0.28]	0.20	+0.126 $\pm$ 0.098 $^\circ\text{C}$ per +10 pp
Mean building height (m)	+0.277	0.107	[+0.06, +0.49]	0.01	+0.056 $\pm$ 0.022 $^\circ\text{C}$ per +1m
Sky view factor (0–1)	-0.031	0.076	[-0.18, +0.12]	0.68	+0.012 $\pm$ 0.030 $^\circ\text{C}$ per -

Term	Estimate per SD (°C)	SE	95% CI	p	Effect in natural units
			+0.12]		0.1
Canyon aspect ratio (H/W)	-0.081	0.145	[-0.37, +0.21]	0.58	-0.024 ± 0.043°C per +0.1
Distance to coast (km)	-0.196	0.080	[-0.36, - 0.04]	0.02	-0.177 ± 0.072°C per +1km

4.7 District models and the transferability test

Fitted separately (Table 10; Figure 16b, c), the two district models explain more of the variance than the pooled model and identify different drivers. In Honolulu the full model has a marginal R^2 of 0.39 and a conditional R^2 of 0.83 ($\chi^2(8) = 28.1$, $p < 0.001$); distance to the coast ($-0.53 \pm 0.15^\circ\text{C}$ per kilometer; $p = 0.002$) and impervious fraction ($+0.18 \pm 0.07^\circ\text{C}$ per 10 percentage points; $p = 0.02$) are its significant terms, aspect ratio is marginal ($+0.11 \pm 0.06^\circ\text{C}$ per 0.1; $p = 0.07$), building footprint again takes a negative conditional coefficient ($p = 0.03$), and height, tree canopy, water and sky view factor have no conditional effect. In ‘Ewa the full model has a marginal R^2 of 0.33 and a conditional R^2 of 0.39 ($\chi^2(8) = 41.5$, $p < 0.001$); building height ($+0.12 \pm 0.03^\circ\text{C}$ per meter; $p < 0.001$) and impervious fraction ($+0.13 \pm 0.05^\circ\text{C}$ per 10 percentage points; $p = 0.01$) are its significant terms, aspect ratio is negative given height ($-0.19 \pm 0.09^\circ\text{C}$ per 0.1; $p = 0.05$), tree canopy is marginally cooling ($p = 0.07$), and distance to the coast has no effect ($p = 0.52$).

The exhaustive comparison of small models gives the same result more simply (Tables 11 and 12). In Honolulu the best model of at most three descriptors is impervious fraction plus distance to the coast (marginal $R^2 = 0.36$), with six other models within 2 AIC units of it, every one of which contains coastal distance; over all candidate models coastal distance has a summed Akaike weight of 1.00 and impervious fraction 0.72, and no other descriptor exceeds 0.23. In that model each kilometer inland cools a site by $0.62 \pm 0.13^\circ\text{C}$ and each 10 percentage points of

impervious cover warms it by $0.10 \pm 0.04^{\circ}\text{C}$. In ‘Ewa the best model is impervious fraction, building height and aspect ratio (marginal $R^2 = 0.31$, Akaike weight 0.66, no competitor within 2 AIC units); height (summed weight 1.00) and impervious fraction (0.98) are essential, aspect ratio supporting (0.67), and coastal distance is irrelevant (0.02). In that model each meter of mean building height warms an ‘Ewa site by $0.13 \pm 0.03^{\circ}\text{C}$, each 10 percentage points of impervious cover by $0.17 \pm 0.04^{\circ}\text{C}$, and, given height and cover, each 0.1 of aspect ratio cools it by $0.17 \pm 0.06^{\circ}\text{C}$.

The formal test of whether the drivers differ between districts (Table 13; Figure 17) confirms the difference. Adding all eight district $\times$ descriptor interactions to the pooled model improves it markedly ($\chi^2(8) = 30.4$, $p < 0.001$; marginal R^2 rises from 0.25 to 0.41), and the interaction model has by far the lowest AIC of all candidate structures (Table 14). Two interactions are individually significant: building height (slope $+0.72$ per SD in ‘Ewa against $+0.09$ in Honolulu; $p = 0.004$) and distance to the coast (-0.05 per SD in ‘Ewa against -0.56 in Honolulu; $p = 0.004$). The impervious-fraction effect is positive in both districts and does not differ significantly ($p = 0.18$). The single-descriptor models, the best-subset comparison and the interaction test agree: impervious cover warms both districts, building height warms ‘Ewa but not Honolulu, and coastal distance cools inland sites in Honolulu but not in ‘Ewa. A single pooled model without interactions is therefore a poor description of either district, and Table 14 shows that the data prefer the interaction model to the pooled additive model.

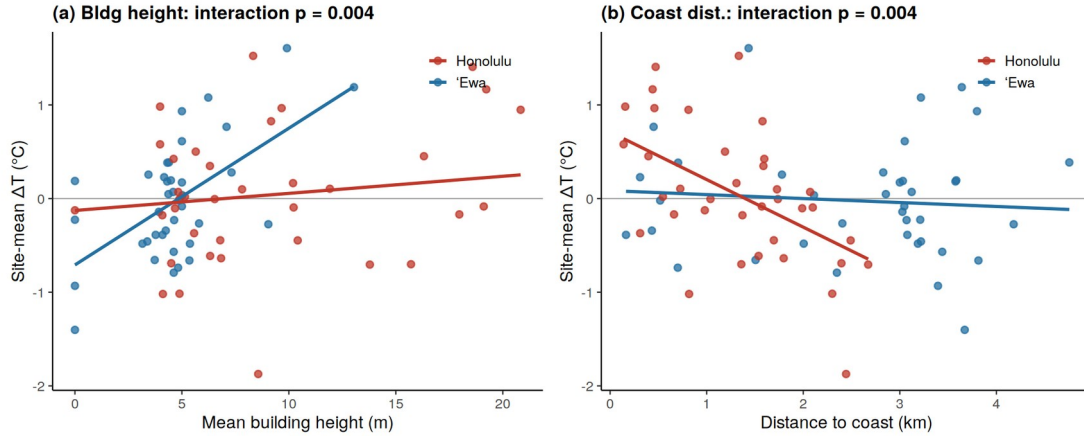


Figure 17. The two descriptors whose effects differ significantly between districts: site-mean ΔT against (a) mean building height and (b) distance to the coast, with the district-specific slopes from the interaction model. This demonstrates the transferability failure directly (Research Question 3).

Table 10. District full mixed models (eight standardized predictors, random intercept per site, REML; Satterthwaite p-values). Honolulu: $R^2_m = 0.39$, $R^2_c = 0.83$, LRT vs null $\chi^2(8) = 28.1$, $p < 0.001$. 'Ewa: $R^2_m = 0.33$, $R^2_c = 0.39$, $\chi^2(8) = 41.5$, $p < 0.001$. Standardization uses the pooled standard deviations so that coefficients are comparable between areas. Bold: $p < 0.05$.

Term	Honolulu: β per SD (°C)	Honolulu: u: p	Honolulu: natural units	'Ewa: β per SD (°C)	'Ewa: p	'Ewa: natural units
Intercept	-0.223 $\pm$ 0.125	0.09		+0.092 $\pm$ 0.159	0.57	
Impervious surface fraction (%)	+0.456 $\pm$ 0.178	0.02	+0.182 per +10 pp	+0.326 $\pm$ 0.122	0.01	+0.130 per +10 pp
Tree canopy fraction (%)	+0.049 $\pm$ 0.121	0.69	+0.043 per +10 pp	-0.143 $\pm$ 0.077	0.07	-0.125 per +10 pp
Building footprint fraction (%)	-0.519 $\pm$ 0.220	0.03	-0.450 per +10 pp	+0.089 $\pm$ 0.179	0.62	+0.077 per +10 pp
Water surface fraction (%)	+0.035 $\pm$ 0.107	0.75	+0.040 per +10 pp	-0.195 $\pm$ 0.222	0.39	-0.222 per +10 pp
Mean building height (m)	-0.112 $\pm$ 0.144	0.44	-0.023 per +1m	+0.615 $\pm$ 0.152	< 0.001	+0.125 per +1m
Sky view factor (0–1)	-0.017 $\pm$ 0.087	0.84	+0.007 per -0.1	-0.082 $\pm$ 0.120	0.50	+0.032 per -0.1
Canyon aspect ratio (H/W)	+0.367 $\pm$ 0.195	0.07	+0.110 per +0.1	-0.634 $\pm$ 0.309	0.05	-0.189 per +0.1
Distance to coast (km)	-0.587 $\pm$ 0.167	0.002	-0.530 per +1km	-0.050 $\pm$ 0.078	0.52	-0.045 per +1km

Table 11. The five best subsets of at most three predictors in each study area, ranked by maximum-likelihood AIC among all 93 candidate models (null model included). Akaike weights are relative to the whole candidate set.

Study area	Rank	Predictors	k	ΔAIC	Akaike weight	R ² m
Honolulu	1	impervious + coast distance	2	0.0	0.18	0.36
	2	impervious + footprint + coast distance	3	0.1	0.17	0.38
	3	impervious + aspect + coast distance	3	1.6	0.08	0.36
	4	impervious + height + coast distance	3	1.8	0.07	0.36
	5	impervious + water + coast distance	3	1.9	0.07	0.36
'Ewa	1	impervious + height + aspect	3	0.0	0.66	0.31
	2	impervious + tree + height	3	3.4	0.12	0.29
	3	impervious + footprint + height	3	5.2	0.05	0.28
	4	impervious + height	2	5.5	0.04	0.27
	5	impervious + height + sky view	3	5.5	0.04	0.28

Table 12. Relative importance of each predictor: sum of Akaike weights of all candidate models (≤ 3 predictors) that contain it, by study area.

Predictor	Importance Honolulu	Importance 'Ewa
Impervious surface fraction (%)	0.72	0.98
Tree canopy fraction (%)	0.12	0.13
Building footprint fraction (%)	0.23	0.06
Water surface fraction (%)	0.13	0.04
Mean building height (m)	0.14	1.00
Sky view factor (0–1)	0.11	0.05
Canyon aspect ratio (H/W)	0.22	0.67
Distance to coast (km)	1.00	0.02

Table 13. Study-area $\times$ predictor interactions, each added on its own to the full pooled model and tested by likelihood-ratio test (ML). Slopes are from the model with that interaction (REML). Joint test of all eight interactions: $\chi^2(8) = 30.4$, $p < 0.001$; the interaction model has $R^2m = 0.41$.

Predictor	Slope 'Ewa (per SD)	Slope Honolulu (per SD)	Difference $\pm$ SE	LRT $\chi^2(1)$	p
Impervious surface fraction (%)	+0.49	+0.29	-0.20 $\pm$ 0.16	1.8	0.18
Tree canopy fraction (%)	-0.23	-0.06	+0.17 $\pm$ 0.15	1.5	0.21
Building footprint fraction (%)	-0.17	-0.44	-0.27 $\pm$ 0.18	2.5	0.11
Water surface fraction (%)	-0.12	+0.13	+0.25 $\pm$ 0.32	0.7	0.40
Mean building height (m)	+0.72	+0.09	-0.63 $\pm$ 0.23	8.4	0.004
Sky view factor (0–1)	-0.17	+0.01	+0.18 $\pm$ 0.18	1.1	0.29
Canyon aspect ratio (H/W)	-0.14	-0.08	+0.06 $\pm$ 0.32	0.0	0.84
Distance to coast (km)	-0.05	-0.56	-0.52 $\pm$ 0.19	8.3	0.004

Table 14. Maximum-likelihood AIC of the candidate model structures (all with the same 400 sensor-nights; the parameter count includes the variance components).

Model	Parameters	AIC	Δ AIC
OLS: study area + 8 predictors (no random effect)	11	863.7	156.3
Null: random intercept only	3	738.9	31.5
Random intercept + study area	4	740.8	33.3
Random intercept + 8 predictors	11	723.9	16.5
Random intercept + study area + impervious + height	6	727.1	19.7
Random intercept + study area + 8 predictors	12	721.8	14.4
Random intercept + study area $\times$ 8 predictors (interactions)	20	707.4	0.0

4.8 Robustness

Night-selection thresholds. Table 15 and Figure 18 show the pooled results under each alternative selection rule. Across wind thresholds of 8 to 20km h⁻¹ and cloud thresholds of 15 to 50%, the impervious effect ranged from +0.15 to +0.25°C per SD and the height effect from +0.15 to +0.20°C per SD, always significant; the marginal R² of the full model ranged from 0.14 to 0.34. Relaxing the wind threshold attenuates the effects and the explained variance; relaxing the cloud threshold to 35% or the required share of bins to 60% changes them little, and only the loosest rules (cloud below 50%, wind up to 15–20km h⁻¹) reduce them appreciably. The site pattern remains almost unchanged throughout (correlation of site means with the baseline selection 0.94–0.99). Without any weather filter (all 42 complete nights per district, 3,087 sensor-nights) the effects are about a third smaller (+0.14 and +0.13°C per SD), the spread of the site means shrinks from 0.67 to 0.47°C, and the full model explains 13% of the variance; the complementary set of windy or cloudy nights gives the weakest signal of all. Filtering to calm, clear nights therefore sharpens a pattern that is present, attenuated, on every night, and the conclusions do not hinge on the exact thresholds.

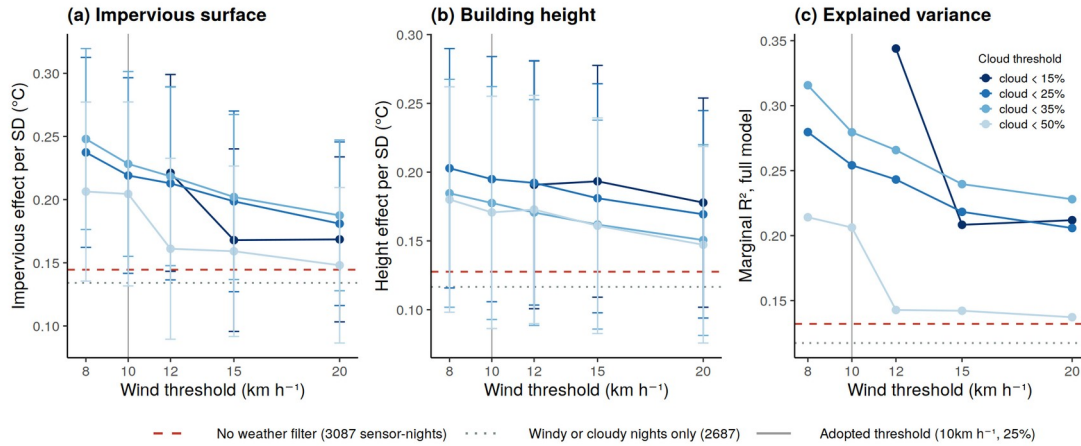


Figure 18. Sensitivity to the night-selection thresholds: (a) the impervious-fraction effect, (b) the building-height effect and (c) the marginal R^2 of the full pooled model, against the wind threshold for four cloud thresholds; the dashed line is the value with no weather filter and the dotted line the value on windy or cloudy nights only. This shows that the filtering refines thermal observations.

Table 15. Sensitivity of the results to the night-selection thresholds. Nights were re-selected from the complete record under each rule and the pooled models refitted. β : effect per pooled SD ($\pm$ SE) in the parsimonious model $\Delta T \sim$ study area + impervious + height; R^2_m (full): marginal R^2 of the full pooled model; SD sites: standard deviation of site-mean ΔT ; r vs base: correlation of site means with those of the adopted rule (wind $< 10\text{ km h}^{-1}$, cloud $< 25\%$, $\geq 75\%$ of bins).

Selection rule	Nights Hon/'Ewa	Sensor- nights	Imperv. β	Height β	R^2_m (full)	SD sites (°C)	r vs base
wind $< 8\text{ km h}^{-1}$, cloud $< 25\%$, $\geq 75\%$ of bins	5 / 4	331	$+0.24 \pm 0.08$	$+0.20 \pm 0.09$	0.28	0.67	0.99
wind $< 10\text{ km h}^{-1}$, cloud $< 25\%$, $\geq 60\%$ of bins	12 / 9	768	$+0.24 \pm 0.07$	$+0.18 \pm 0.09$	0.28	0.65	0.98
wind $< 10\text{ km h}^{-1}$, cloud $< 25\%$, $\geq 75\%$ of bins	7 / 4	400	$+0.22 \pm 0.08$	$+0.19 \pm 0.09$	0.25	0.67	1.00
wind $< 10\text{ km h}^{-1}$, cloud $< 35\%$, $\geq 75\%$ of bins	10 / 6	583	$+0.23 \pm 0.07$	$+0.18 \pm 0.08$	0.28	0.64	0.99
wind $< 10\text{ km h}^{-1}$, cloud $< 50\%$, $\geq 75\%$ of bins	13 / 11	877	$+0.20 \pm 0.07$	$+0.17 \pm 0.08$	0.21	0.63	0.98
wind $< 12\text{ km h}^{-1}$, cloud $< 25\%$, $\geq 75\%$ of bins	7 / 7	514	$+0.21 \pm 0.08$	$+0.19 \pm 0.09$	0.24	0.66	0.99
wind $< 15\text{ km h}^{-1}$, cloud $< 25\%$, $\geq 75\%$ of bins	9 / 11	737	$+0.20 \pm 0.07$	$+0.18 \pm 0.08$	0.22	0.62	0.98
wind $< 20\text{ km h}^{-1}$, cloud $< 25\%$, $\geq 75\%$ of bins	11 / 14	923	$+0.18 \pm 0.06$	$+0.17 \pm 0.08$	0.21	0.57	0.97
no weather filter (all complete nights)	42 / 42	3087	$+0.14 \pm 0.05$	$+0.13 \pm 0.06$	0.13	0.47	0.96
windy or cloudy nights only (complement)	35 / 38	2687	$+0.13 \pm 0.05$	$+0.12 \pm 0.06$	0.12	0.44	0.95

Calm-core refit. Refitting the district best-subset models on the calmest nights only (night-mean wind below 5 km h^{-1} : five Honolulu and three ‘Ewa nights) strengthened rather than weakened them. In Honolulu the marginal R^2 rose from 0.36 to 0.42 (impervious $+0.11^\circ\text{C}$ per 10 points, coast -0.60°C per km, both $p \leq 0.005$); in ‘Ewa it rose from 0.31 to 0.49 (impervious $+0.22^\circ\text{C}$ per 10 points, height $+0.13^\circ\text{C}$ per meter, $p < 0.001$ and $p = 0.002$; aspect ratio $p = 0.08$), and the ‘Ewa conditional ICC rose from 0.10 to 0.72 once the anomalous night of 25 November was excluded.

Terrain. Because the Honolulu coastal gradient runs up-slope toward the valley mouths, the models were refitted with site elevation (Section 3.6) as an additional standardized term (Table 16; Figure 19). On its own, elevation is a weaker version of distance to the coast in Honolulu ($-0.32 \pm 0.14^\circ\text{C}$ per 10m, $p = 0.03$, $R^2_m = 0.11$, against $R^2_m = 0.26$ for coastal distance), and it has no effect in ‘Ewa ($p = 0.56$). Given distance to the coast, elevation adds nothing in Honolulu (likelihood-ratio $p = 0.82$; $\Delta\text{AIC} +1.9$), whereas given elevation, distance to the coast remains ($p = 0.003$) with its coefficient unchanged (-0.55 ± 0.18 against $-0.57 \pm 0.14^\circ\text{C}$ per km). Added to the Honolulu best-subset and full models, elevation is not significant (likelihood-ratio $p = 0.67$ and 0.63), raises the AIC by 1.8, and leaves the coast coefficient within one standard error of its value (-0.57 and -0.49°C per km); added to the ‘Ewa models it changes nothing (likelihood-ratio $p \geq 0.32$). With elevation among the candidates, the exhaustive search returns the same best subsets in both districts (elevation first appears in the fifth-ranked Honolulu model, $\Delta\text{AIC} = 1.8$, and in no ‘Ewa model within 7 AIC units), and adding it to the best-subset models makes the leave-one-site-out predictions slightly worse (Honolulu $r = 0.57$ instead of 0.58 ; ‘Ewa 0.68 instead of 0.71). Only in the pooled parsimonious model, which has no coastal

term, does elevation enter with a marginal effect ($-0.18 \pm 0.08^{\circ}\text{C}$ per 10m, $p = 0.04$; $\Delta\text{AIC} = -2.7$), standing in for the coast term; in the pooled full model the two share the weak pooled coastal signal (each $p > 0.1$ in the presence of the other), and the elevation slope does not differ between districts (interaction $p = 0.35$). The coastal effect in Honolulu is therefore not an elevation effect in disguise.

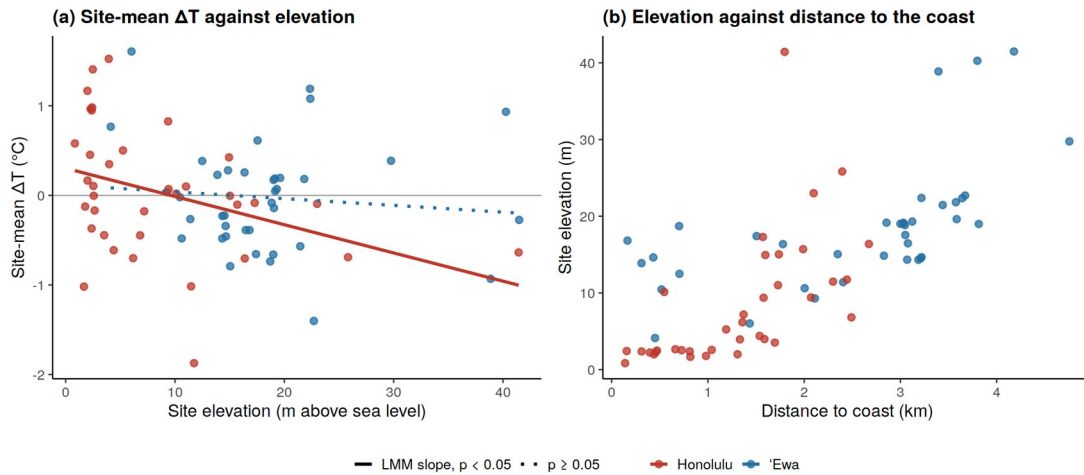


Figure 19. The terrain check. (a) Site-mean ΔT against the bare-earth elevation of each site (USGS 3DEP 1m DEM), by district, with the slope of the elevation-only mixed model (solid where $p < 0.05$, dotted otherwise). (b) Elevation against distance to the coast, by district ($r = 0.61$ in both). This shows that the Honolulu coastal gradient is not an elevation gradient: elevation spans only about 40m and, given distance to the coast, explains nothing (Table 16).

Table 16. Terrain check: the models of Sections 4.5–4.7 refitted with the bare-earth elevation of each site (USGS 3DEP 1m lidar DEM) as an additional standardized term. Coefficients are REML estimates in natural units ($\pm$ SE; Satterthwaite p -values); ΔAIC and the likelihood-ratio p -value refer to adding elevation to the same model without it (ML fits; for the elevation-only rows, to the null model). R^2_m is that of the REML fit, so the best-subset rows read 0.34 and 0.30 where Table 11, which ranks ML fits, gives 0.36 and 0.31. With elevation in the candidate pool (130 models of ≤ 3 terms per study area), the best subsets of Table 11 remain the best; the summed Akaike weight of models containing elevation is 0.11 in Honolulu and 0.02 in 'Ewa, against 0.99 and 0.02 for distance to the coast.

Study area	Model	Coast ($^{\circ}\text{C}$ per km)	Elevation ($^{\circ}\text{C}$ per 10m)	Marginal R^2	ΔAIC (adding elevation)	LRT p (adding elevation)
Honolulu	coast only	-0.57 ± 0.14 ($p < 0.001$)		0.26		
	elevation only		-0.32 ± 0.14 ($p = 0.03$)	0.11	-3.3	0.02
	coast + elevation	-0.55 ± 0.18 ($p = 0.005$)	-0.03 ± 0.15 ($p = 0.83$)	0.25	+1.9	0.82
	best subset (impervious +	-0.62 ± 0.13 (p		0.34		

Study area	Model	Coast (°C per km)	Elevation (°C per 10m)	Marginal R ²	ΔAIC (adding elevation)	LRT p (adding elevation)
'Ewa	coast)	< 0.001)				
	best subset (impervious + coast) + elevation	-0.57 ± 0.17 (p = 0.002)	-0.06 ± 0.14 (p = 0.69)	0.34	+1.8	0.67
	full (eight descriptors)	-0.53 ± 0.15 (p = 0.002)		0.39		
	full (eight descriptors) + elevation	-0.49 ± 0.18 (p = 0.01)	-0.06 ± 0.15 (p = 0.68)	0.39	+1.8	0.63
	coast only	-0.02 ± 0.08 (p = 0.84)		0.00		
	elevation only		-0.08 ± 0.13 (p = 0.56)	0.00	+1.6	0.54
	best subset (impervious + height + aspect)			0.30		
	best subset (impervious + height + aspect) + elevation		-0.07 ± 0.09 (p = 0.44)	0.31	+1.3	0.41
	full (eight descriptors)	-0.05 ± 0.07 (p = 0.52)		0.33		
	full (eight descriptors) + elevation	-0.01 ± 0.08 (p = 0.87)	-0.09 ± 0.11 (p = 0.39)	0.33	+1.0	0.32
Pooled (with study area)	elevation only		-0.20 ± 0.09 (p = 0.03)	0.04	-2.8	0.03
	parsimonious (study area + impervious + height)			0.15		
	parsimonious (study area + impervious + height) + elevation		-0.18 ± 0.08 (p = 0.04)	0.18	-2.7	0.03
	full (study area + eight descriptors)	-0.18 ± 0.07 (p = 0.02)		0.25		
	full (study area + eight descriptors) + elevation	-0.14 ± 0.09 (p = 0.12)	-0.08 ± 0.10 (p = 0.43)	0.26	+1.3	0.39

Sensor positions inside building footprints. The five sites whose recorded positions fall inside a building footprint (Section 3.6) have sky view factors below 0.1. Recomputed from a point 1m outside the nearest footprint edge, their sky view factors are 0.52–0.65. With these values the sky view factor remains non-significant in the pooled and district full models ($p \geq 0.37$), the other coefficients change by 0.01°C or less in their natural units, and the best subsets are unchanged. On its own, the recomputed sky view factor has a larger warming effect in 'Ewa ($+0.37 \pm 0.14^\circ\text{C}$ per 0.1 decrease, $p = 0.01$, against $+0.14 \pm 0.06^\circ\text{C}$ with the original values) and still none in Honolulu ($p = 0.34$).

Buildings without a recorded height. 23% of the buildings on O‘ahu have no height in the FEMA/ORNL inventory and were given a height of 5m (Section 3.6). Such buildings make up 11% of the building area within 100m of the ‘Ewa sensors and 2% in Honolulu. Giving them 3m instead leaves the best subsets unchanged and the ‘Ewa height effect at $+0.14^{\circ}\text{C}$ per meter ($p < 0.001$, against $+0.13^{\circ}\text{C}$ with 5m). Refitting the ‘Ewa models without the six sites where such buildings make up more than half of the building area gives the same result ($+0.14^{\circ}\text{C}$ per meter, with the same best subset of impervious fraction, height and aspect ratio).

Predictive skill. Leave-one-site-out cross-validation (Table 17; Figure 20) measures how well the models predict a site they have not seen. The pooled full model predicts site means poorly ($r = 0.39$, $\text{RMSE} = 0.63^{\circ}\text{C}$, cross-validated $R^2 = 0.09$), because its many terms overfit the pooled data, and a model that knows only the district is no better than the mean. The district models do better, and the parsimonious ones do best: impervious fraction plus coastal distance predicts held-out Honolulu sites with $r = 0.58$ and $\text{RMSE} = 0.59^{\circ}\text{C}$ (cross-validated $R^2 = 0.33$), and impervious fraction, height and aspect ratio predict held-out ‘Ewa sites with $r = 0.71$ and $\text{RMSE} = 0.44^{\circ}\text{C}$ (cross-validated $R^2 = 0.49$). Within-district models with two or three terms therefore transfer to new sites in the same district about as well as their in-sample fit suggests, whereas the pooled model does not.

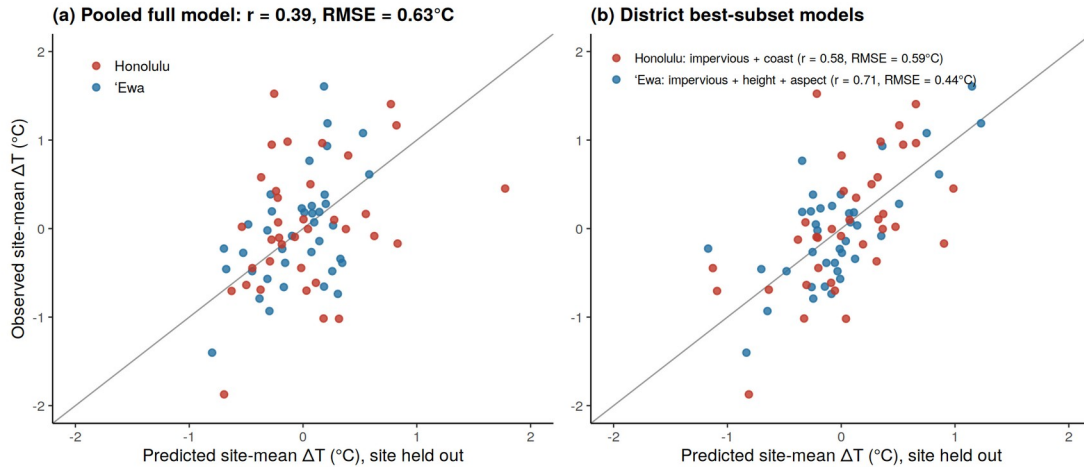


Figure 20. Leave-one-site-out cross-validation: observed against predicted site-mean ΔT for (a) the pooled full model and (b) the district best-subset models, each site predicted by a model fitted without it. This measures how well the models predict sites they have not seen.

Table 17. Leave-one-site-out cross-validation of site-mean ΔT (each site predicted from the fixed effects of a model fitted without it). R^2_{cv} is $1 - SSE/SST$ over the held-out sites (negative values mean worse than the mean).

Model	RMSE (°C)	MAE (°C)	r	R^2_{cv}
Pooled: study area + 8 predictors	0.63	0.49	0.39	0.09
Pooled: study area + impervious + height	0.63	0.49	0.36	0.11
Pooled: study area only	0.68	0.53	-0.47	-0.05
Honolulu: 8 predictors	0.64	0.53	0.54	0.22
Honolulu: impervious + coast distance	0.59	0.47	0.58	0.33
'Ewa: 8 predictors	0.47	0.39	0.65	0.41
'Ewa: impervious + height + aspect ratio	0.44	0.36	0.71	0.49

4.9 Unexplained site effects

After district and the eight descriptors, the site random intercepts of the full pooled model, the persistent, repeatable part of each site's departure that the descriptors do not explain, range from -0.86 to $+1.47^\circ\text{C}$ with a standard deviation of 0.45°C (0.53°C among Honolulu sites, 0.38°C among 'Ewa sites; Figure 21). Their conditional 95% intervals are about $\pm 0.4^\circ\text{C}$ wide, so individual values should be read as indications rather than measurements, but the pattern is informative. The two most negative intercepts belong to HNL01 and HNL04 at the mouth of Mānoa Valley (-0.86 and -0.84°C), sites that are cooler than even their coastal distance and low

cover would predict; they are followed by HNL38 in Kaka‘ako (-0.81°C), which its 95% impervious cover over-predicts, and by Hnl15a and HNL14 (-0.78 and -0.76°C). The most positive are HNL08 ($+1.47^{\circ}\text{C}$), a sparsely built, well-treed site north of Kapi‘olani Park that is the warmest site in Honolulu and among the more variable from night to night (SD 1.00°C), and two sites on the shoreline of Ala Moana (HNL26, ID06; $+0.84$ and $+0.73^{\circ}\text{C}$) that are warmer than their open, grassy surroundings would predict. A generator next to the HNL08 station was likely the source of its extra warmth (Section 5.4). Refitting the models without this site changes none of the conclusions. The Honolulu best-subset model is still impervious fraction and distance to the coast (-0.62°C per km), and the share of the Honolulu variance explained by the descriptors rises (marginal R^2 0.45 instead of 0.36 for the best subset, and 0.49 instead of 0.39 for the full model). In the pooled full model the impervious, height and coastal effects change by less than 0.01°C , and the tree-canopy effect becomes slightly larger and statistically significant ($-0.14 \pm 0.06^{\circ}\text{C}$ per 10 percentage points, $p = 0.02$). Leaving the logger out of the Honolulu network median moves the median by 0.03°C or less on any night. In ‘Ewa the intercepts are smaller and show no clear geography, the largest being the paved Campbell Industrial Park site (EWA26, $+0.83^{\circ}\text{C}$), which is warmer even than its 99% impervious cover predicts, and EWA32 in Kapolei ($+0.67^{\circ}\text{C}$).

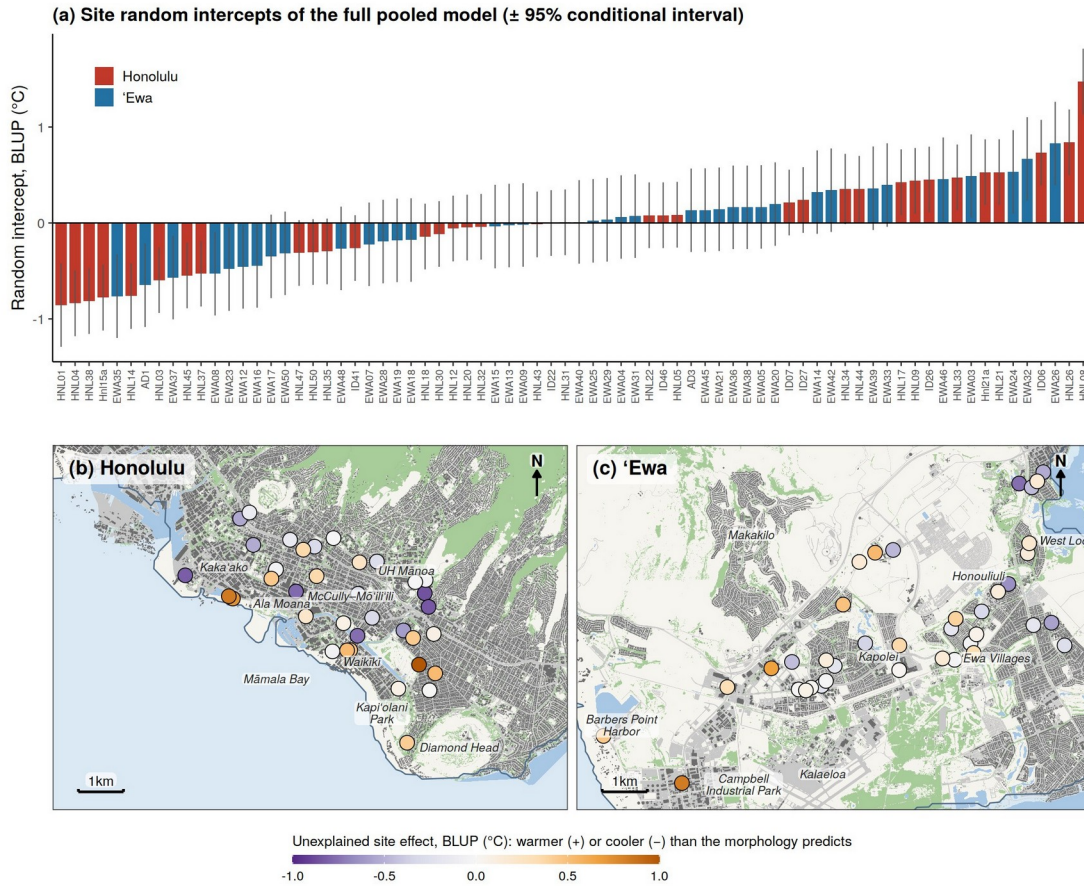


Figure 21. The unexplained site effects: (a) random intercepts (BLUPs) of the full pooled model with 95% conditional intervals, sorted, and (b, c) the same values mapped. Positive values are sites warmer than their descriptors predict. This shows how much persistent variation the descriptors leave unexplained and where (Research Question 4).

4.10 Absolute comparison of the two districts on shared nights

On the two calm, clear nights that both networks share, the absolute night-mean temperatures can be compared directly (Table 18; Figure 22). On 1 December 2024 the Honolulu network median was 21.7 $^{\circ}$ C and the 'Ewa median 20.1 $^{\circ}$ C, a difference of 1.6 $^{\circ}$ C; on 21 December the medians were 22.0 and 20.9 $^{\circ}$ C, a difference of 1.1 $^{\circ}$ C. On both nights the Honolulu median exceeded the 'Ewa median at every half-hour from 18:00 to 06:00, and the 10th–90th percentile band of Honolulu sites lay higher than that of 'Ewa (both edges of the band) at 22 of the 24 half-hours on

1 December and at all 24 on 21 December. The within-network spread of the two districts was similar (standard deviation of site means 0.6–0.8°C in Honolulu and 0.9°C in ‘Ewa). Two nights are too few to establish the typical, long-term difference between the districts, but on these two nights the Honolulu network was warmer than the ‘Ewa network by 1–1.6°C. This is consistent with the expected effect of Honolulu’s denser development, its position on a narrower coastal plain and, possibly, the greater exposure of the ‘Ewa plain to drainage and trade-wind flow.

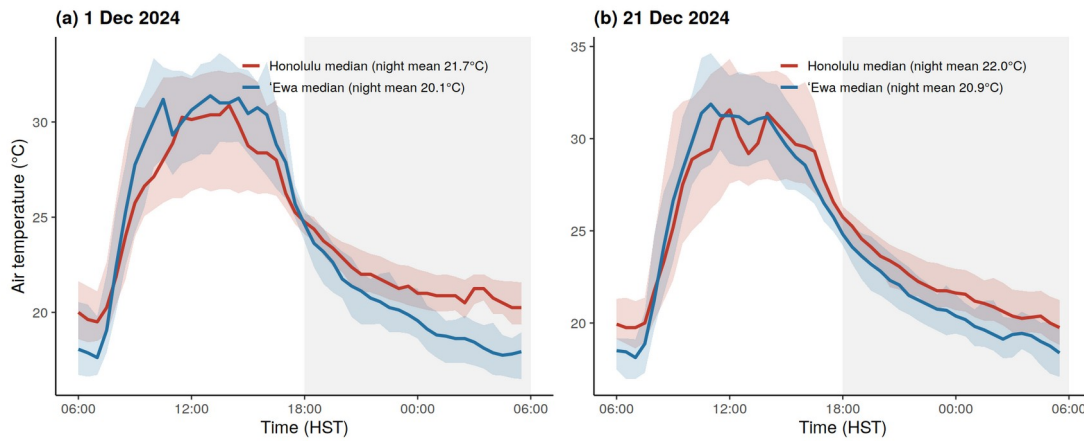


Figure 22. Absolute comparison of the two networks on the shared calm, clear nights of (a) 1 December and (b) 21 December 2024: network median air temperature (lines) and the 10th–90th percentile range of sites (bands) through the 24-hour cycle. This compares the districts on a common absolute scale, which the within-district reference removes.

Table 18. Absolute night-mean (18:00–06:00 HST) air temperatures of the two networks on the two calm, clear nights they share.

Night	Study area	Sensors	Network median (°C)	10th–90th percentile of sites (°C)	Coolest – warmest site (°C)	Wind (km h ⁻¹) / cloud (%)
1 Dec 2024	Honolulu	35	21.7	21.0 – 22.7	20.5 – 23.1	1.7 / 5
1 Dec 2024	‘Ewa	38	20.1	18.9 – 21.1	18.5 – 22.2	1.9 / 16
21 Dec 2024	Honolulu	36	22.0	21.2 – 23.2	19.9 – 23.5	3.4 / 5
21 Dec 2024	‘Ewa	38	20.9	19.8 – 21.9	19.1 – 23.2	2.7 / 10

5 Discussion

5.1 Answers to the research questions

The first question asked how large and how persistent the within-district variation of nighttime air temperature is on calm, clear nights. It is large: site means span 3.4°C in the Honolulu core and 3.0°C on the 'Ewa plain, and individual sensor-nights range over 6°C (a single night spans up to 5°C). On the calmest nights it is also highly persistent. The Honolulu pattern repeats from night to night with correlations of 0.90–0.98, and 81% of the variance of ΔT in Honolulu is between sites rather than between nights. These contrasts are of the same order as the whole urban–rural heat island of many tropical coastal cities ([Anjos & Lopes, 2017](#); [Murphy et al., 2011](#); [Roth, 2007](#)), and they occur within a few kilometers inside one district.

The second question asked which properties of form and setting explain the variation and at what scale. A 100m source area was the best or an equivalent choice in every comparison, in agreement with the block scale identified in Madison ([Ziter et al., 2019](#)), the 180m upwind source area in San Juan ([Murphy et al., 2011](#)) and the 60m buffers of the Denver network ([Ibsen et al., 2022](#)). Impervious fraction was the one descriptor that warmed both districts, by 0.10 – 0.18°C per 10 percentage points depending on the model, and it is the property most directly under the control of land-use decisions. Building height warmed the 'Ewa plain by about 0.13°C per meter; distance to the coast cooled the Honolulu core by about 0.5 – 0.6°C per kilometer inland; tree canopy cooled only weakly and marginally; and sky view factor, water and canyon aspect ratio added little once the other descriptors were known.

The third question asked whether the drivers differ between the two districts, and they do (joint interaction test $p < 0.001$); a model fitted to one district predicts the other poorly. The fourth asked how much persistent between-site variation remains unexplained. The answer is about 0.45°C (one standard deviation of the site intercepts), concentrated in Honolulu and organized in space: cool at the valley mouths, warm at the Ala Moana shoreline and at one anomalous site near Kapi‘olani Park.

5.2 The drivers in the context of the literature

Impervious surface behaved as the literature predicts. Its effect was positive in every model, in both districts and under every night-selection rule, and its size, of the order of $0.1\text{--}0.2^{\circ}\text{C}$ per 10 percentage points, or $1\text{--}2^{\circ}\text{C}$ across the full 0–100% range, is of the same order as the 0.5°C contrast between green and impervious-dominated sites in Baltimore ([Scott, Zaitchik, et al., 2017](#)); in Denver, impervious and building cover warmed the air by day and irrigated turf cooled it by night ([Ibsen et al., 2022](#)). The physical reason was set out in the Introduction: impervious materials store heat by day and release it by night, and suppress the evaporative cooling that pervious ground provides. The effect held in both the compact high-rise core and the low-rise suburban plain, and under every weather filter tested, so it is the result most likely to transfer.

Building height warmed the ‘Ewa plain strongly but had no conditional effect in the Honolulu core, and the interaction was significant. Two explanations fit the data, and they are not mutually exclusive. One is statistical. In Honolulu, height is correlated with aspect ratio ($r = 0.68$) and the tallest sites are also the most coastal, so its conditional effect is shared with those terms; on its own, height was marginally warming in Honolulu ($p = 0.10$). The other is physical. On the ‘Ewa plain, height varies from open ground to two-story housing and 10–13m

commercial and industrial buildings, and each added meter adds thermal mass and shelter to an otherwise exposed, well-ventilated site. In the Honolulu core, the sensors stood at 1.5m in streets whose buildings range up to 20m in mean height and much more individually; at that height the air of a high-rise street is not necessarily warmer for being surrounded by taller walls, because tall buildings also shade the street by day, drive stronger canyon ventilation, and in Waikīkī stand directly on the shore. The mixed evidence in the literature, in which height cools daytime surface temperature through shading in several Chinese cities ([Huang & Wang, 2019](#); [B. Yuan et al., 2024](#); [Zhu et al., 2023](#)) but three-dimensional form adds to nighttime air-temperature warmth in residential neighborhoods ([Tian et al., 2019](#)), while in Hong Kong’s high-rise estates it was reduced sky view rather than height that warmed the night air ([Giridharan et al., 2007](#)), reflects the same competition, and the Honolulu result sits within it. The compact high-rise sites were warm (the Waikīkī and Ala Moana sites were among the warmest in the district), but once impervious cover and position were in the model their warmth was captured by those terms and not by height.

Tree canopy cooled only weakly and marginally at night, by about 0.13°C per 10 percentage points in the pooled full model (0.14°C, and statistically significant, when the anomalous site HNL08 is left out; Section 4.9) and 0.12 in the ‘Ewa full model. This is what the humidity dependence reported by Ibsen et al. ([2021](#)) and the limited nocturnal effect reported by Ziter et al. ([2019](#)) lead one to expect in a maritime tropical city. With high humidity damping the radiative cooling of all surfaces, the nighttime advantage of a vegetated surface over a paved one is small. Trees did cool the night air in this record, but only slightly, and the size of the effect could not be estimated precisely. Their daytime cooling, which is large and well established, was

not measured here because the shields do not permit reliable daytime readings, and the tree fraction within 100m of a street-level sensor is a coarse description of the shade a person experiences. How much tree planting can lower nighttime air temperature on O‘ahu is therefore a question for further research, and a summer record is the obvious next test. For now, the case for planting rests mainly on daytime shade and on the impervious cover that trees replace.

Sky view factor and canyon aspect ratio contributed little. The building-only sky view factor at 1.5m was above 0.9 at most sites in both districts (the five sites with values below 0.1 are an artifact of positions that fall inside building footprints, and correcting them does not change this result; Section 4.8), because randomized siting placed most sensors on posts along streets rather than in the deep canyons where the classic relationship ([Oke, 1981](#)) is strongest; its variance was small and its single-descriptor effect was null in Honolulu and modest in ‘Ewa. The aspect ratio, which combines height and spacing, carried a positive marginal effect in Honolulu and a negative conditional effect given height in ‘Ewa; the latter means that among ‘Ewa sites with the same building height, those with more closely spaced buildings were slightly cooler, plausibly through mutual shading by day. The modest role of enclosure is consistent with mid-latitude field studies: sky view factor explained about half of the areal-mean nocturnal variation in Szeged ([Unger, 2004](#)) and was fairly strongly related to air temperature in Göteborg, where thermal admittance also mattered ([Svensson, 2004](#)), while a two-canyon study in Adelaide attributed nocturnal differences mainly to thermal mass ([Erell & Williamson, 2007](#)); and it suggests that canyon geometry will not be a strong lever in a city whose residential areas are mostly low-rise and open.

Water had no effect. Only a handful of Honolulu sites had appreciable water within 100m (the Ala Wai Canal, the Ala Moana lagoon and the Kewalo basin), and their departures were in different directions.

Distance to the coast is the finding that most distinguishes this island setting. In the Honolulu core, nighttime air was warmer by about half a degree for every kilometer closer to the shore, the coast was the single strongest descriptor, and it remained so in every model; on the ‘Ewa plain it had no effect. The sign is the opposite of the daytime expectation from the sea-breeze literature, in which the shore is the cool end of the gradient ([Emmanuel & Johansson, 2006](#); [Y. Zhou et al., 2021](#)), and consistent with the behavior of coastal water at night in winter. On the study nights the water at the Honolulu tide station stayed between 25.7 and 26.5°C ([NOAA Center for Operational Oceanographic Products and Services, n.d.](#)), 3–6°C warmer than the night air over the land, so marine air reaching the shoreline sites warms them rather than cooling them, whether through residual onshore flow, turbulent mixing at the coast or long-wave radiation from the nearby water surface. The role of distance from the Tagus estuary in Lisbon’s nocturnal heat island ([Alcoforado & Andrade, 2006](#)) and the seasonally reversing effect of water bodies on surface temperature on Xiamen Island ([Zhu et al., 2023](#)) are analogous. At the inland end of the Honolulu network the gradient is reinforced by topography: the coolest sites, with the most negative unexplained intercepts, lie at the mouth of Mānoa Valley, where cold air drains from the Ko‘olau slopes on calm nights. The terrain check (Section 4.8) shows that this is not simply an effect of height above sea level: the Honolulu sites span only about 40m of elevation, elevation on its own explains less than half of what coastal distance explains, and once coastal distance is in the model elevation adds nothing while the coast coefficient is unchanged.

Distance to the coast in Honolulu may still be partly a proxy for distance from the valley mouths, because drainage flow depends on a site's position relative to the valley axes and on the slopes upwind of it, not on its own elevation, and those two effects cannot be separated on a plain only 2–3km wide without local wind observations. Land use at the valley mouth may also play a part. The two coolest sites lie near Ka Papa Lo'i o Kānewai, the taro terraces (lo'i kalo) of the university's Hawai'i inuiākea School of Hawaiian Knowledge at the foot of the slopes of Wa'ahila, watered by an 'auwai (irrigation channel) from Mānoa Stream ([Hawai'i inuiākea School of Hawaiian Knowledge, n.d.](#)). Whether the lo'i add to the cooling, through evaporation from their flooded surface or by keeping an open, unbuilt path for cold air along the stream, was not tested here; standing water can also release stored heat at night, so even the sign of any effect is uncertain. A few loggers placed around the lo'i would answer the question directly. On the 'Ewa plain, which is about 5km wide, whose network does not reach the mountain foot, and which is exposed to the same flow from end to end, neither mechanism operates, and the coast term vanishes. For several 'Ewa sites, moreover, the nearest shoreline is the sheltered West Loch of Pearl Harbor rather than the open coast. The 'Ewa result is useful too: it shows that coastal proximity does not transfer as a single variable, because the physical process behind it differs from place to place.

5.3 Transferability between districts

The interaction test, the model comparison and the cross-validation agree. The relationship between urban form and nighttime heat is not the same in two districts 25km apart on the same coast. Each district is well described by a model of two or three terms, and each such model predicts unseen sites in its own district with a correlation of 0.6–0.7, whereas the pooled model

with all descriptors predicts unseen sites poorly ($r = 0.39$). Two things follow. For O‘ahu, it means that heat-adaptation mapping should be done at the scale of the district, using the drivers that operate there (impervious cover and position relative to the shore and the valleys in the core; impervious cover and building mass on the plain) instead of one island-wide surface. For the wider literature, it adds an island-scale example to the growing evidence that the effect of urban form on temperature depends on the background climate and setting ([Manoli et al., 2019](#); [Zhao et al., 2014](#); [Zhu et al., 2023](#)). The descriptors are useful in both settings, but a coefficient estimated in one kind of urban development and climate should not be applied to another.

5.4 What the unexplained component may be

Even with the descriptors in the model, a persistent between-site component remains that they do not capture: the site intercepts of the pooled model have a standard deviation of 0.45°C , and the between-site standard deviations of the Honolulu and ‘Ewa full models are 0.55 and 0.22°C . Four contributions are likely. The first is real microclimate that the descriptors do not describe: nocturnal drainage flows from the valleys, the marine influence at the shoreline, the sheltering or exposure of a particular street to the prevailing flow, and local heat sources such as air-conditioning exhausts, traffic and cooking. The valley-mouth and shoreline intercepts are of this kind and are spatially coherent, which suggests that a terrain variable would explain part of them; site elevation alone does not (Section 4.8), so the candidate is distance to the valley axis or a modeled drainage-flow index rather than height above sea level. The second is the coarseness of the descriptors themselves: a 100m circle of land cover says nothing about what is immediately next to the sensor, such as the pavement beneath it or a building wall a few meters away that gives off heat at night, and the building heights come from a national inventory (the

FEMA/ORNL USA Structures data set) rather than from surveys of each building. About a quarter of the O‘ahu buildings in that inventory have no recorded height and were given 5m, although neither a lower value nor leaving out the sites where such buildings dominate changes the height effect (Section 4.8). The third is measurement: the DS1921H specification allows each logger to read up to 1°C too high or too low, and because the loggers were not compared side by side before deployment, such offsets cannot be told apart from real site effects; the standard deviation of the intercepts (0.45°C) is within the range that logger offsets alone could produce, so part of the unexplained component may not be climate at all. The fourth is the one anomalous site, HNL08, which is the warmest in Honolulu, variable from night to night, and unexplained by its descriptors. A generator located next to the station was likely the source of its extra warmth, which would make it a local heat source rather than an effect of urban form. The site was kept in the main analysis because it belongs to the randomized network and no rule for excluding sites had been set in advance; leaving it out strengthens the Honolulu models without changing any conclusion (Section 4.9). The between-night stability of the pattern in Honolulu argues that most of the unexplained component is real and persistent, whatever its cause; a side-by-side comparison of the loggers and a terrain descriptor are the two additions that would resolve most of it (Section 5.8).

The ‘Ewa network adds a different lesson. Its pattern on 25 November 2024, the windiest of its four nights, bore no relation to its pattern on the three calmer nights, and its night-to-night residual standard deviation was twice that of Honolulu. The ‘Ewa plain is open and exposed, so on a night with even light regional wind the spatial pattern is governed by ventilation and fetch rather than by surface properties; only on the calmest nights does the surface-controlled pattern

emerge, and when it does it is strongly related to impervious cover and building mass (marginal R^2 of 0.49 on the three calm nights). For an exposed, low-rise landscape the calm, clear criterion may need to be stricter than for a sheltered urban core, and four nights is a thin basis for the plain. A longer record would help.

5.5 Methodological lessons

Several lessons of this study concern method rather than climate, and are recorded here because low-cost networks of this kind are proliferating ([Muller et al., 2013](#)).

Clock alignment before referencing. Loggers started by hand do not share a clock. If a network median is computed at each exact time stamp, loggers whose stamps differ by a minute or two are each compared with only part of the network rather than with the whole of it. Readings should therefore be binned to a common time step before any reference is computed, as was done here (Section 3.3).

An internal reference has consequences. Using the district median as the reference removes the night and the district from every value. That makes ΔT a clean within-district quantity and avoids the arbitrariness of a rural station, but it also means that a district term in a pooled model measures how the networks are centered, not which district is warmer. Here the district term of the pooled model pointed toward a cooler Honolulu, whereas the direct comparison on the two shared nights shows the opposite (Section 4.10). Studies that pool several networks centered on their own medians should not interpret network terms as temperature differences.

Night selection strengthens a signal that is already present. The calm, clear criterion adopted from the literature is justified in this record by the collapse of the between-site spread with wind, and the sensitivity analysis shows that the conclusions survive any reasonable choice of threshold and even the absence of a filter. The numbers, however, do depend on the thresholds (the impervious effect is a third smaller on all nights than on calm, clear ones), so a heat-island coefficient should always be reported together with the weather conditions under which it was estimated ([Stewart, 2011](#)).

Reanalysis winds classify nights, not streets. ERA5 winds served well to identify synoptically calm nights, but they say nothing about ventilation in a given street, and the ‘Ewa experience suggests that a district’s own exposure modulates how calm a night must be for the surface-controlled pattern to appear. A few anemometers installed within each network would allow the criterion to be set on local rather than regional wind.

Air temperature, not surface temperature. None of this analysis could have been done with satellite land-surface temperature. The quantity of interest is air temperature at 1.5m at night, whose relationships to form differ from those of the surface ([Venter et al., 2021](#); [Zhang et al., 2022](#)) and which is the quantity closest to what residents experience ([Clement et al., 2023](#)).

5.6 Limitations

The study covers one season, November to January, the coolest and among the wettest months on the leeward coast, so the magnitudes reported are those of winter nights with air temperatures of about 20–23°C, and the summer heat island, which matters most for exposure, is likely to be larger and may have different drivers. The ‘Ewa record contributes only four nights, one of them atypical. Seventy-four sites over two districts are enough to estimate two or three effects per

district but not eight; the full models are over-parameterized and their conditional coefficients for collinear descriptors (building footprint, aspect ratio) should not be interpreted individually. The loggers were not compared side by side before deployment, and because each can read up to 1°C too high or too low, the value for any single site should be interpreted with caution. The site descriptors carry their own errors: building heights are the values recorded in the FEMA/ORNL building inventory, averaged over each circle, rather than heights surveyed for this study, and the inventory predates the measurements (its heights were measured by lidar in 2010–2011, its newest footprints date from 2016, and buildings completed since then, most of them on the ‘Ewa plain, are absent); the sky view factor omits trees, the descriptors are circular averages that ignore wind direction, and terrain enters only through the elevation check of Section 4.8; slope, aspect and position relative to the valley axes, which govern drainage flow, were not examined. The night-selection thresholds are study-specific choices; their effect on the results is reported in Section 4.8. The two networks’ references are their own medians, so no absolute comparison between districts is possible beyond the two shared nights. Finally, each network median is a mixture of built and unbuilt sites, so “ $\Delta T = 0$ ” means “as warm as the district as a whole” and not “no heat island”.

5.7 Implications for heat adaptation on O‘ahu

With those limitations in view, four implications for the *Climate Ready O‘ahu* strategy ([City and County of Honolulu, 2024](#)) and for the allocation of tree-canopy and cooling investments follow from the results. They are offered as evidence for the strategy to draw on, not as recommendations. First, impervious cover is the one driver of nighttime heat that operated in both districts and under all conditions; where the strategy’s actions include permeable surfaces

and the removal of unnecessary pavement, this study supplies a local, measured basis for expecting a nighttime benefit of the order of 0.1–0.2°C per 10 percentage points of cover removed within a block. Second, the nighttime heat of the low-rise ‘Ewa plain is not small just because the district is suburban: its paved commercial and industrial sites were as far above their district median as Waikīkī was above Honolulu’s, and building mass on the plain adds to it; heat planning for the growing districts should not assume that low-rise development is thermally benign. Third, tree canopy delivered only modest nighttime cooling at 1.5m in this humid winter record. More research is needed, particularly in summer, when the effect of trees on nighttime heat may differ; until then, the case for canopy expansion rests mainly on daytime shade, on the impervious surface that trees replace, and on benefits beyond temperature, and heat-mapping tools that prioritize canopy investment by nighttime or all-hours heat should be validated against air-temperature measurements before they are used to rank neighborhoods. Fourth, in the Honolulu core the shoreline and the valley mouths differ by more than a degree at night independently of cover, so vulnerability mapping should represent the coast and the valleys as well as the buildings, and should be done district by district. All four would be strengthened by a summer deployment.

5.8 Future work

The most valuable next steps are a side-by-side test in which all loggers record together in one place for two days, so that each logger’s constant offset from the others can be measured and removed; a few loggers around the lo‘i at Kānewai, to test whether the taro terraces cool the night air at the mouth of Mānoa Valley; a summer deployment of the same networks with the same processing, to obtain the season that matters most for exposure; a terrain descriptor beyond

elevation (distance to the valley axes or a modeled drainage-flow index) and a wind-direction-aware source area in the models; a few anemometers and humidity sensors in each district so that night selection and interpretation can use local rather than regional weather; and extension of the network to the windward side and to the central plain, where the trade-wind exposure differs again.

6 Conclusions

This thesis set out to measure the nocturnal canopy-layer heat island of leeward O‘ahu for the first time with a fixed sensor network, to identify the properties of urban form and setting that explain its within-district pattern, and to test whether those relationships transfer between two districts of the same island and, by implication, from the continental literature to a tropical maritime city. Six conclusions follow.

1. On calm, clear winter nights, air temperature at 1.5m differs by about 3°C between the warmest and coolest sites within each district, as much as the whole heat island of many tropical coastal cities, and in the Honolulu core the pattern is highly repeatable from night to night.
2. Urban form explains a substantial but partial share of that pattern: 25% of the variance pooled, 39% in Honolulu and 33% in ‘Ewa with all descriptors, and 31–36% with two or three. A 100m source area is the appropriate scale.
3. Impervious surface fraction is the one driver common to both districts, warming the night air by about 0.1–0.2°C per 10 percentage points; building height warms the low-rise ‘Ewa plain by about 0.13°C per meter; tree canopy cooled the night air only slightly in this humid winter record, and its effect in summer remains to be measured; enclosure and water contribute little.
4. Position within the Honolulu core has an effect that the continental literature would not predict. Night air is warmer by about half a degree per kilometer toward the shore, an effect that site elevation does not account for, with the coolest sites at the mouths of the Ko‘olau valleys, whereas on the ‘Ewa plain distance to the coast has no effect. The

drivers differ significantly between the two districts, and a model fitted to one predicts the other poorly. Heat mapping on O‘ahu should therefore be done district by district with locally estimated relationships.

5. The results hold under every alternative night-selection threshold tested, and they are present, although weaker, on all nights, so they are not created by the choice of calm, clear nights. They do depend on careful processing: the loggers’ clocks must be aligned before the network reference is computed, and because each network is centered on its own median, differences between the districts must be compared on absolute temperatures rather than read from model terms.
6. Canopy-layer air temperature can be measured across a district with low-cost loggers and a few weeks of volunteer effort, and it yields air temperature at 1.5m, which heat-adaptation planning needs and satellite surface temperature cannot provide. A summer deployment, a side-by-side comparison of the loggers and a terrain descriptor are the additions that would most increase the value of the record.

Declaration of the use of artificial intelligence

Large language models were used in preparing this thesis, principally Claude (Anthropic).

ChatGPT (OpenAI), Gemini (Google) and Grok (xAI) were also used occasionally, to explore the literature, to plan the structure of the document, to clarify statistical and thermodynamic concepts and to copy-edit drafts. Under the author's direction, Claude was used to write and check the data-processing and analysis code in R and its independent replication in Python, to generate the figures and tables, to check the bibliography against the Crossref database, and to draft and edit text. The author designed the study, deployed the sensor network and collected the data, defined the research questions, reviewed every analytical choice and every number, and takes full responsibility for the content.

Appendix A. Site table

Table 19. All 74 sites: coordinates (WGS84), number of analyzed nights, site-mean ΔT with its night-to-night standard deviation, and the random intercept (BLUP) of the full pooled model.

Area	Site	Lat (°N)	Lon (°E)	Nights	ΔT (°C)	SD (°C)	BLUP (°C)
'Ewa	AD1	21.3498	-158.0301	4	-0.66	+1.01	-0.65
'Ewa	AD3	21.3348	-158.0692	4	+0.18	+0.33	+0.13
'Ewa	EWA03	21.3459	-158.0655	4	+0.39	+0.25	+0.49
'Ewa	EWA04	21.3398	-158.0370	4	+0.28	+0.57	+0.06
'Ewa	EWA05	21.3704	-158.0238	4	+0.23	+0.69	+0.17
'Ewa	EWA07	21.3296	-158.0702	4	-0.14	+0.46	-0.22
'Ewa	EWA08	21.3722	-158.0225	4	-0.39	+0.72	-0.53
'Ewa	EWA09	21.3306	-158.0692	4	+0.07	+0.61	-0.02
'Ewa	EWA12	21.3692	-158.0250	4	-0.34	+0.64	-0.46
'Ewa	EWA13	21.3347	-158.0417	4	-0.48	+0.98	-0.03
'Ewa	EWA14	21.3295	-158.0904	4	+1.08	+0.68	+0.32
'Ewa	EWA15	21.3380	-158.0383	4	-0.23	+0.34	-0.04
'Ewa	EWA16	21.3345	-158.0765	4	-1.40	+0.45	-0.45
'Ewa	EWA17	21.3381	-158.0608	4	-0.66	+0.97	-0.35
'Ewa	EWA18	21.3416	-158.0248	4	+0.04	+0.75	-0.18
'Ewa	EWA19	21.3410	-158.0424	4	-0.39	+0.94	-0.18
'Ewa	EWA20	21.3580	-158.0256	4	-0.02	+0.56	+0.20
'Ewa	EWA21	21.3560	-158.0259	4	+0.38	+0.65	+0.14
'Ewa	EWA23	21.3568	-158.0547	4	-0.93	+1.15	-0.48
'Ewa	EWA24	21.3563	-158.0586	4	+0.93	+0.29	+0.53
'Ewa	EWA25	21.3327	-158.0536	4	+0.61	+0.47	+0.02
'Ewa	EWA26	21.3103	-158.1002	4	+1.61	+0.77	+0.83
'Ewa	EWA28	21.3336	-158.0674	4	-0.57	+0.39	-0.19
'Ewa	EWA29	21.3293	-158.0724	4	+0.19	+0.60	+0.03
'Ewa	EWA31	21.3287	-158.0735	4	+0.17	+0.44	+0.07
'Ewa	EWA32	21.3332	-158.0809	4	+1.19	+0.75	+0.67
'Ewa	EWA33	21.3429	-158.0414	4	+0.05	+0.61	+0.40
'Ewa	EWA35	21.3699	-158.0277	4	-0.74	+0.56	-0.76
'Ewa	EWA36	21.3544	-158.0619	4	-0.27	+0.63	+0.16
'Ewa	EWA37	21.3420	-158.0209	4	-0.48	+0.75	-0.57
'Ewa	EWA38	21.3350	-158.0443	4	-0.46	+0.90	+0.16
'Ewa	EWA39	21.3377	-158.0535	4	+0.19	+0.59	+0.36
'Ewa	EWA40	21.3290	-158.0751	4	-0.08	+0.34	+0.01
'Ewa	EWA42	21.3361	-158.0376	4	-0.23	+0.53	+0.34
'Ewa	EWA45	21.3483	-158.0323	4	+0.26	+0.69	+0.13
'Ewa	EWA46	21.3199	-158.1169	4	+0.77	+0.84	+0.46
'Ewa	EWA48	21.3375	-158.0182	4	-0.26	+0.64	-0.27
'Ewa	EWA50	21.3444	-158.0358	4	-0.79	+0.99	-0.32
Honolulu	HNL01	21.2929	-157.8131	4	-1.87	+0.23	-0.86
Honolulu	HNL03	21.2855	-157.8178	7	-0.61	+0.21	-0.60
Honolulu	HNL04	21.2902	-157.8123	7	-1.02	+0.33	-0.84
Honolulu	HNL05	21.2847	-157.8113	7	-0.10	+0.19	+0.09
Honolulu	HNL08	21.2786	-157.8145	7	+1.52	+1.00	+1.47

Area	Site	Lat (°N)	Lon (°E)	Nights	ΔT (°C)	SD (°C)	BLUP (°C)
Honolulu	HNL09	21.2840	-157.8157	7	+0.35	+0.15	+0.44
Honolulu	HNL12	21.2814	-157.8330	7	-0.37	+0.15	-0.06
Honolulu	HNL14	21.2845	-157.8276	7	-1.02	+0.14	-0.76
Honolulu	HNL17	21.2630	-157.8171	7	+0.02	+0.10	+0.42
Honolulu	HNL18	21.3038	-157.8418	7	+0.10	+0.24	-0.14
Honolulu	HNL20	21.2956	-157.8129	7	-0.70	+0.27	-0.05
Honolulu	HNL21	21.2816	-157.8298	7	+0.97	+0.19	+0.53
Honolulu	HNL22	21.2738	-157.8189	7	+0.10	+0.18	+0.08
Honolulu	HNL26	21.2926	-157.8550	7	+0.98	+0.29	+0.84
Honolulu	HNL30	21.3094	-157.8505	7	-0.64	+0.20	-0.11
Honolulu	HNL31	21.2735	-157.8123	7	-0.18	+0.18	+0.01
Honolulu	HNL32	21.2979	-157.8449	7	-0.00	+0.10	-0.04
Honolulu	HNL33	21.2816	-157.8292	7	+1.41	+0.17	+0.47
Honolulu	HNL34	21.2965	-157.8362	6	+0.50	+0.18	+0.35
Honolulu	HNL35	21.3024	-157.8366	7	-0.01	+0.21	-0.30
Honolulu	HNL37	21.3029	-157.8497	7	-0.70	+0.24	-0.53
Honolulu	HNL38	21.2969	-157.8643	7	+0.45	+0.17	-0.81
Honolulu	HNL43	21.3041	-157.8326	7	-0.09	+0.18	-0.02
Honolulu	HNL44	21.3018	-157.8390	7	+0.83	+0.18	+0.36
Honolulu	HNL45	21.3082	-157.8525	7	-0.08	+0.18	-0.55
Honolulu	HNL47	21.2881	-157.8244	7	+0.17	+0.07	-0.31
Honolulu	HNL50	21.2929	-157.8274	7	-0.44	+0.08	-0.30
Honolulu	Hnl15a	21.2935	-157.8407	7	-0.17	+0.13	-0.78
Honolulu	Hnl21a	21.2769	-157.8110	7	+0.42	+0.27	+0.53
Honolulu	ID06	21.2921	-157.8542	7	+0.58	+1.46	+0.73
Honolulu	ID07	21.2885	-157.8386	7	+1.17	+0.13	+0.21
Honolulu	ID22	21.2952	-157.8151	7	-0.45	+0.13	-0.00
Honolulu	ID26	21.2961	-157.8459	7	+0.95	+0.13	+0.45
Honolulu	ID27	21.2992	-157.8270	7	+0.07	+0.10	+0.24
Honolulu	ID41	21.2993	-157.8233	7	-0.69	+0.12	-0.26
Honolulu	ID46	21.2870	-157.8306	7	-0.12	+0.13	+0.08

Table 20. All 74 sites: the eight descriptors at the 100m source-area radius (sky view factor and coastal distance are point properties).

Area	Site	Imperv. %	Tree %	Bldg %	Water %	Height m	SVF	H/W	Coast km
'Ewa	AD1	56.2	1.7	17.3	0.0	3.7	1.00	0.21	1.51
'Ewa	AD3	72.3	2.2	18.4	0.0	4.3	0.99	0.25	3.58
'Ewa	EWA03	71.1	3.9	17.5	0.0	4.4	0.94	0.30	4.76
'Ewa	EWA04	63.0	15.2	10.4	0.0	7.3	0.99	0.17	2.83
'Ewa	EWA05	69.1	27.9	26.0	0.0	4.2	0.98	0.36	0.31
'Ewa	EWA07	54.5	0.2	6.7	0.0	3.9	1.00	0.10	3.03
'Ewa	EWA08	68.1	11.5	25.3	0.0	4.1	0.96	0.35	0.16
'Ewa	EWA09	70.2	5.3	15.8	0.0	4.6	0.99	0.27	3.12
'Ewa	EWA12	69.4	33.5	15.0	0.0	4.2	1.00	0.23	0.43
'Ewa	EWA13	23.8	9.8	2.4	0.0	3.2	1.00	0.04	3.19
'Ewa	EWA14	79.8	1.1	9.2	0.0	6.2	0.04	0.07	3.22

Area	Site	Imperv. %	Tree %	Bldg %	Water %	Height m	SVF	H/W	Coast km
'Ewa	EWA15	53.3	18.9	10.1	0.0	4.6	1.00	0.17	3.07
'Ewa	EWA16	18.1	19.1	0.0	0.0	0.0	1.00	0.00	3.67
'Ewa	EWA17	37.8	8.5	12.2	17.1	5.4	0.98	0.23	3.81
'Ewa	EWA18	68.4	12.5	12.7	0.0	5.0	1.00	0.26	2.11
'Ewa	EWA19	62.1	13.3	16.3	0.0	3.8	0.98	0.22	3.08
'Ewa	EWA20	50.7	40.8	19.3	0.0	4.9	0.98	0.27	0.52
'Ewa	EWA21	70.1	6.1	27.5	0.0	4.3	0.98	0.40	0.70
'Ewa	EWA23	31.9	0.1	0.0	0.0	0.0	1.00	0.00	3.40
'Ewa	EWA24	62.3	2.9	1.1	0.0	5.0	1.00	0.03	3.80
'Ewa	EWA25	88.5	3.8	11.8	0.0	5.0	0.99	0.06	3.05
'Ewa	EWA26	99.0	2.9	42.6	0.0	9.9	0.97	0.28	1.43
'Ewa	EWA28	43.0	25.2	3.7	0.0	4.6	1.00	0.08	3.44
'Ewa	EWA29	58.7	0.2	0.0	0.0	0.0	1.00	0.00	3.03
'Ewa	EWA31	41.4	4.7	0.3	0.0	5.0	1.00	0.03	3.00
'Ewa	EWA32	78.3	5.4	26.1	0.0	13.0	0.89	0.34	3.64
'Ewa	EWA33	33.4	31.5	1.9	0.0	4.4	1.00	0.07	2.86
'Ewa	EWA35	33.5	13.2	2.5	0.0	4.8	1.00	0.06	0.70
'Ewa	EWA36	5.7	0.0	2.4	0.0	9.0	1.00	0.09	4.18
'Ewa	EWA37	75.3	0.6	25.7	0.0	5.4	0.93	0.50	2.01
'Ewa	EWA38	12.1	11.7	2.4	0.0	3.4	1.00	0.06	3.22
'Ewa	EWA39	39.1	0.2	8.9	0.0	4.5	1.00	0.16	3.58
'Ewa	EWA40	62.3	10.1	19.7	0.0	5.0	0.99	0.09	3.05
'Ewa	EWA42	12.1	0.9	0.0	0.0	0.0	1.00	0.00	3.21
'Ewa	EWA45	76.3	2.8	29.2	0.0	3.4	0.99	0.32	1.78
'Ewa	EWA46	17.6	0.0	4.9	0.0	7.1	1.00	0.10	0.45
'Ewa	EWA48	51.1	2.8	12.7	0.0	5.8	0.99	0.37	2.41
'Ewa	EWA50	33.5	25.5	8.2	0.0	4.6	1.00	0.13	2.35
Honolulu	HNL01	28.1	22.6	16.7	0.0	8.6	0.99	0.35	2.44
Honolulu	HNL03	61.2	2.4	15.3	0.0	6.3	1.00	0.19	1.54
Honolulu	HNL04	80.8	10.1	24.7	0.0	4.9	0.08	0.35	2.30
Honolulu	HNL05	80.4	0.9	30.8	0.0	4.7	0.94	0.52	1.99
Honolulu	HNL08	44.2	25.0	8.8	3.8	8.3	0.94	0.12	1.33
Honolulu	HNL09	90.6	0.3	40.8	0.0	6.3	0.97	0.74	1.59
Honolulu	HNL12	13.6	11.5	1.5	0.0	5.6	1.00	0.04	0.31
Honolulu	HNL14	16.2	11.4	11.8	40.3	4.1	0.98	0.08	0.82
Honolulu	HNL17	55.3	39.8	17.8	0.0	5.1	0.99	0.22	0.55
Honolulu	HNL18	92.4	8.6	23.6	0.0	7.8	0.94	0.59	1.73
Honolulu	HNL20	37.8	52.3	10.7	0.0	13.8	0.98	0.31	2.67
Honolulu	HNL21	74.1	8.9	26.7	0.0	9.7	0.01	0.44	0.46

Area	Site	Imperv. %	Tree %	Bldg %	Water %	Height m	SVF	H/W	Coast km
u									
Honolulu	HNL22	52.9	5.2	26.8	0.0	11.9	0.85	0.59	0.73
u									
Honolulu	HNL26	28.6	21.8	0.2	20.5	4.0	1.00	0.02	0.16
u									
Honolulu	HNL30	55.4	23.9	21.4	0.0	6.8	0.97	0.42	1.80
u									
Honolulu	HNL31	79.9	1.0	33.4	0.0	4.1	0.98	0.48	1.37
u									
Honolulu	HNL32	76.6	9.2	21.8	0.0	4.9	0.99	0.38	1.04
u									
Honolulu	HNL33	78.9	7.5	20.8	0.0	18.6	0.90	0.91	0.47
u									
Honolulu	HNL34	78.7	15.0	19.4	0.0	5.6	0.99	0.27	1.19
u									
Honolulu	HNL35	76.8	10.7	8.4	0.0	6.5	0.99	0.21	1.74
u									
Honolulu	HNL37	52.6	33.3	22.8	0.0	15.7	0.97	0.34	1.36
u									
Honolulu	HNL38	94.6	3.6	14.1	0.0	16.3	0.98	0.21	0.40
u									
Honolulu	HNL43	86.6	14.3	36.8	0.0	10.2	0.04	0.84	2.10
u									
Honolulu	HNL44	78.7	0.7	18.4	0.0	9.2	0.70	0.42	1.58
u									
Honolulu	HNL45	73.8	2.7	27.4	0.0	19.1	0.92	0.93	1.57
u									
Honolulu	HNL47	94.6	3.5	24.7	0.0	10.2	0.88	0.72	1.31
u									
Honolulu	HNL50	60.4	12.2	19.3	0.0	6.8	0.07	0.39	1.70
u									
Honolulu	Hnl15a	97.7	3.8	40.5	0.0	18.0	0.86	0.97	0.66
u									
Honolulu	Hnl21a	80.1	0.5	32.0	0.0	4.6	0.97	0.54	1.60
u									
Honolulu	ID06	20.0	19.3	0.0	6.2	4.0	1.00	0.02	0.14
u									
Honolulu	ID07	59.8	11.0	20.9	27.9	19.2	0.85	0.87	0.44
u									
Honolulu	ID22	27.6	19.9	8.2	3.1	10.4	0.99	0.20	2.49
u									
Honolulu	ID26	85.4	0.4	41.2	0.0	20.8	0.78	1.97	0.81
u									
Honolulu	ID27	76.2	5.4	26.1	0.0	4.8	0.92	0.41	2.07
u									
Honolulu	ID41	60.3	8.3	19.6	0.0	4.5	0.98	0.27	2.40
u									
Honolulu	ID46	1.3	0.0	0.0	42.0	0.0	0.99	0.00	0.98
u									

Appendix B. Supplementary maps and diagrams

This appendix documents three pieces of the study that are summarized in the text: the land cover of O‘ahu and of the two study domains (Figure B1), the topography of the island and of the Honolulu district (Figure B2), and how the sky view factor was calculated at each sensor (Figure B3).

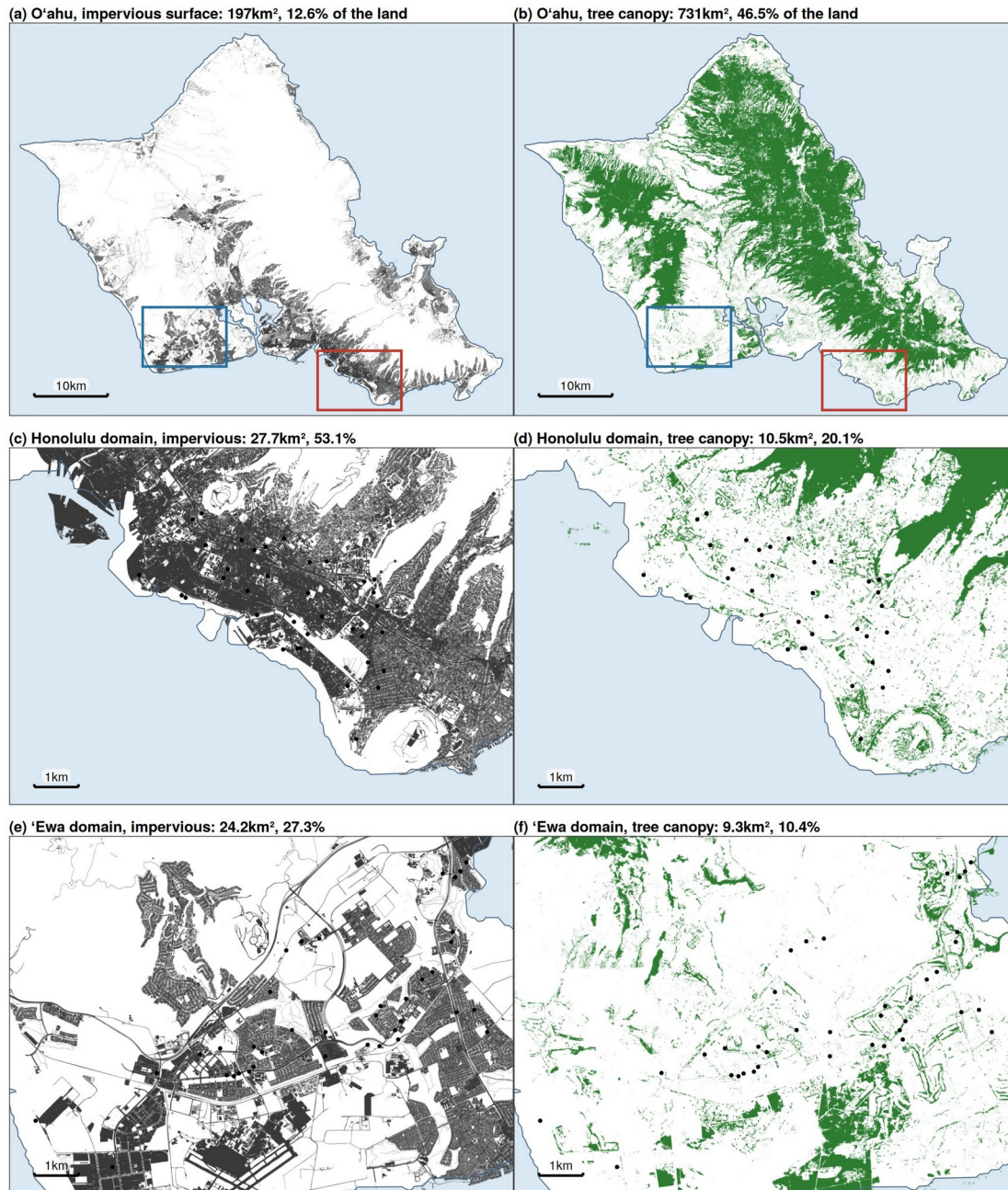
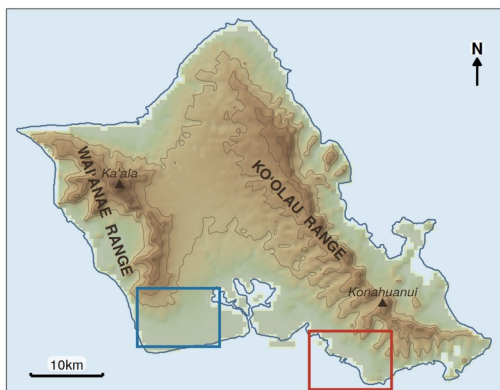


Figure B1. Land cover from the 2021 NOAA C-CAP 1m product: impervious surface and tree canopy on (a, b) O'ahu, with the two study domains outlined, and in (c, d) the Honolulu and (e, f) the Ewa study domains, with the sensor sites. Shading gives the share of each cell (60m on the island, 12m in the domains) covered by the class; the areas and shares in the panel titles are those quoted in Section 1.2. This documents the land-cover figures given in the text.

(a) O'ahu (500m grid; contours every 200m)



(b) Honolulu district (150m grid; contours at 20, 50, 100m, then every 100m)

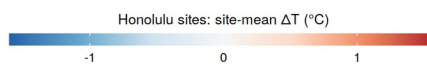
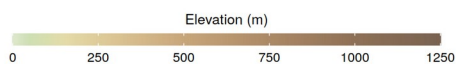
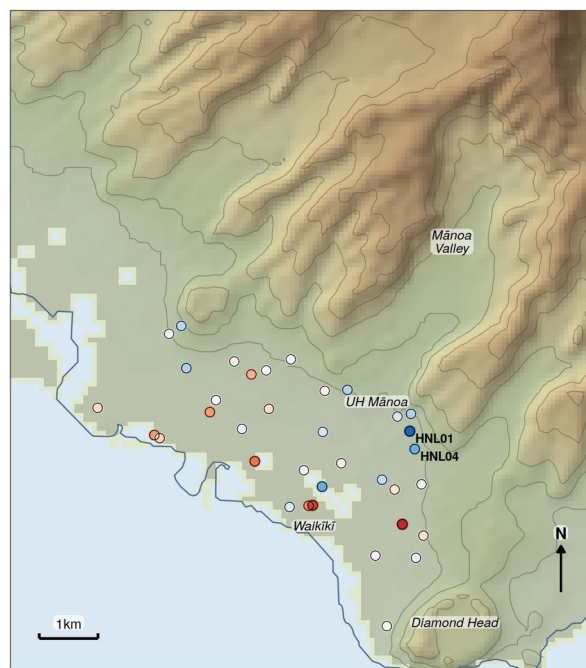
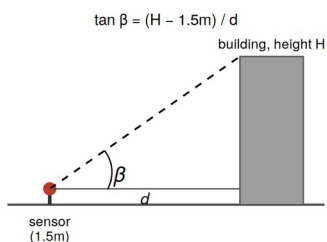
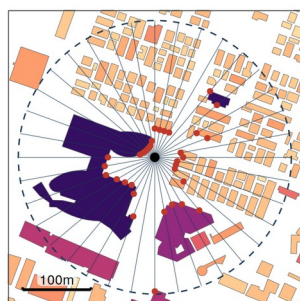


Figure B2. Terrain from the USGS 10m digital elevation model of O'ahu (served by PacIOOS), sampled every 500m for (a) the island, with the study domains outlined, and every 150m for (b) the Honolulu district, with the sensor sites colored by their site-mean ΔT . Because the grid is sampled, the highest summits (Ka'ala, 1,227m) appear slightly lower than their true height. This shows how steeply the Ko'olau and Wai'anae ranges rise behind the leeward coast and where the Honolulu sensors sit relative to the valley mouths.

(a) The obstruction angle β



(b) Site ID26: 36 directions to 200m



(c) β in each direction

SVF = mean of $\cos^2 \beta$ over the 36 directions = 0.78

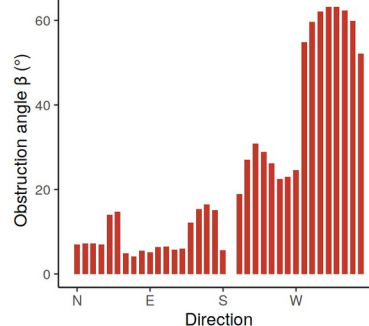


Figure B3. How the sky view factor is calculated, for site ID26 in Honolulu. (a) Along one direction, β is the steepest angle from the sensor (1.5m above the ground) to the top of a building. (b) The 36 directions traced from

the sensor to 200m, with the building footprints colored by height and the building that sets β in each direction marked in red. (c) β in each of the 36 directions; the sky view factor is the mean of $\cos^2\beta$, here 0.78. This shows the calculation behind the sky view factor values used in the models.

Appendix C. Data and code

Every result in this thesis is generated by a scripted pipeline in R, published with the data and results in the project’s repository (<https://github.com/justinkellum-ship-it/JustinKellumThesis>; folder analysis). The scripts, in the order they run, are:

`00a_compile_logger_readings.R` (the loggers’ export tables, one per district as produced by the manufacturer’s application, compiled into one record of readings and one table of logger metadata, with a check that the stored record agrees with the exports reading for reading);
`00_prepare_inputs.R` (time binning, network median, ERA5 join, night inventory, sensor-night table); `01_site_predictors.R` (site descriptors at 50, 100 and 200m from the C-CAP rasters, the FEMA footprints and the GSHHG shoreline, including the sky view factor);
`02_models.R` (all statistical analyses, writing `thesis_results.json` and the CSV tables);
`03_figures.R` (all figures); `04_replication_check.R` (comparison with the Python replication); `05_thesis_tables.R` (the tables in this document); `06_elevation_check.R` (the terrain check of Section 4.8, which adds the USGS 3DEP elevation of each site, `data/descriptors/site_elevation.csv`, to the models); `07_sensitivity_checks.R` (the site checks of Sections 4.8 and 4.9: the models refitted without HNL08, with 3m instead of 5m for the buildings without a recorded height and without the sites dominated by such buildings, and with the sky view factor of the positions inside footprints recomputed); and
`08_appendix_figures.R` (the maps and the diagram of Appendix B, including the terrain from subsamples of the USGS 10m digital elevation model of O‘ahu stored in `data/external`). `run_all.R` runs them in order in about thirty minutes, and `requirements.R` installs the packages. Inputs are the loggers’ export tables

(data/raw/exports, unedited) and the record compiled from them (data/raw/logger_readings.csv: district, logger, time stamp and temperature of every reading), the logger positions (data/raw/sites.csv), the ERA5 hourly wind and cloud series of the two districts (data/external/era5_hourly_series.csv), the hourly water temperature at the NOAA Honolulu tide station on the analyzed nights (data/external/noaa_1612340_water_temperature.csv), the C-CAP 2021 rasters for the study domains, the FEMA USA Structures footprints for O‘ahu and the GSHHG shoreline. The repository’s README gives the data sources with their URLs, and the R Markdown document walkthrough/Analysis_walkthrough.Rmd (rendered as HTML and PDF) walks through the chain step by step, showing the code of every processing rule, model and check together with its output. Table 21 lists the packages.

Table 21. Software used for the analysis, with the role of each package and the functions called. Versions are those of the run that produced this document (results/sessionInfo.txt).

Package	Version	Role	Functions used	Reference
R	4.3.3	the language: arithmetic, median(), cor(), lm(), wilcox.test(), cor.test()		(R Core Team, 2024)
dplyr, tidyr, tibble, purrr, readr	1.1.4 / 1.3.1 / 2.1.5	grouped summaries, joins and reshaping of the record; typed CSV input/output	group_by(), summarize(), inner_join(), pivot_wider(), read_csv()	(Wickham et al., 2019)
lubridate	1.9.3	rounding of the loggers’ time stamps to the nearest 30 minutes	round_date(), ymd_hms()	(Grolemund & Wickham, 2011)
jsonlite	1.8.8	the results file read by the tables and figures	write_json(), fromJSON()	(Ooms, 2014)
sf	1.0.15	vector geometry: projection, buffers, footprint clipping, sight lines for the sky view factor, shoreline distance	st_transform(), st_buffer(), st_intersects(), st_intersection(), st_distance()	(Pebesma, 2018)
terra	1.7.65	the 1m C-CAP masks as virtual mosaics; pixel extraction inside circles; aggregation for base maps	vrt(), extract(), crop(), aggregate()	(Hijmans, 2024)
lme4	1.1.35.1	the mixed-effects models (random intercept per	lmer(), fixef(), ranef(), VarCorr(), logLik(),	(Bates et al., 2015)

Package	Version	Role	Functions used	Reference
		site; REML and ML fits; BLUPs)	AIC(), predict()	
lmerTest	3.1.3	Satterthwaite degrees of freedom and t tests for the coefficients	lmer(), summary()	(Kuznetsova et al., 2017)
performance	0.10.8	check of the marginal and conditional R ² and ICC computed from the variance components	r2_nakagawa(), icc()	(Lüdtke et al., 2021)
car	3.1.2	variance inflation factors	vif()	(Fox & Weisberg, 2019)
ggplot2, patchwork, scales, cowplot	3.4.4 / 1.2.0	all figures; multi-panel layout; color scales	ggplot(), geom_sf(), geom_raster(), coord_sf(), wrap_plots()	(Pedersen, 2024; Wickham, 2016)
rmarkdown, knitr	2.25 / 1.45	the analysis walkthrough document	render()	
Python replication: pandas, NumPy, SciPy	3.0.2 / 2.4.4 / 1.17.1	record processing and statistics in the independent replication		(Harris et al., 2020; McKinney, 2010; Virtanen et al., 2020)
Python replication: statsmodels	0.15.0	mixed models (MixedLM) in the replication		(Seabold & Perktold, 2010)
Python replication: rasterio, Shapely, GeoPandas	1.4.4 / 2.1.2 / 1.1.4	rasters and vector geometry in the replication		(Gillies et al., 2007, 2013)

Independent replication. The same pipeline was implemented a second time in Python 3.11 (pandas 3.0 for the record, rasterio 1.4 and Shapely 2.1 for the descriptors, statsmodels 0.15 for the mixed models) and run on the same inputs; the Python scripts are kept in `replication/python` and their outputs in `replication/results`.

`04_replication_check.R` makes 142 comparisons between the two implementations: 117 paired quantities and 25 maximum-difference checks over the 74 sites (Table 22). The sensor-night departures agree to machine precision; the raster fractions differ by at most 0.2 percentage points at 50m and about 0.03 at 100m because the two raster libraries make different choices about pixels on the edge of a circle; every coefficient and variance component agrees to three decimals and every AIC and R² to two. The few differences larger than 0.01 are either optimizer tolerance (a Δ AIC of 12.34 against 12.32; a joint likelihood-ratio χ^2 of 30.43 against 30.42) or documented implementation choices: statsmodels' Wald z tests give slightly smaller p-values

than lmerTest's Satterthwaite t tests for the same estimates; and one leave-one-site-out fold of the over-parameterized 'Ewa full model converged to a poor optimum in statsmodels (the R values are reported). None of the conclusions depends on which implementation fitted the models.

Table 22. Agreement between the R analysis (lme4 / lmerTest) and the independent Python replication (statsmodels MixedLM) on the same inputs: a selection of the 117 paired quantities in `replication_check.csv` (which also holds 25 maximum-difference checks over the 74 sites). Differences larger than 0.01 are optimizer tolerance (Δ AIC, joint χ^2) or a documented implementation choice (one unstable leave-one-site-out fold of the 'Ewa full model in statsmodels).

Quantity	R	Python	Difference
Scale rule: max Δ AIC at 100m	1.671	1.674	-0.003
Scale rule: max Δ AIC at 200m	10.196	10.206	-0.010
Scale rule: max Δ AIC at 50m	12.344	12.324	+0.020
Null model ICC (pooled)	0.626	0.626	+0.000
Null model ICC (Honolulu)	0.810	0.810	+0.000
Null model ICC (Ewa)	0.365	0.365	+0.000
Full pooled: imperv_z estimate	0.403	0.403	-0.000
Full pooled: tree_z estimate	-0.143	-0.143	+0.000
Full pooled: bldg_z estimate	-0.254	-0.255	+0.000
Full pooled: water_z estimate	0.111	0.111	+0.000
Full pooled: height_z estimate	0.277	0.277	-0.000
Full pooled: svf_point_z estimate	-0.031	-0.031	-0.000
Full pooled: aspect_z estimate	-0.081	-0.081	+0.000
Full pooled: coast_km_z estimate	-0.196	-0.196	+0.000
Full pooled: R2m	0.254	0.254	+0.000
Full pooled: R2c	0.652	0.652	+0.000
Full pooled: LRT vs null, chi2	35.083	35.093	-0.009
Honolulu best subset: R2m (imperv + coast_km)	0.356	0.355	+0.001
Ewa best subset: R2m (imperv + height + aspect)	0.312	0.311	+0.001
Interactions: joint LRT chi2	30.429	30.419	+0.010
BLUP SD	0.453	0.453	+0.000
LOSO full: r	0.391	0.391	-0.000
LOSO parsimonious: r	0.356	0.357	-0.000
LOSO region only: r	-0.467	-0.467	-0.000
LOSO Honolulu full: r	0.538	0.538	+0.000
LOSO Honolulu best subset (imperv + coast_km): r	0.578	0.578	-0.000
LOSO Ewa full: r	0.649	0.615	+0.034
LOSO Ewa best subset (imperv + height + aspect): r	0.705	0.705	-0.000

Glossary

Akaike information criterion (AIC). A measure of model quality that balances fit against the number of parameters; differences of less than 2 between models indicate equivalent support.

Best linear unbiased prediction (BLUP). The estimated random intercept of a site, i.e. its persistent departure from the value predicted by the fixed effects.

Calm, clear night. In this thesis, a night on which ERA5 10m wind was below 10 km h^{-1} and cloud cover below 25% in at least 75% of the 24 half-hourly bins between 18:00 and 06:00 HST.

Canyon aspect ratio (H/W). Mean building height divided by an estimate of mean street width; a measure of enclosure.

Conditional and marginal R^2 . The share of variance explained by fixed and random effects together (conditional) or by fixed effects alone (marginal).

Diurnal date. The label of a 24-hour cycle beginning at 06:00 HST; a “night” is the 18:00–06:00 window that ends the cycle.

ERA5. The fifth-generation global atmospheric reanalysis of the European Centre for Medium-Range Weather Forecasts, used here for hourly wind and cloud cover.

Fixed effect. A model term whose coefficient applies to all sites (e.g. the effect of impervious fraction).

Intraclass correlation coefficient (ICC). The share of the total variance of ΔT that is between sites (persistent) rather than between nights.

Linear mixed-effects model. A regression model containing both fixed effects and random effects, used here to account for repeated measurements at the same sites.

Local Climate Zone (LCZ). A standardized classification of urban and rural landscapes by surface structure and cover ([Stewart & Oke, 2012](#)).

Network median. The median air temperature of all sensors in a district at a given half-hour; the reference from which ΔT is measured.

Pseudoreplication. Treating non-independent observations (here, several nights at the same site) as independent replicates, which overstates the effective sample size.

Random effect. A model term that varies by group (here, a random intercept per site) and is assumed to be drawn from a distribution.

Sky view factor (SVF). The fraction of the sky hemisphere visible from a point; here computed from building obstructions within 200m of the sensor at 1.5m height.

Source area. The area around a sensor whose surface properties influence the air temperature it measures; here represented by circles of 50, 100 and 200m radius.

Trade winds. The prevailing north-easterly winds of the tropical North Pacific, which blow over O‘ahu on most days.

Urban canopy layer. The layer of air between the ground and the mean roof height, in which canopy-layer air temperature is measured.

Urban heat island (UHI). The excess temperature of urban areas over their non-urban surroundings; measured here as the within-district departure ΔT .

ΔT . The departure of a sensor's air temperature from its district network median, averaged over a night; the response variable of this thesis.

References

- Alcoforado, M.-J., & Andrade, H. (2006). Nocturnal urban heat island in Lisbon (Portugal): main features and modelling attempts. *Theoretical and Applied Climatology*, 84(1–3), 151–159. <https://doi.org/10.1007/s00704-005-0152-1>
- Anjos, M., & Lopes, A. (2017). Urban Heat Island and Park Cool Island Intensities in the Coastal City of Aracaju, North-Eastern Brazil. *Sustainability*, 9(8), 1379. <https://doi.org/10.3390/su9081379>
- Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software*, 67(1), 1–48. <https://doi.org/10.18637/jss.v067.i01>
- Bolker, B. M., Brooks, M. E., Clark, C. J., Geange, S. W., Poulsen, J. R., Stevens, M. H. H., & White, J.-S. S. (2009). Generalized linear mixed models: a practical guide for ecology and evolution. *Trends in Ecology & Evolution*, 24(3), 127–135. <https://doi.org/10.1016/j.tree.2008.10.008>
- Bowler, D. E., Buyung-Ali, L., Knight, T. M., & Pullin, A. S. (2010). Urban greening to cool towns and cities: A systematic review of the empirical evidence. *Landscape and Urban Planning*, 97(3), 147–155. <https://doi.org/10.1016/j.landurbplan.2010.05.006>
- Burnham, K. P., & Anderson, D. R. (2002). *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach* (2nd ed.). Springer. <https://doi.org/10.1007/b97636>
- Cao, G., Giambelluca, T. W., Stevens, D. E., & Schroeder, T. A. (2007). Inversion Variability in the Hawaiian Trade Wind Regime. *Journal of Climate*, 20(7), 1145–1160. <https://doi.org/10.1175/JCLI4033.1>
- CAPA Strategies. (2019). *Heat Watch Report: Honolulu, HI*. CAPA Strategies, LLC. <https://osf.io/tdngm/>
- Chapman, S., Thatcher, M., Salazar, A., Watson, J. E. M., & McAlpine, C. A. (2018). The Effect of Urban Density and Vegetation Cover on the Heat Island of a Subtropical City. *Journal of Applied Meteorology and Climatology*, 57(11), 2531–2550. <https://doi.org/10.1175/JAMC-D-17-0316.1>
- Cheval, S., Amihăesei, V.-A., Chitu, Z., Dumitrescu, A., Falcescu, V., Iraşoc, A., Micu, D. M., Mihuleţ, E., Ontel, I., Paraschiv, M.-G., & Tudose, N. C. (2024). A systematic review of urban heat island and heat waves research (1991–2022). *Climate Risk Management*, 44, 100603. <https://doi.org/10.1016/j.crm.2024.100603>
- Chow, W. T. L., & Roth, M. (2006). Temporal dynamics of the urban heat island of Singapore. *International Journal of Climatology*, 26(15), 2243–2260. <https://doi.org/10.1002/joc.1364>
- City and County of Honolulu. (2024). *Climate ready O‘ahu: A climate adaptation strategy*. Office of Climate Change, Sustainability and Resiliency, Honolulu, HI. Adopted by the City Council in February 2024. <https://www.resilientoahu.org/climatereadyoahu>
- City and County of Honolulu, Department of Parks and Recreation. (n.d.). *Kapi‘olani Regional Park*. Retrieved September 26, 2026, from <https://www.honolulu.gov/dpr/kapiolani-regional-park/>
- Clement, A., Troxler, T., Keefe, O., Arcodia, M., Cruz, M., Hernandez, A., Moanga, D., Adefris, Z., Brown, N., & Jacobson, S. (2023). Hyperlocal Observations Reveal Persistent Extreme Urban Heat in Southeast Florida. *Journal of Applied Meteorology and Climatology*, 62(7), 863–872. <https://doi.org/10.1175/JAMC-D-22-0165.1>

- Comarazamy, D. E., González, J. E., Luvall, J. C., Rickman, D. L., & Mulero, P. J. (2010). A Land–Atmospheric Interaction Study in the Coastal Tropical City of San Juan, Puerto Rico. *Earth Interactions*, 14(16), 1–24. <https://doi.org/10.1175/2010EI309.1>
- Emmanuel, R., & Johansson, E. (2006). Influence of urban morphology and sea breeze on hot humid microclimate: the case of Colombo, Sri Lanka. *Climate Research*, 30, 189–200. <https://doi.org/10.3354/cr030189>
- Erell, E., & Williamson, T. (2007). Intra-urban differences in canopy layer air temperature at a mid-latitude city. *International Journal of Climatology*, 27(9), 1243–1255. <https://doi.org/10.1002/joc.1469>
- Federal Emergency Management Agency and Oak Ridge National Laboratory. (2025). *USA structures: Building footprints and attributes for Hawaii*. <https://gis-fema.hub.arcgis.com/pages/usa-structures>.
- Fernandes, R., Nascimento, V., Freitas, M., & Ometto, J. (2023). Local climate zones to identify surface urban heat islands: A systematic review. *Remote Sensing*, 15(4), 884. <https://doi.org/10.3390/rs15040884>
- Fox, J., & Weisberg, S. (2019). *An R companion to applied regression* (3rd ed.). Sage. <https://www.john-fox.ca/Companion/>
- Freitas, E. D., Rozoff, C. M., Cotton, W. R., & Silva Dias, P. L. (2007). Interactions of an urban heat island and sea-breeze circulations during winter over the metropolitan area of São Paulo, Brazil. *Boundary-Layer Meteorology*, 122(1), 43–65. <https://doi.org/10.1007/s10546-006-9091-3>
- Garza, J. A., Chu, P.-S., Norton, C. W., & Schroeder, T. A. (2012). Changes of the prevailing trade winds over the islands of Hawaii and the North Pacific. *Journal of Geophysical Research: Atmospheres*, 117(D11), D11109. <https://doi.org/10.1029/2011JD016888>
- Giambelluca, T. W., Chen, Q., Frazier, A. G., Price, J. P., Chen, Y.-L., Chu, P.-S., Eischeid, J. K., & Delparte, D. M. (2013). Online rainfall atlas of Hawaii. *Bulletin of the American Meteorological Society*, 94(3), 313–316. <https://doi.org/10.1175/BAMS-D-11-00228.1>
- Giambelluca, T. W., Diaz, H. F., & Luke, M. S. A. (2008). Secular temperature changes in Hawai‘i. *Geophysical Research Letters*, 35(12), L12702. <https://doi.org/10.1029/2008GL034377>
- Gillies, S. et al. (2007). *Shapely: Manipulation and analysis of geometric objects*. <https://github.com/shapely/shapely>
- Gillies, S. et al. (2013). *Rasterio: Geospatial raster I/O for python programmers*. Mapbox. <https://github.com/rasterio/rasterio>
- Giridharan, R., Lau, S. S. Y., & Ganesan, S. (2005). Nocturnal heat island effect in urban residential developments of Hong Kong. *Energy and Buildings*, 37(9), 964–971. <https://doi.org/10.1016/j.enbuild.2004.12.005>
- Giridharan, R., Lau, S. S. Y., Ganesan, S., & Givoni, B. (2007). Urban design factors influencing heat island intensity in high-rise high-density environments of Hong Kong. *Building and Environment*, 42(10), 3669–3684. <https://doi.org/10.1016/j.buildenv.2006.09.011>
- Grolemund, G., & Wickham, H. (2011). Dates and times made easy with lubridate. *Journal of Statistical Software*, 40(3), 1–25. <https://doi.org/10.18637/jss.v040.i03>

- Guo, A., Yue, W., Yang, J., He, T., Zhang, M., & Li, M. (2022). Divergent impact of urban 2D/3D morphology on thermal environment along urban gradients. *Urban Climate*, 45, 101278. <https://doi.org/10.1016/j.uclim.2022.101278>
- Handy, E. S. C., Handy, E. G., & Pukui, M. K. (1972). *Native Planters in Old Hawaii: Their Life, Lore, and Environment*. Bishop Museum Press. <https://bishopmuseumpress.org/products/native-planters-in-old-hawaii-their-life-lore-and-environment-ebook>
- Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk, M. H. van, Brett, M., Haldane, A., Fernández del Río, J., Wiebe, M., Peterson, P., ... Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, 585(7825), 357–362. <https://doi.org/10.1038/s41586-020-2649-2>
- Harrison, X. A., Donaldson, L., Correa-Cano, M. E., Evans, J., Fisher, D. N., Goodwin, C. E. D., Robinson, B. S., Hodgson, D. J., & Inger, R. (2018). A brief introduction to mixed effects modelling and multi-model inference in ecology. *PeerJ*, 6, e4794. <https://doi.org/10.7717/peerj.4794>
- Hart, M. A., & Sailor, D. J. (2009). Quantifying the influence of land-use and surface characteristics on spatial variability in the urban heat island. *Theoretical and Applied Climatology*, 95(3–4), 397–406. <https://doi.org/10.1007/s00704-008-0017-5>
- Hawai‘inuiākea School of Hawaiian Knowledge. (n.d.). *History: Ka Papa Lo‘i o Kānewai*. University of Hawai‘i at Mānoa. Retrieved September 26, 2026, from <https://manoa.hawaii.edu/hshk/ka-papa-loi-o-kanewai/history/>
- Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., et al. (2020). The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730), 1999–2049. <https://doi.org/10.1002/qj.3803>
- Heusinkveld, B. G., Steeneveld, G. J., Hove, L. W. A. van, Jacobs, C. M. J., & Holtslag, A. A. M. (2014). Spatial variability of the Rotterdam urban heat island as influenced by urban land use. *Journal of Geophysical Research: Atmospheres*, 119(2), 677–692. <https://doi.org/10.1002/2012JD019399>
- Hijmans, R. J. (2024). *Terra: Spatial data analysis*. <https://CRAN.R-project.org/package=terra>
- Holden, Z. A., Klene, A. E., Keefe, R. F., & Moisen, G. G. (2013). Design and evaluation of an inexpensive radiation shield for monitoring surface air temperatures. *Agricultural and Forest Meteorology*, 180, 281–286. <https://doi.org/10.1016/j.agrformet.2013.06.011>
- Howard, L. (1833). *The Climate of London, Deduced from Meteorological Observations* (2nd ed.). Harvey and Darton. <https://archive.org/details/climatelondonde01howagoog>
- Hu, Y., Dai, Z., & Guldman, J.-M. (2020). Modeling the impact of 2D/3D urban indicators on the urban heat island over different seasons: A boosted regression tree approach. *Journal of Environmental Management*, 266, 110424. <https://doi.org/10.1016/j.jenvman.2020.110424>
- Huang, X., & Wang, Y. (2019). Investigating the effects of 3D urban morphology on the surface urban heat island effect in urban functional zones by using high-resolution remote sensing data: A case study of Wuhan, Central China. *ISPRS Journal of Photogrammetry and Remote Sensing*, 152, 119–131. <https://doi.org/10.1016/j.isprsjprs.2019.04.010>

- Hubbart, J., Link, T., Campbell, C., & Cobos, D. (2005). Evaluation of a low-cost temperature measurement system for environmental applications. *Hydrological Processes*, 19(7), 1517–1523. <https://doi.org/10.1002/hyp.5861>
- Hurlbert, S. H. (1984). Pseudoreplication and the design of ecological field experiments. *Ecological Monographs*, 54(2), 187–211. <https://doi.org/10.2307/1942661>
- Ibsen, P. C., Borowy, D., Dell, T., Greydanus, H., Gupta, N., Hondula, D. M., Meixner, T., Santelmann, M. V., Shiflett, S. A., Sukop, M. C., Swan, C. M., Talal, M. L., Valencia, M., Wright, M. K., & Jenerette, G. D. (2021). Greater aridity increases the magnitude of urban nighttime vegetation-derived air cooling. *Environmental Research Letters*, 16(3), 034011. <https://doi.org/10.1088/1748-9326/abdf8a>
- Ibsen, P. C., Jenerette, G. D., Dell, T., Bagstad, K. J., & Diffendorfer, J. E. (2022). Urban landcover differentially drives day and nighttime air temperature across a semi-arid city. *Science of The Total Environment*, 829, 154589. <https://doi.org/10.1016/j.scitotenv.2022.154589>
- Imhoff, M. L., Zhang, P., Wolfe, R. E., & Bounoua, L. (2010). Remote sensing of the urban heat island effect across biomes in the continental USA. *Remote Sensing of Environment*, 114(3), 504–513. <https://doi.org/10.1016/j.rse.2009.10.008>
- Johnson, G. T., & Watson, I. D. (1984). The determination of view-factors in urban canyons. *Journal of Climate and Applied Meteorology*, 23(2), 329–335. [https://doi.org/10.1175/1520-0450\(1984\)023<0329:TDOVFI>2.0.CO;2](https://doi.org/10.1175/1520-0450(1984)023<0329:TDOVFI>2.0.CO;2)
- Kim, Y.-H., & Baik, J.-J. (2002). Maximum Urban Heat Island Intensity in Seoul. *Journal of Applied Meteorology*, 41(6), 651–659. [https://doi.org/10.1175/1520-0450\(2002\)041<0651:MUHIII>2.0.CO;2](https://doi.org/10.1175/1520-0450(2002)041<0651:MUHIII>2.0.CO;2)
- Kurashima, N., Fortini, L., & Ticktin, T. (2019). The potential of indigenous agricultural food production under climate change in Hawai‘i. *Nature Sustainability*, 2(3), 191–199. <https://doi.org/10.1038/s41893-019-0226-1>
- Kuznetsova, A., Brockhoff, P. B., & Christensen, R. H. B. (2017). lmerTest package: Tests in linear mixed effects models. *Journal of Statistical Software*, 82(13), 1–26. <https://doi.org/10.18637/jss.v082.i13>
- Laaidi, K., Zeghnoun, A., Dousset, B., Bretin, P., Vandentorren, S., Giraudet, E., & Beaudeau, P. (2012). The Impact of Heat Islands on Mortality in Paris during the August 2003 Heat Wave. *Environmental Health Perspectives*, 120(2), 254–259. <https://doi.org/10.1289/ehp.1103532>
- Lefevre, A., Malet-Damour, B., Boyer, H., & Rivière, G. (2025). Urban heat island in the tropics: A review of advances, challenges, and future directions. *City and Environment Interactions*, 28, 100265. <https://doi.org/10.1016/j.cacint.2025.100265>
- Leopold, L. B. (1949). The interaction of trade wind and sea breeze, Hawaii. *Journal of Meteorology*, 6(5), 312–320. [https://doi.org/10.1175/1520-0469\(1949\)006<0312:TIOTWA>2.0.CO;2](https://doi.org/10.1175/1520-0469(1949)006<0312:TIOTWA>2.0.CO;2)
- Li, Y., Schubert, S., Kropp, J. P., & Rybski, D. (2020). On the influence of density and morphology on the Urban Heat Island intensity. *Nature Communications*, 11(1), 2647. <https://doi.org/10.1038/s41467-020-16461-9>
- Lüdecke, D., Ben-Shachar, M. S., Patil, I., Waggoner, P., & Makowski, D. (2021). Performance: An R package for assessment, comparison and testing of statistical models. *Journal of Open Source Software*, 6(60), 3139. <https://doi.org/10.21105/joss.03139>

- Manoli, G., Fatichi, S., Schläpfer, M., Yu, K., Crowther, T. W., Meili, N., Burlando, P., Katul, G. G., & Bou-Zeid, E. (2019). Magnitude of urban heat islands largely explained by climate and population. *Nature*, 573(7772), 55–60. <https://doi.org/10.1038/s41586-019-1512-9>
- McKenzie, M. M., Giambelluca, T. W., & Diaz, H. F. (2019). Temperature trends in Hawai‘i: A century of change, 1917–2016. *International Journal of Climatology*, 39(10), 3987–4001. <https://doi.org/10.1002/joc.6053>
- McKinney, W. (2010). Data structures for statistical computing in python. *Proceedings of the 9th Python in Science Conference*, 56–61. <https://doi.org/10.25080/Majora-92bf1922-00a>
- Mora, C., Dousset, B., Caldwell, I. R., Powell, F. E., Geronimo, R. C., Bielecki, C. R., Counsell, C. W., Dietrich, B. S., Johnston, E. T., Louis, L. V., Lucas, M. P., McKenzie, M. M., Shea, A. G., Tseng, H., Giambelluca, T. W., Leon, L. R., Hawkins, E., & Trauernicht, C. (2017). Global risk of deadly heat. *Nature Climate Change*, 7(7), 501–506. <https://doi.org/10.1038/nclimate3322>
- Morris, C. J. G., Simmonds, I., & Plummer, N. (2001). Quantification of the influences of wind and cloud on the nocturnal urban heat island of a large city. *Journal of Applied Meteorology*, 40(2), 169–182. [https://doi.org/10.1175/1520-0450\(2001\)040<0169:QOTIOW>2.0.CO;2](https://doi.org/10.1175/1520-0450(2001)040<0169:QOTIOW>2.0.CO;2)
- Muller, C. L., Chapman, L., Grimmond, C. S. B., Young, D. T., & Cai, X. (2013). Sensors and the city: A review of urban meteorological networks. *International Journal of Climatology*, 33(7), 1585–1600. <https://doi.org/10.1002/joc.3678>
- Murage, P., Hajat, S., & Kovats, R. S. (2017). Effect of night-time temperatures on cause and age-specific mortality in London. *Environmental Epidemiology*, 1(2), e005. <https://doi.org/10.1097/EE9.0000000000000005>
- Murphy, D. J., Hall, M. H., Hall, C. A. S., Heisler, G. M., Stehman, S. V., & Anselmi-Molina, C. (2011). The relationship between land cover and the urban heat island in northeastern Puerto Rico. *International Journal of Climatology*, 31(8), 1222–1239. <https://doi.org/10.1002/joc.2145>
- Muse, N., Clement, A., & Mach, K. J. (2024). Daytime land surface temperature and its limits as a proxy for surface air temperature in a subtropical, seasonally wet region. *PLOS Climate*, 3(10), e0000278. <https://doi.org/10.1371/journal.pclm.0000278>
- Nakagawa, S., & Schielzeth, H. (2013). A general and simple method for obtaining R^2 from generalized linear mixed-effects models. *Methods in Ecology and Evolution*, 4(2), 133–142. <https://doi.org/10.1111/j.2041-210x.2012.00261.x>
- NOAA Center for Operational Oceanographic Products and Services. (n.d.). *Water temperature at station 1612340, Honolulu, HI: hourly observations, November 2024 to January 2025*. NOAA Tides and Currents. Retrieved September 26, 2026, from <https://tidesandcurrents.noaa.gov/stationhome.html?id=1612340>
- NOAA National Centers for Environmental Information. (2021). *U.S. Climate Normals 1991–2020: monthly normals for Honolulu International Airport (station USW00022521)*. NCEI Climate Data Online. <https://www.ncei.noaa.gov/access/us-climate-normals/>
- NOAA Office for Coastal Management. (2024). *C-CAP high-resolution land cover: Hawaii 2021 impervious surface, tree canopy and water (version 2)*. Coastal Change Analysis Program (C-CAP), Charleston, SC. <https://coast.noaa.gov/digitalcoast/data/ccaphighres.html>
- O’Brien, R. M. (2007). A Caution Regarding Rules of Thumb for Variance Inflation Factors. *Quality & Quantity*, 41(5), 673–690. <https://doi.org/10.1007/s11135-006-9018-6>

- Ojeh, V., Balogun, A., & Okhimamhe, A. (2016). Urban-Rural Temperature Differences in Lagos. *Climate*, 4(2), 29. <https://doi.org/10.3390/cli4020029>
- Oke, T. R. (1973). City size and the urban heat island. *Atmospheric Environment*, 7(8), 769–779. [https://doi.org/10.1016/0004-6981\(73\)90140-6](https://doi.org/10.1016/0004-6981(73)90140-6)
- Oke, T. R. (1976). The distinction between canopy and boundary-layer urban heat islands. *Atmosphere*, 14(4), 268–277. <https://doi.org/10.1080/00046973.1976.9648422>
- Oke, T. R. (1981). Canyon geometry and the nocturnal urban heat island: Comparison of scale model and field observations. *Journal of Climatology*, 1(3), 237–254. <https://doi.org/10.1002/joc.3370010304>
- Oke, T. R. (1982). The energetic basis of the urban heat island. *Quarterly Journal of the Royal Meteorological Society*, 108(455), 1–24. <https://doi.org/10.1002/qj.49710845502>
- Oke, T. R. (1988). Street design and urban canopy layer climate. *Energy and Buildings*, 11(1–3), 103–113. [https://doi.org/10.1016/0378-7788\(88\)90026-6](https://doi.org/10.1016/0378-7788(88)90026-6)
- Oke, T. R. (2002). *Boundary Layer Climates* (2nd ed.). Routledge. <https://doi.org/10.4324/9780203407219>
- Oke, T. R. (2006). *Initial guidance to obtain representative meteorological observations at urban sites* (Instruments and Observing Methods Report 81, WMO/TD-No. 1250). World Meteorological Organization. <https://library.wmo.int/viewer/35333>
- Oke, T. R., Johnson, G. T., Steyn, D. G., & Watson, I. D. (1991). Simulation of surface urban heat islands under “ideal” conditions at night. Part 2: Diagnosis of causation. *Boundary-Layer Meteorology*, 56(4), 339–358. <https://doi.org/10.1007/BF00119211>
- Oke, T. R., Mills, G., Christen, A., & Voogt, J. A. (2017). *Urban climates*. Cambridge University Press. <https://doi.org/10.1017/9781139016476>
- Ooms, J. (2014). *The jsonlite package: A practical and consistent mapping between JSON data and R objects*. arXiv:1403.2805. <https://arxiv.org/abs/1403.2805>
- Pebesma, E. (2018). Simple features for R: Standardized support for spatial vector data. *The R Journal*, 10(1), 439–446. <https://doi.org/10.32614/RJ-2018-009>
- Pedersen, T. L. (2024). *Patchwork: The composer of plots*. <https://CRAN.R-project.org/package=patchwork>
- R Core Team. (2024). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. <https://www.R-project.org/>
- Ramamurthy, P., Pena, J. C., Liu, S., & Gonzalez-Cruz, J. (2025). The interaction between urban heat island intensity and sea-breeze effect. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 383(2308), 20240578. <https://doi.org/10.1098/rsta.2024.0578>
- Roth, M. (2007). Review of urban climate research in (sub)tropical regions. *International Journal of Climatology*, 27(14), 1859–1873. <https://doi.org/10.1002/joc.1591>
- Runnalls, K. E., & Oke, T. R. (2000). Dynamics and controls of the near-surface heat island of Vancouver, British Columbia. *Physical Geography*, 21(4), 283–304. <https://doi.org/10.1080/02723646.2000.10642711>

- Sanderson, M. (Ed.). (1993). *Prevailing trade winds: Weather and climate in Hawai‘i*. University of Hawai‘i Press. <https://uhpress.hawaii.edu/title/prevailing-trade-winds-weather-and-climate-in-hawaii/>
- Schroeder, T. A. (1993). Climate controls. In M. Sanderson (Ed.), *Prevailing trade winds: Weather and climate in Hawai‘i* (pp. 12–36). University of Hawai‘i Press. <https://doi.org/10.1515/9780824841775-005>
- Schwaab, J., Meier, R., Mussetti, G., Seneviratne, S., Bürgi, C., & Davin, E. L. (2021). The role of urban trees in reducing land surface temperatures in European cities. *Nature Communications*, 12(1), 6763. <https://doi.org/10.1038/s41467-021-26768-w>
- Scott, A. A., Misiani, H., Okoth, J., Jordan, A., Gohlke, J., Ouma, G., Arrighi, J., Zaitchik, B. F., Jjemba, E., Verjee, S., & Waugh, D. W. (2017). Temperature and heat in informal settlements in Nairobi. *PLOS ONE*, 12(11), e0187300. <https://doi.org/10.1371/journal.pone.0187300>
- Scott, A. A., Zaitchik, B., Waugh, D. W., & O’Meara, K. (2017). Intraurban temperature variability in Baltimore. *Journal of Applied Meteorology and Climatology*, 56(1), 159–171. <https://doi.org/10.1175/JAMC-D-16-0232.1>
- Seabold, S., & Perktold, J. (2010). Statsmodels: Econometric and statistical modeling with Python. *Proceedings of the 9th Python in Science Conference*, 92–96. <https://doi.org/10.25080/Majora-92bf1922-011>
- Seto, K. C., Güneralp, B., & Hutyra, L. R. (2012). Global forecasts of urban expansion to 2030 and direct impacts on biodiversity and carbon pools. *Proceedings of the National Academy of Sciences*, 109(40), 16083–16088. <https://doi.org/10.1073/pnas.1211658109>
- Steenekveld, G. J., Koopmans, S., Heusinkveld, B. G., Hove, L. W. A. van, & Holtslag, A. A. M. (2011). Quantifying urban heat island effects and human comfort for cities of variable size and urban morphology in the Netherlands. *Journal of Geophysical Research*, 116(D20), D20129. <https://doi.org/10.1029/2011JD015988>
- Stewart, I. D. (2011). A systematic review and scientific critique of methodology in modern urban heat island literature. *International Journal of Climatology*, 31(2), 200–217. <https://doi.org/10.1002/joc.2141>
- Stewart, I. D., & Oke, T. R. (2012). Local climate zones for urban temperature studies. *Bulletin of the American Meteorological Society*, 93(12), 1879–1900. <https://doi.org/10.1175/BAMS-D-11-00019.1>
- Svensson, M. K. (2004). Sky view factor analysis – implications for urban air temperature differences. *Meteorological Applications*, 11(3), 201–211. <https://doi.org/10.1017/S1350482704001288>
- Tarara, J. M., & Hoheisel, G.-A. (2007). Low-cost Shielding to Minimize Radiation Errors of Temperature Sensors in the Field. *HortScience*, 42(6), 1372–1379. <https://doi.org/10.21273/HORTSCI.42.6.1372>
- Terando, A. J., Youngsteadt, E., Meineke, E. K., & Prado, S. G. (2017). Ad hoc instrumentation methods in ecological studies produce highly biased temperature measurements. *Ecology and Evolution*, 7(23), 9890–9904. <https://doi.org/10.1002/ece3.3499>
- Tian, Y., Zhou, W., Qian, Y., Zheng, Z., & Yan, J. (2019). The effect of urban 2D and 3D morphology on air temperature in residential neighborhoods. *Landscape Ecology*, 34(5), 1161–1178. <https://doi.org/10.1007/s10980-019-00834-7>

- Unger, J. (2004). Intra-urban relationship between surface geometry and urban heat island: review and new approach. *Climate Research*, 27, 253–264. <https://doi.org/10.3354/cr027253>
- United Nations, Department of Economic and Social Affairs, Population Division. (2019). *World Urbanization Prospects: The 2018 Revision* (ST/ESA/SER.A/420). United Nations. <https://population.un.org/wup/assets/WUP2018-Report.pdf>
- U.S. Census Bureau. (2021). *2020 Census Redistricting Data (Public Law 94-171) Summary File: Honolulu County, Hawaii*. QuickFacts: Honolulu County, Hawaii. <https://www.census.gov/quickfacts/honolulucountyhawaii>
- U.S. Department of Agriculture, National Agricultural Statistics Service. (2024). *2022 Census of Agriculture: Hawaii State and County Data* (Volume 1, Geographic Area Series Part 11). U.S. Department of Agriculture, National Agricultural Statistics Service. https://www.nass.usda.gov/Publications/AgCensus/2022/Full_Report/Volume_1,_Chapter_2_County_Level/Hawaii/hiv1.pdf
- U.S. Geological Survey. (n.d.). *3D Elevation Program 1-meter resolution digital elevation model*. The National Map; accessed through the Elevation Point Query Service. Retrieved September 19, 2026, from <https://epqs.nationalmap.gov/>
- Velazquez-Lozada, A., Gonzalez, J. E., & Winter, A. (2006). Urban heat island effect analysis for San Juan, Puerto Rico. *Atmospheric Environment*, 40(9), 1731–1741. <https://doi.org/10.1016/j.atmosenv.2005.09.074>
- Venter, Z. S., Chakraborty, T., & Lee, X. (2021). Crowdsourced air temperatures contrast satellite measures of the urban heat island and its mechanisms. *Science Advances*, 7(22), eabb9569. <https://doi.org/10.1126/sciadv.abb9569>
- Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., Walt, S. J. van der, Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... SciPy 1.0 Contributors. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in python. *Nature Methods*, 17(3), 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- Wessel, P., & Smith, W. H. F. (1996). A global, self-consistent, hierarchical, high-resolution shoreline database. *Journal of Geophysical Research: Solid Earth*, 101(B4), 8741–8743. <https://doi.org/10.1029/96JB00104>
- Western Regional Climate Center. (n.d.). *Climate of Hawaii*. Desert Research Institute. Retrieved September 26, 2026, from https://wrcc-archive.dri.edu/Climate/narrative_hi.php
- Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer International Publishing. <https://doi.org/10.1007/978-3-319-24277-4>
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., ... Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686. <https://doi.org/10.21105/joss.01686>
- World Meteorological Organization. (2024). *Guide to Instruments and Methods of Observation, Volume I – Measurement of Meteorological Variables* (WMO-No. 8, 2024 edition). World Meteorological Organization. <https://library.wmo.int/records/item/68695-guide-to-instruments-and-methods-of-observation>

- Yang, J., Xin, J., Zhang, Y., Xiao, X., & Xia, J. C. (2022). Contributions of sea–land breeze and local climate zones to daytime and nighttime heat island intensity. *Npj Urban Sustainability*, 2, 12. <https://doi.org/10.1038/s42949-022-00055-z>
- Yang, Q., Huang, X., Yang, J., & Liu, Y. (2021). The relationship between land surface temperature and artificial impervious surface fraction in 682 global cities: spatiotemporal variations and drivers. *Environmental Research Letters*, 16(2), 024032. <https://doi.org/10.1088/1748-9326/abdae>
- Yoshikado, H. (1992). Numerical Study of the Daytime Urban Effect and Its Interaction with the Sea Breeze. *Journal of Applied Meteorology*, 31(10), 1146–1164. [https://doi.org/10.1175/1520-0450\(1992\)031<1146:NSOTDU>2.0.CO;2](https://doi.org/10.1175/1520-0450(1992)031<1146:NSOTDU>2.0.CO;2)
- Yuan, B., Zhou, L., Hu, F., & Wei, C. (2024). Effects of 2D/3D urban morphology on land surface temperature: Contribution, response, and interaction. *Urban Climate*, 53, 101791. <https://doi.org/10.1016/j.uclim.2023.101791>
- Yuan, F., & Bauer, M. E. (2007). Comparison of impervious surface area and normalized difference vegetation index as indicators of surface urban heat island effects in Landsat imagery. *Remote Sensing of Environment*, 106(3), 375–386. <https://doi.org/10.1016/j.rse.2006.09.003>
- Zhang, Z., Paschalis, A., Mijic, A., Meili, N., Manoli, G., Reeuwijk, M. van, & Fatichi, S. (2022). A mechanistic assessment of urban heat island intensities and drivers across climates. *Urban Climate*, 44, 101215. <https://doi.org/10.1016/j.uclim.2022.101215>
- Zhao, L., Lee, X., Smith, R. B., & Oleson, K. (2014). Strong contributions of local background climate to urban heat islands. *Nature*, 511(7508), 216–219. <https://doi.org/10.1038/nature13462>
- Zhou, B., Rybski, D., & Kropp, J. P. (2017). The role of city size and urban form in the surface urban heat island. *Scientific Reports*, 7(1), 4791. <https://doi.org/10.1038/s41598-017-04242-2>
- Zhou, Y., Guan, H., Gharib, S., Batelaan, O., & Simmons, C. T. (2021). Cooling power of sea breezes and its inland penetration in dry-summer Adelaide, Australia. *Atmospheric Research*, 250, 105409. <https://doi.org/10.1016/j.atmosres.2020.105409>
- Zhu, Z., Shen, Y., Fu, W., Zheng, D., Huang, P., Li, J., Lan, Y., Chen, Z., Liu, Q., Xu, X., & Yao, X. (2023). How does 2D and 3D of urban morphology affect the seasonal land surface temperature in island city? A block-scale perspective. *Ecological Indicators*, 150, 110221. <https://doi.org/10.1016/j.ecolind.2023.110221>
- Zippenfenig, P. (2023). *Open-meteo.com weather API*. Zenodo. <https://doi.org/10.5281/zenodo.7970649>
- Ziter, C. D., Pedersen, E. J., Kucharik, C. J., & Turner, M. G. (2019). Scale-dependent interactions between tree canopy cover and impervious surfaces reduce daytime urban heat during summer. *Proceedings of the National Academy of Sciences*, 116(15), 7575–7580. <https://doi.org/10.1073/pnas.1817561116>
- Zuur, A. F., Ieno, E. N., Walker, N., Saveliev, A. A., & Smith, G. M. (2009). *Mixed Effects Models and Extensions in Ecology with R*. Springer. <https://doi.org/10.1007/978-0-387-87458-6>